MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
3370 FLOTATION PROCESS WATER STORAGE AND DISTRIBUTION
DETAILS OF STAIR FOUNDATIONS

Document Number: **MD-216-3370-EG-CV-CAL-1233**
Revision: **D01**
Revision Date: **17.11.2023**
Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	AG01	CED01	33702	OU100681342	2
L&T Document Number:	O19045	450	C	K	24103	002	2

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	18.04.2020	FIRST ISSUE	MMJ	AVG	SJM	
C01	15.05.2020	ISSUED FOR CONSTRUCTION	MMJ	AVG	SJM	
D01	17.11.2023	AS BUILT	MMJ	AVG	SJM	

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS ……4

1. INTENT……5

2. EXECUTIVE SUMMARY……5

3. SCOPE……6

4. DESIGN BASIS

4.1 Materials……6
4.2 Loading……6
4.3 Load Combinations……7
4.4 Minimum Reinforcing Steel (Rebar) Size……8
4.5 Minimum Concrete Cover……8
4.6 Allowable Soil Bearing Capacity……8
4.7 Vertical Modulus of Subgrade Reaction……8

5. DESIGN CALCULATION

5.1 Design of foundation raft F1……9
5.2 Design calculation of Footing F2……12
5.3 Design of column C1……17
5.4 Design of column C2……18

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

References……..……19

ATTACHMENT 1 STAAD INPUT & OUTPUT FILE FOR F1 DESIGN……..……20

ATTACHMENT 2 STAAD INPUT & OUTPUT FILE FOR F2 DESIGN……...….82

ATTACHMENT 3 STAAD INPUT & OUTPUT FILE FOR F2 STABILITY……107

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

1. INTENT

The purpose of this document is to analysis and design of **Floatation process water storage and distribution stair** foundations. In this document design & analysis of foundation for staircase have been covered. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

3. SCOPE

The analysis has been carried out in Staad pro & designed by Excel Sheet. For member strength calculation force values are directly taken from STAAD Pro & designed in excel sheet. Load data for foundation design taken from document no. **MD-216-3370-EG-ST-CAL-1002**.

4. DESIGN BASIS

4.1 Material

A. Concrete Strength of 28 days

Foundation and Structural Members: 35 MPa

Blinding Concrete below foundation bases: 15 MPa

B. Reinforcement Grade

Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)

Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

C. Grouting

Grout shall be 50 mm cementitious of Strength 40 MPa.

4.2 Loading

A. Primary Load

1 DL= Dead load (self weight)

2 LL= Live Load

3 WL +X = Wind load in +X direction

4 WL -X = Wind load in -X direction

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5 WL +Z = Wind load in +Z direction

6 WL -Z = Wind load in -Z direction

7 HL = Soil Load

4.3 Load combination

A. The following load combination are considered for base pressure, sliding & overturning check.

201 1.0DL+1.0HL+1.0LL

202 1.0DL+1.0HL+0.75LL

203 1.0DL+1.0HL+1.0WLX1

204 1.0DL+1.0HL+1.0WLX2

205 1.0DL+1.0HL+1.0WLZ1

206 1.0DL+1.0HL+1.0WLZ2

207 1.0DL+1.0HL+0.75LL+0.75WLX1

208 1.0DL+1.0HL+0.75LL+0.75WLX2

209 1.0DL+1.0HL+0.75LL+0.75WLZ1

210 1.0DL+1.0HL+0.75LL+0.75WLZ2

211 0.6DL+1.0HL+1.0WLX1

212 0.6DL+1.0HL+1.0WLX2

213 0.6DL+1.0HL+1.0WLZ1

214 0.6DL+1.0HL+1.0WLX1

B. The following load combination are considered for member strength calculation.

101 1.4DL

102 1.4DL+1.7LL

103 1.2DL+1.0LL+1.6WLX1

104 1.2DL+1.0LL+1.6WLX2

105 1.2DL+1.0LL+1.6WLZ1

106 1.2DL+1.0LL+1.6WLZ2

107 0.9DL+1.6HL+1.6WLX1

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

108 0.9DL+1.6HL+1.6WLX2

109 0.9DL+1.6HL+1.6WLZ1

110 0.9DL+1.6HL+1.6WLZ2

4.4 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm except where welded wire fabric is used for ties and spirals the minimum rebar size shall be 10 mm.

4.5 Minimum concrete cover

	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation & Columns	50

4.6 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 375 kN/m2 at 0.6m depth below EGL (RL 912.000) refer Table 5 - BH-26 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.7 Vertical Modulus of Subgrade Reaction

The Vertical Modulus of Subgrade Reaction has been considered as 33750 kN/m^2/m at 0.6m depth below EGL (RL 912.000) refer Table 5 - BH-26 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5.1 Design of Foundation Raft F1 :

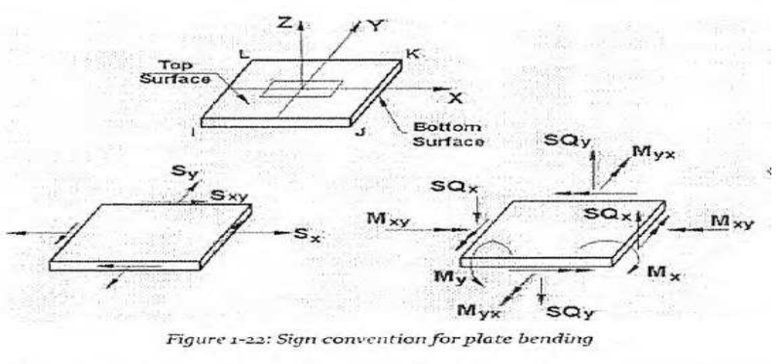
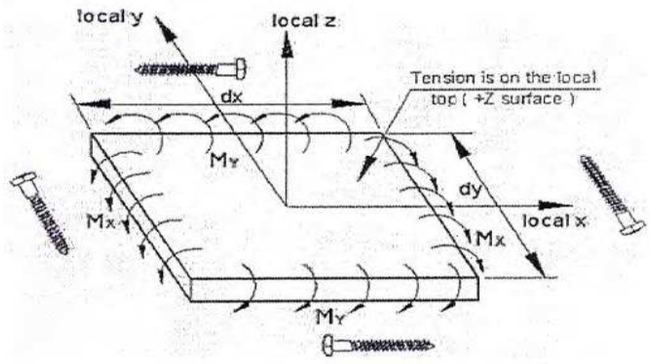


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; : FDN_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	780	0.326	-0.103	0	0	0	-41.3	-50.109	11.996	-53.296	-62.105	106 1.2DL+1.0LL+1.6WLZ2
Min Qx	1725	-0.399	0.052	0	0	0	-35.55	-54.763	0	-35.55	-54.763	105 1.2DL+1.0LL+1.6WLZ1
Max Qy	776	-0.075	0.411	0	0	0	-55.4	-50.356	1.917	-57.315	-52.273	106 1.2DL+1.0LL+1.6WLZ2
Min Qy	1706	-0.075	-0.409	0	0	0	-54.93	-50.66	-2.034	-56.962	-52.694	105 1.2DL+1.0LL+1.6WLZ1
Max Sx	2451	0.012	0.015	0	0	0	-3.073	-1.308	-1.856	-4.929	-3.164	101 1.4DL
Min Sx	2451	0.012	0.015	0	0	0	-3.073	-1.308	-1.856	-4.929	-3.164	101 1.4DL
Max Sy	2451	0.012	0.015	0	0	0	-3.073	-1.308	-1.856	-4.929	-3.164	101 1.4DL
Min Sy	2451	0.012	0.015	0	0	0	-3.073	-1.308	-1.856	-4.929	-3.164	101 1.4DL
Max Sxy	2451	0.012	0.015	0	0	0	-3.073	-1.308	-1.856	-4.929	-3.164	101 1.4DL
Min Sxy	2451	0.012	0.015	0	0	0	-3.073	-1.308	-1.856	-4.929	-3.164	101 1.4DL
Max Mx	194	0.001	-0.005	0	0	0	26.704	0.475	-0.183	26.887	0.658	102 1.4DL+1.7LL
Min Mx	783	0.037	0.401	0	0	0	-57.15	-51.269	13.619	-70.764	-64.888	106 1.2DL+1.0LL+1.6WLZ2
Max My	783	-0.034	-0.124	0	0	0	9.256	30.658	-1.422	10.678	32.08	109 0.9DL+1.6HL+1.6WLZ1
Min My	1725	-0.399	0.052	0	0	0	-35.55	-54.763	-3.583	-39.133	-58.346	105 1.2DL+1.0LL+1.6WLZ1
Max Mxy	625	0.003	-0.047	0	0	0	-21.65	-41.665	27.098	-48.745	-68.763	106 1.2DL+1.0LL+1.6WLZ2
Min Mxy	2273	0.003	0.05	0	0	0	-21.55	-42.048	-27.137	-48.687	-69.185	105 1.2DL+1.0LL+1.6WLZ1

PLATE CENTER STRESS ; FDN_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	178	0.203	0.01	0	0	0	0.078	-1.215	-6.844	6.922	-8.059	106 1.2DL+1.0LL+1.6WLZ2
Min Qx	1724	-0.242	0.005	0	0	0	-15.69	-42.328	0	-15.69	-42.328	105 1.2DL+1.0LL+1.6WLZ1
Max Qy	621	0.022	0.249	0	0	0	-38.49	-27.757	11.485	-49.976	-39.242	106 1.2DL+1.0LL+1.6WLZ2
Min Qy	2271	0.021	-0.247	0	0	0	-38.19	-28.228	-11.613	-49.807	-39.841	105 1.2DL+1.0LL+1.6WLZ1
Max Sx	2451	0.012	0.015	0	0	0	-3.073	-1.308	-1.856	-4.929	-3.164	101 1.4DL
Min Sx	2451	0.012	0.015	0	0	0	-3.073	-1.308	-1.856	-4.929	-3.164	101 1.4DL
Max Sy	2451	0.012	0.015	0	0	0	-3.073	-1.308	-1.856	-4.929	-3.164	101 1.4DL
Min Sy	2451	0.012	0.015	0	0	0	-3.073	-1.308	-1.856	-4.929	-3.164	101 1.4DL
Max Sxy	2451	0.012	0.015	0	0	0	-3.073	-1.308	-1.856	-4.929	-3.164	101 1.4DL
Min Sxy	2451	0.012	0.015	0	0	0	-3.073	-1.308	-1.856	-4.929	-3.164	101 1.4DL
Max Mx	194	0.001	-0.005	0	0	0	26.704	0.475	-0.183	26.887	0.658	102 1.4DL+1.7LL
Min Mx	1452	0.032	0.159	0	0	0	-40.47	-7.467	0.209	-40.679	-7.676	102 1.4DL+1.7LL
Max My	255	0.005	-0.047	0	0	0	0.408	25.587	-3.355	3.763	28.942	106 1.2DL+1.0LL+1.6WLZ2
Min My	1295	-0.055	0.018	0	0	0	-1.011	-48.232	-0.333	-1.344	-48.565	105 1.2DL+1.0LL+1.6WLZ1
Max Mxy	625	0.003	-0.047	0	0	0	-21.65	-41.665	27.098	-48.745	-68.763	106 1.2DL+1.0LL+1.6WLZ2
Min Mxy	2273	0.003	0.05	0	0	0	-21.55	-42.048	-27.137	-48.687	-69.185	105 1.2DL+1.0LL+1.6WLZ1

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

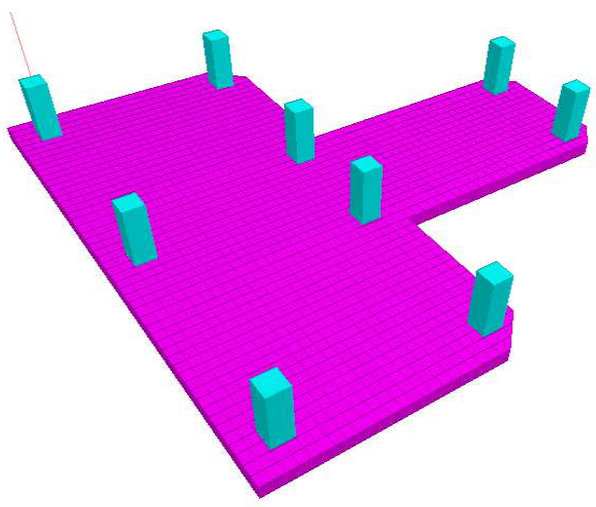
Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	26.89	top face	50	344	26.887	0.30	0.0007	249.4	720	12	157.14	150	754.3	10.64	13.29	0.075	tension control
Min Mx	-70.76	bottom face	50	344	70.764	0.80	0.0019	662.0		12	157.14	150	754.3	10.64	13.29	0.075	tension control
Max Mxy	-48.75	bottom face	50	344	48.745	0.55	0.0013	454.1		12	157.14	150	754.3	10.64	13.29	0.075	tension control
Min Mxy	-48.69	bottom face	50	344	48.687	0.55	0.0013	453.5		12	157.14	150	754.3	10.64	13.29	0.075	tension control

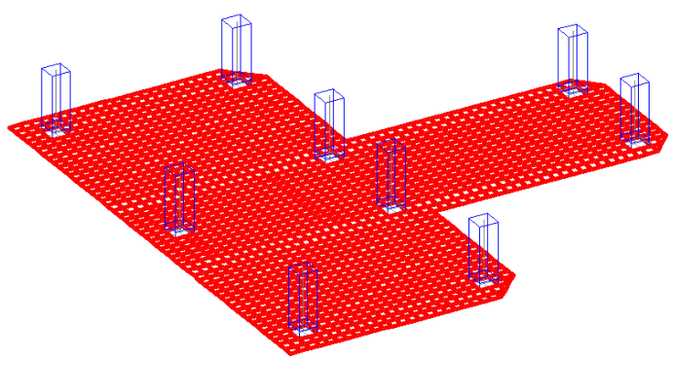
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.203	1	81.2	6.922	0.002	0.344	377.367	338.444	590.188	338.44	1261	> 81.2	ok	253.8	φVc>Vu	not required
	Min Qx	0.242	1	96.8	15.69			352.836			338.44	1261	> 96.8	ok	253.8	φVc>Vu	not required

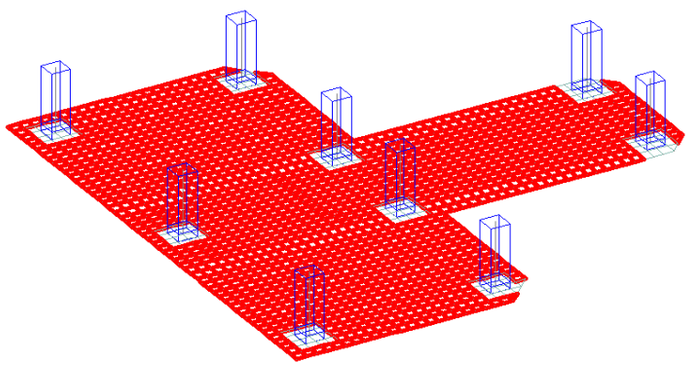
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	32.08	top face	50	344	32.08	0.36	0.0009	297.9	720	12	157.14	150	754.3	10.64	13.29	0.075	tension control
Min My	-58.35	bottom face	50	344	58.346	0.66	0.0016	544.5		12	157.14	150	754.3	10.64	13.29	0.075	tension control
Max Mxy	-68.76	bottom face	50	344	68.763	0.77	0.0019	643.1		12	157.14	150	754.3	10.64	13.29	0.075	tension control
Min Mxy	-69.19	bottom face	50	344	69.185	0.78	0.0019	647.1		12	157.14	150	754.3	10.64	13.29	0.075	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.249	1	99.6	39.242	0.002	0.344	fc'336.8	fc'338.4	fc'590.2	336.82	353	> 99.6	ok	252.6	φVc>Vu	not required
	Min Qy	0.247	1	98.8	39.841			336.56			336.56	353	> 98.8	ok	252.4	φVc>Vu	not required







3D Rendered View of foundation

Highlighted plate for moment (at column face)

Highlighted plate for shear (at d distance from column face)

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	12ϕT	**150**	c/c
At Top	12ϕT	**150**	c/c
Along Z Direction :			
At Bottom	12ϕT	**150**	c/c
At Top	12ϕT	**150**	c/c

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Base pressure check of Footing F1 :

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	1	201	0	31.629	0
Min Px	1	201	0	31.629	0
Max Py	39	210	0	45.511	0
Min Py	1878	214	0	0.365	0
Max Pz	1	201	0	31.629	0
Min Pz	1	201	0	31.629	0

Gross Maximum Base pressure, p_{max} = 45.5 KN/m^2

Net Allowable Safe Bearing Capacity = 375 KN/m^2 (Bore hole 26)
Depth of soil above footing = 0.2 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 378.6 KN/m^2
> 45.5 KN/m^2

Hence OK

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5.2 DESIGN CALCULATION OF FOOTING F2

Size of Columns at node	Along-X	Along-Z
1	0.50m	0.50m
2	0.50m	0.50m
Size of fdn. raft:	4.50m	2.00m
Thickness of raft:	0.40	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	1.25	m
U/S level of fdn:	-1.50	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	18	kN/m^3

Calculation of Weight of Foundation :

Weight of raft = (4.5*2*0.4*25) = 90.00KN

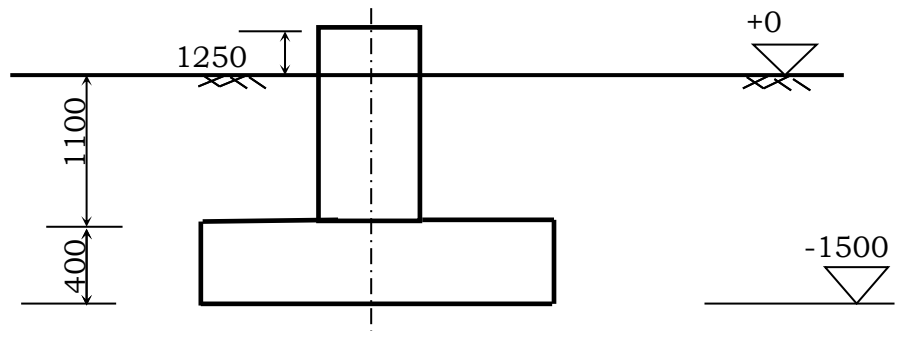
Weight of soil = ((4.5*2)-2*(0.5*0.5))*1.1*18 = 168.30KN

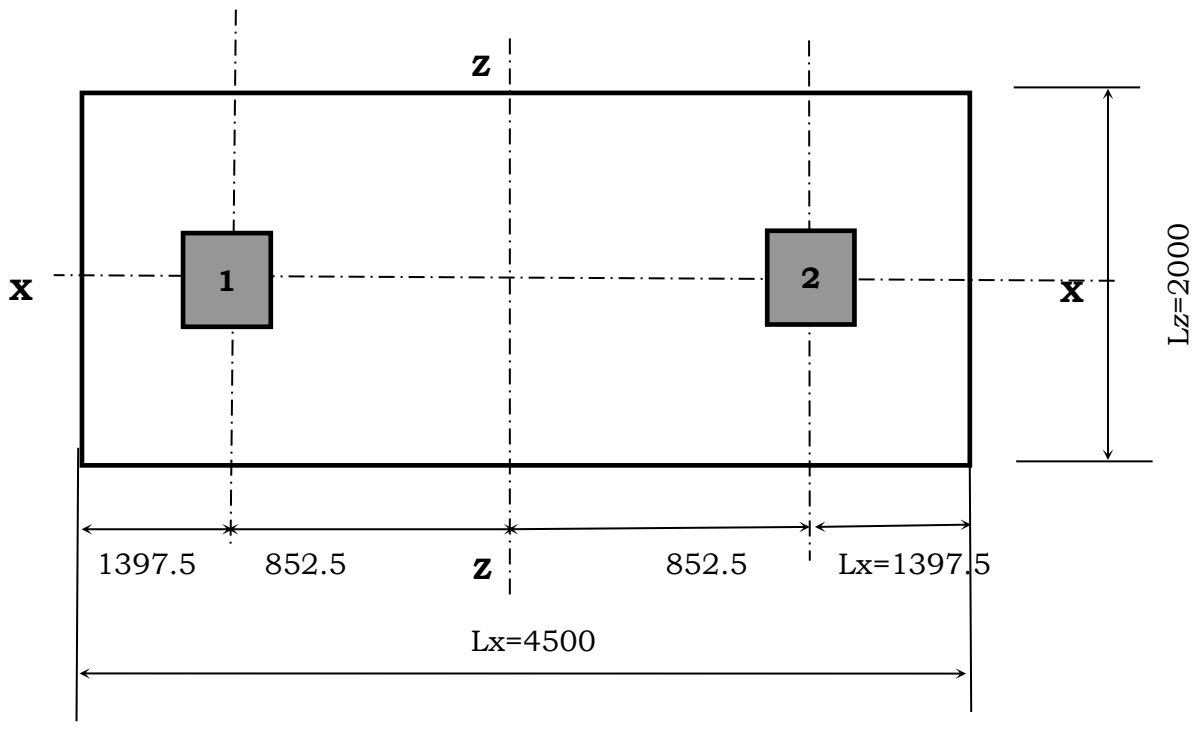
Total = 258.30KN

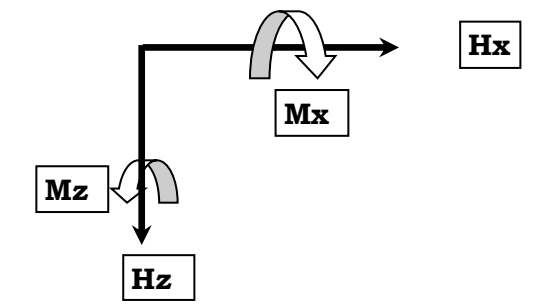
Submerged wt. = 258.3-4.5*2*1.5*10 = 123.30KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm







H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Check for overturning and sliding of footing F2

Load at bottom of fooring	Node 1					Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL+1.0LL	60.0	10.0	-27.5	-10.0	-27.5	50.0	-10.0	27.5	10.0	27.5
202 1.0DL+0.75LL	50.0	7.5	-20.6	-7.5	-20.6	42.5	-7.5	20.6	7.5	20.6
203 1.0DL+1.0WLX1	-40.0	-10.0	27.5	10.0	27.5	80.0	-10.0	27.5	10.0	27.5
204 1.0DL+1.0WLX2	80.0	10.0	-27.5	-10.0	-27.5	-40.0	10.0	-27.5	-10.0	-27.5
205 1.0DL+1.0WLZ1	10.0	-10.0	27.5	-10.0	-27.5	10.0	-10.0	27.5	-10.0	-27.5
206 1.0DL+1.0WLZ2	30.0	10.0	-27.5	10.0	27.5	30.0	10.0	-27.5	10.0	27.5
207 1.0DL+0.75LL+0.75WLX1	5.0	0.0	0.0	0.0	0.0	87.5	-15.0	41.3	15.0	41.3
208 1.0DL+0.75LL+0.75WLX2	95.0	15.0	-41.3	-15.0	-41.3	-2.5	0.0	0.0	0.0	0.0
209 1.0DL+0.75LL+0.75WLZ1	42.5	0.0	0.0	-15.0	-41.3	35.0	-15.0	41.3	0.0	0.0
210 1.0DL+0.75LL+0.75WLZ2	57.5	15.0	-41.3	0.0	0.0	50.0	0.0	0.0	15.0	41.3
211 0.6DL+1.0WLX1	-48.0	-10.0	27.5	10.0	27.5	72.0	-10.0	27.5	10.0	27.5
212 0.6DL+1.0WLX2	72.0	10.0	-27.5	-10.0	-27.5	-48.0	10.0	-27.5	-10.0	-27.5
213 0.6DL+1.0WLZ1	2.0	-10.0	27.5	-10.0	-27.5	2.0	-10.0	27.5	-10.0	-27.5
214 0.6DL+1.0WLZ2	22.0	10.0	-27.5	10.0	27.5	22.0	10.0	-27.5	10.0	27.5

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL+1.0LL	368.3	0.00	-8.5	0.0	0.0	97.2	NA	NA	NA
202 1.0DL+0.75LL	350.8	0.00	-6.4	0.0	0.0	123.4	NA	NA	NA
203 1.0DL+1.0WLX1	298.3	-20.00	157.3	20.0	55.0	4.3	5.4	6.0	6.0
204 1.0DL+1.0WLX2	298.3	20.00	-157.3	-20.0	-55.0	4.3	5.4	6.0	6.0
205 1.0DL+1.0WLZ1	278.3	-20.00	55.0	-20.0	-55.0	11.4	5.1	5.6	5.6
206 1.0DL+1.0WLZ2	318.3	20.00	-55.0	20.0	55.0	13.0	5.8	6.4	6.4
207 1.0DL+0.75LL+0.75WLX1	350.8	-15.00	111.6	15.0	41.3	7.1	8.5	9.4	9.4
208 1.0DL+0.75LL+0.75WLX2	350.8	15.00	-124.4	-15.0	-41.3	6.3	8.5	9.4	9.4
209 1.0DL+0.75LL+0.75WLZ1	335.8	-15.00	34.9	-15.0	-41.3	21.6	8.1	9.0	9.0
210 1.0DL+0.75LL+0.75WLZ2	365.8	15.00	-47.7	15.0	41.3	17.3	8.9	9.8	9.8
211 0.6DL+1.0WLX1	282.3	-20.00	157.3	20.0	55.0	4.0	5.1	5.6	5.6
212 0.6DL+1.0WLX2	282.3	20.00	-157.3	-20.0	-55.0	4.0	5.1	5.6	5.6
213 0.6DL+1.0WLZ1	262.3	-20.00	55.0	-20.0	-55.0	10.7	4.8	5.2	5.2
214 0.6DL+1.0WLZ2	302.3	20.00	-55.0	20.0	55.0	12.4	5.5	6.0	6.0
					Min.=	4.0	4.8	5.2	5.2

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Base pressure check of footing F2

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	1	201	0	51.33	0
Min Px	1	201	0	51.33	0
Max Py	165	203	0	101.58	0
Min Py	6	203	0	0.00	0
Max Pz	1	201	0	51.33	0
Min Pz	1	201	0	51.33	0

Gross Maximum Base pressure, p_{max} = 101.6 KN/m^2

Net Allowable Safe Bearing Capacity = 375 KN/m^2
Depth of soil above footing = 1.1 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 394.8 KN/m^2
> 101.6 KN/m^2
Hence OK

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Design of foundation raft F2

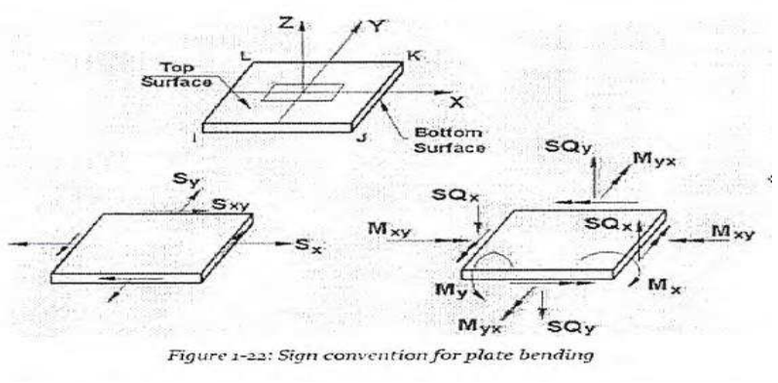
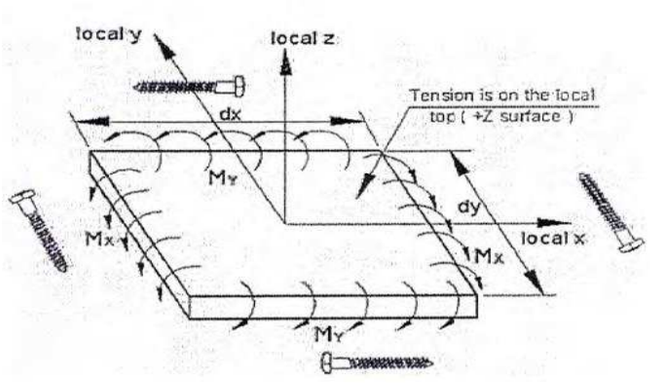


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F12_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	953	0.274	-0.044	0	0	0	-15.78	-31.588	2.808	-18.585	-34.396	103 1.2DL+1.0LL+1.6WLX1
Min Qx	972	-0.282	0.046	0	0	0	-16.1	-32.667	0	-16.102	-32.667	104 1.2DL+1.0LL+1.6WLX2
Max Qy	48	-0.092	0.341	0	0	0	-4.264	-32.402	1.498	-5.762	-33.9	103 1.2DL+1.0LL+1.6WLX1
Min Qy	227	0.086	-0.356	0	0	0	-5.325	-33.668	1.386	-6.711	-35.054	104 1.2DL+1.0LL+1.6WLX2
Max Sx	1012	0.001	-0.012	0	0	0	-0.188	-1.521	-0.318	-0.506	-1.839	101 1.4DL
Min Sx	1012	0.001	-0.012	0	0	0	-0.188	-1.521	-0.318	-0.506	-1.839	101 1.4DL
Max Sy	1012	0.001	-0.012	0	0	0	-0.188	-1.521	-0.318	-0.506	-1.839	101 1.4DL
Min Sy	1012	0.001	-0.012	0	0	0	-0.188	-1.521	-0.318	-0.506	-1.839	101 1.4DL
Max Sxy	1012	0.001	-0.012	0	0	0	-0.188	-1.521	-0.318	-0.506	-1.839	101 1.4DL
Min Sxy	1012	0.001	-0.012	0	0	0	-0.188	-1.521	-0.318	-0.506	-1.839	101 1.4DL
Max Mx	69	-0.019	-0.062	0	0	0	42.051	-3.672	5.001	47.052	-8.673	106 1.2DL+1.0LL+1.6WLZ2
Min Mx	763	0.14	0.023	0	0	0	-21.92	-5.384	5.623	-27.541	-11.007	109 0.9DL+1.6HL+1.6WLZ1
Max My	235	0.04	-0.194	0	0	0	2.888	17.763	0.908	3.796	18.671	108 0.9DL+1.6HL+1.6WLX2
Min My	971	-0.088	0.034	0	0	0	-2.09	-35.207	1.27	-3.36	-36.477	104 1.2DL+1.0LL+1.6WLX2
Max Mxy	94	-0.057	-0.09	0	0	0	34.848	5.914	20.115	54.963	26.029	104 1.2DL+1.0LL+1.6WLX2
Min Mxy	347	-0.05	0.068	0	0	0	22.701	5.197	-11.787	34.488	16.984	106 1.2DL+1.0LL+1.6WLZ2

PLATE CENTER STRESS : FDN_F12_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	374	0.14	-0.004	0	0	0	-3.83	-12.579	6.658	-10.488	-19.237	105 1.2DL+1.0LL+1.6WLZ1
Min Qx	910	-0.146	0.018	0	0	0	7.222	-1.03	0	7.222	-1.03	106 1.2DL+1.0LL+1.6WLZ2
Max Qy	951	-0.005	0.198	0	0	0	-0.944	-9.656	2.671	-3.615	-12.327	104 1.2DL+1.0LL+1.6WLX2
Min Qy	978	0.004	-0.196	0	0	0	-0.917	-9.282	2.719	-3.636	-12.001	103 1.2DL+1.0LL+1.6WLX1
Max Sx	1012	0.001	-0.012	0	0	0	-0.188	-1.521	-0.318	-0.506	-1.839	101 1.4DL
Min Sx	1012	0.001	-0.012	0	0	0	-0.188	-1.521	-0.318	-0.506	-1.839	101 1.4DL
Max Sy	1012	0.001	-0.012	0	0	0	-0.188	-1.521	-0.318	-0.506	-1.839	101 1.4DL
Min Sy	1012	0.001	-0.012	0	0	0	-0.188	-1.521	-0.318	-0.506	-1.839	101 1.4DL
Max Sxy	1012	0.001	-0.012	0	0	0	-0.188	-1.521	-0.318	-0.506	-1.839	101 1.4DL
Min Sxy	1012	0.001	-0.012	0	0	0	-0.188	-1.521	-0.318	-0.506	-1.839	101 1.4DL
Max Mx	768	0.001	0	0	0	0	39.395	-0.514	11.393	50.788	-11.907	102 1.4DL+1.7LL
Min Mx	924	0.015	-0.1	0	0	0	-15.64	-5.743	5.347	-20.985	-11.09	105 1.2DL+1.0LL+1.6WLZ1
Max My	974	-0.052	0.01	0	0	0	1.353	17.094	0.007	1.36	17.101	108 0.9DL+1.6HL+1.6WLX2
Min My	971	-0.088	0.034	0	0	0	-2.09	-35.207	1.27	-3.36	-36.477	104 1.2DL+1.0LL+1.6WLX2
Max Mxy	96	-0.013	0.003	0	0	0	33.437	-0.396	13.172	46.609	-13.568	104 1.2DL+1.0LL+1.6WLX2
Min Mxy	374	-0.017	-0.014	0	0	0	21.877	0.562	-8.371	30.248	8.933	106 1.2DL+1.0LL+1.6WLZ2

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		400	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

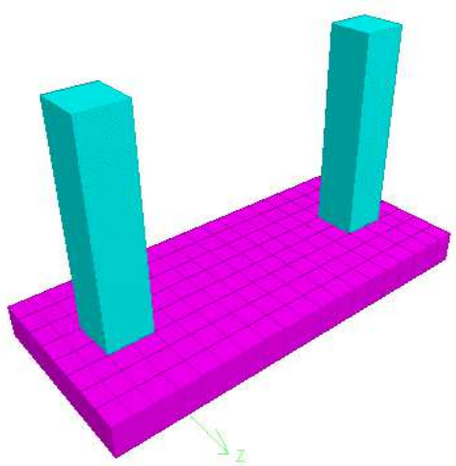
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c$ *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
106 1.2D	47.05	top face	50	344	47.052	0.53	0.0013	438.2	619.2	12	182.72	150	754.3	10.64	13.29	0.075	tension control
109 0.9D	-27.54	bottom face	50	344	27.541	0.31	0.0007	255.5		12	182.72	150	754.3	10.64	13.29	0.075	tension control
104 1.2D	54.96	top face	50	344	54.963	0.62	0.0015	512.6		12	182.72	150	754.3	10.64	13.29	0.075	tension control
106 1.2D	34.49	top face	50	344	34.488	0.39	0.0009	320.4		12	182.72	150	754.3	10.64	13.29	0.075	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.14	1	56	10.488	0.002	0.344	349.174	338.444	590.188	338.44	1261	> 56	ok	253.8	φVc>Vu	not required
	Min Qx	0.146	1	58.4	7.222			361.291			338.44	1261	> 58.4	ok	253.8	φVc>Vu	not required

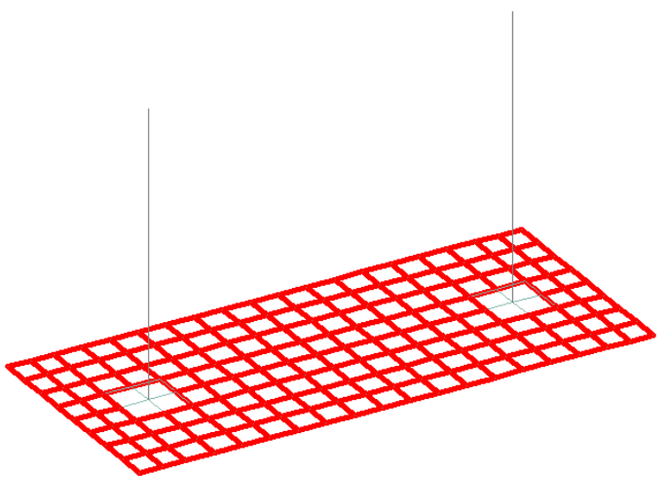
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c$ *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
108 0.9D	18.67	top face	50	344	18.671	0.21	0.0005	172.9	619.2	12	182.72	150	754.3	10.64	13.29	0.075	tension control
104 1.2D	-36.48	bottom face	50	344	36.477	0.41	0.0010	339.0		12	182.72	150	754.3	10.64	13.29	0.075	tension control
104 1.2D	26.03	top face	50	344	26.029	0.29	0.0007	241.4		12	182.72	150	754.3	10.64	13.29	0.075	tension control
106 1.2D	16.98	top face	50	344	16.984	0.19	0.0005	157.2		12	182.72	150	754.3	10.64	13.29	0.075	tension control

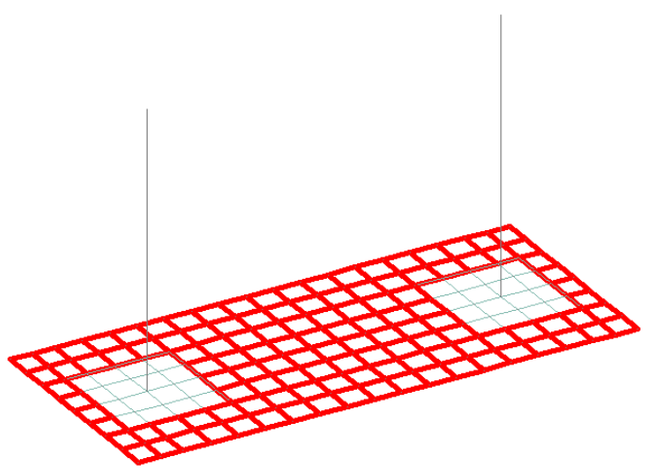
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	104 1.2D	0.198	1	79.2	12.327	0.002	0.344	353.962	338.444	fc'590.2	338.44	355	> 79.2	ok	253.8	φVc>Vu	not required
	103 1.2D	0.196	1	78.4	12.001			354.438			338.44	355	> 78.4	ok	253.8	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	12ϕT	**150**	c/c
At Top	12ϕT	**150**	c/c
Along Z Direction :			
At Bottom	12ϕT	**150**	c/c
At Top	12ϕT	**150**	c/c

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5.3 DESIGN OF COLUMN C1

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 500 mm

Depth parallel to Z-Z axis, b = 500 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.72

Y_b = (b-d')/b = 0.72

Unsupported length of column parallel to X-X axis, l_{ux} = 1.65 m kl_{ux}/r_x = 13.2 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.65 m kl_{uz}/r_z = 13.2

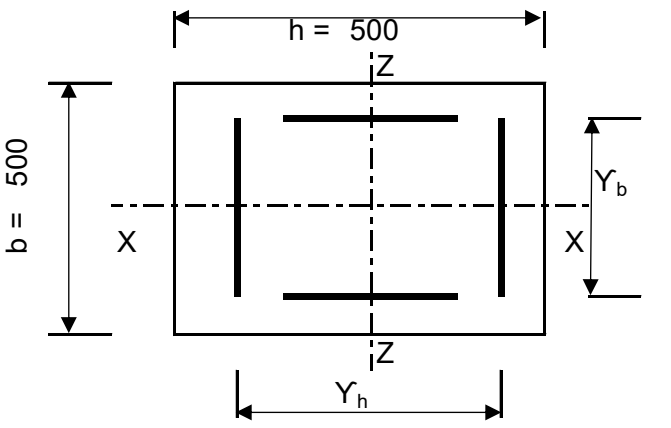
Radius of gyration r_x= 150 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 6845.08 KN safe

Radius of gyration r_z= 150 mm

Factored Force Value				Design force Value			$K_n = P_n/f_cbh$	Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/φ$	$M_{nx}=M_{ux}/φ$	$M_{nz}=M_{uz}/φ$		$R_{nx}= M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz}= M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
105	354.4	79.2	0.0	545	121.8	0.0	0.0623	0.050	218.75	0.56	0.050	218.8	0.00	1.609	0.391	safe
108	23.0	0.0	0.0	35	0.0	0.0	0.0040	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
103	194.4	0.0	52.8	299	0.0	81.2	0.0342	0.050	218.75	0.00	0.050	218.8	0.37		0.204	safe
104	258.4	0.0	52.8	398	0.0	81.2	0.0454	0.050	218.75	0.00	0.050	218.8	0.37		0.204	safe
106	354.4	79.2	0.0	545	121.8	0.0	0.0623	0.050	218.75	0.56	0.050	218.8	0.00		0.391	safe
105	98.4	79.2	0.0	151	121.8	0.0	0.0173	0.050	218.75	0.56	0.050	218.8	0.00		0.391	safe
101	84.4	0.0	0.0	130	0.0	0.0	0.0148	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
101	84.4	0.0	0.0	130	0.0	0.0	0.0148	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
105	98.4	79.2	0.0	151	121.8	0.0	0.0173	0.050	218.75	0.56	0.050	218.8	0.00		0.391	safe
106	354.4	79.2	0.0	545	121.8	0.0	0.0623	0.050	218.75	0.56	0.050	218.8	0.00		0.391	safe
103	194.4	0.0	52.8	299	0.0	81.2	0.0342	0.050	218.75	0.00	0.050	218.8	0.37		0.204	safe
104	258.4	0.0	52.8	398	0.0	81.2	0.0454	0.050	218.75	0.00	0.050	218.8	0.37		0.204	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
2500	4	20	+	8	16	2867	1.15	10	256	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

ACI DESIGN HANDBOOK-SP-17M(09)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5.4 DESIGN OF COLUMN C2

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 500 mm

Depth parallel to Z-Z axis, b = 500 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.72

Y_b = (b-d')/b = 0.72

Unsupported length of column parallel to X-X axis, l_{ux} = 2.55 m kl_{ux}/r_x = 20.4 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 2.55 m kl_{uz}/r_z = 20.4

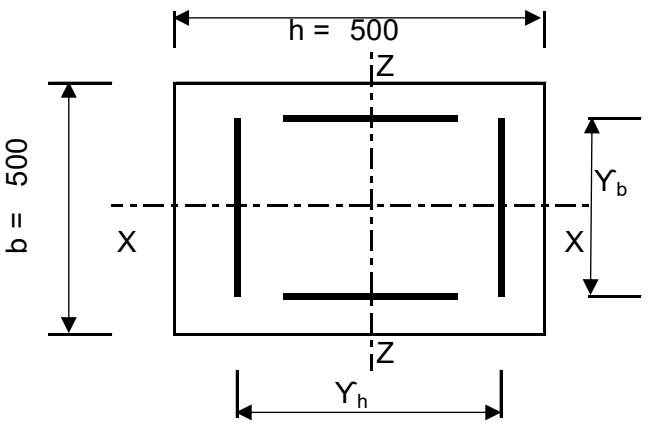
Radius of gyration r_x= 150 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 6845.08 KN safe

Radius of gyration r_z= 150 mm

L/C	Factored Force Value P_u (KN)	M_{ux} (KN-m)	M_{uz} (KN-m)	Design force Value $P_n = P_u/\varphi$ (KN)	$M_{nx}=M_{ux}/\varphi$ (KN-m)	$M_{nz}=M_{uz}/\varphi$ (KN-m)	$K_n = P_n/f_c'bh$	Uniaxial moment capacity about X-X axis: $R_{nx}= M_{nx1}/f_c'b^2h$	M_{nx1} (KN-m)	M_{nx}/M_{nx1}	about ZZ axis: $R_{nz}= M_{nz1}/f_c'h^2b$	M_{nz1} (KN-m)	M_{nz}/M_{nz1}	Check for Interaction Ratio: α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
104	205.1	40.8	40.8	316	62.8	62.8	0.0361	0.050	218.75	0.29	0.050	218.8	0.29	1.609	0.269	safe
107	94.0	0.0	0.0	145	0.0	0.0	0.0165	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
103	169.1	66.3	66.3	260	102.0	102.0	0.0297	0.050	218.75	0.47	0.050	218.8	0.47		0.586	safe
104	179.1	66.3	66.3	276	102.0	102.0	0.0315	0.050	218.75	0.47	0.050	218.8	0.47		0.586	safe
103	169.1	66.3	66.3	260	102.0	102.0	0.0297	0.050	218.75	0.47	0.050	218.8	0.47		0.586	safe
104	179.1	66.3	66.3	276	102.0	102.0	0.0315	0.050	218.75	0.47	0.050	218.8	0.47		0.586	safe
101	50.3	0.0	0.0	77	0.0	0.0	0.0088	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
101	50.3	0.0	0.0	77	0.0	0.0	0.0088	0.050	218.75	0.00	0.050	218.8	0.00		0	safe
104	179.1	66.3	66.3	276	102.0	102.0	0.0315	0.050	218.75	0.47	0.050	218.8	0.47		0.586	safe
103	169.1	66.3	66.3	260	102.0	102.0	0.0297	0.050	218.75	0.47	0.050	218.8	0.47		0.586	safe
103	169.1	66.3	66.3	260	102.0	102.0	0.0297	0.050	218.75	0.47	0.050	218.8	0.47		0.586	safe
104	179.1	66.3	66.3	276	102.0	102.0	0.0315	0.050	218.75	0.47	0.050	218.8	0.47		0.586	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars, db					A_{st} provided (mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
2500	4	20	+	8	16	2867	1.15	10	256	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

ACI DESIGN HANDBOOK-SP-17M(09)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

REFERENCE

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST- CRT-1001) Clause 4.1

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ATTACHMENT 1 STAAD INPUT & OUTPUT FILE FOR F1 DESIGN

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5           *
*           Version  20.07.10.64                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    APR 13, 2020                 *
*           Time=     9:23:45                     *
*                                                *
*      USER ID: LT Construction                  *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F1 DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 09-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 1 0 0 0; 2 4.9 0 0; 3 0 0 4.35; 4 4.9 0 4.35; 5 0 0 5.8; 6 4.9 0 5.8
    10. 7 0 0 7.25; 8 4.9 0 7.25; 9 0 0 11.6; 10 4.9 0 11.6; 11 11.5 0 4.35
    11. 12 11.5 0 5.8; 13 11.5 0 7.25; 14 0 1.85 0; 15 4.9 1.85 0; 16 4.9 1.85 4.35
    12. 17 0 1.85 5.8; 18 4.9 1.85 7.25; 19 0 1.85 11.6; 20 4.9 1.85 11.6
    13. 21 11.5 1.85 4.35; 22 11.5 1.85 7.25; 23 12.35 0 4.35; 24 12.35 0 5.8
    14. 25 12.35 0 7.25; 26 5.7 0 0; 27 5.7 0 4.35; 28 5.7 0 5.8; 29 5.7 0 7.25
    15. 30 5.7 0 11.6; 31 -0.8 0 0; 32 -0.8 0 4.35; 33 -0.8 0 5.8; 34 -0.8 0 7.25
    16. 35 -0.8 0 11.6; 36 0 0 -0.8; 37 4.9 0 -0.8; 39 -0.8 0 -0.8; 40 0 0 12.4
    17. 41 4.9 0 12.4; 43 -0.8 0 12.4; 44 11.5 0 3.75; 46 5.7 0 3.75; 47 11.5 0 7.85
    18. 49 5.7 0 7.85; 50 0 0 3.75; 51 4.9 0 3.75; 52 -0.8 0 3.75; 53 0 0 7.85
    19. 54 4.9 0 7.85; 55 -0.8 0 7.85; 56 -0.533333 0 0; 57 -0.533333 0 -0.266667
    20. 58 -0.8 0 -0.266667; 59 -0.266667 0 0; 60 -0.266667 0 -0.266667
    21. 61 0 0 -0.266667; 62 -0.533333 0 -0.533333; 63 -0.8 0 -0.533333
    22. 64 -0.266667 0 -0.533333; 65 0 0 -0.533333; 66 -0.533333 0 -0.8
    23. 67 -0.266667 0 -0.8; 68 0.245 0 0; 69 0.245 0 -0.266667; 70 0.49 0 0
    24. 71 0.49 0 -0.266667; 72 0.735 0 0; 73 0.735 0 -0.266667; 74 0.98 0 0
    25. 75 0.98 0 -0.266667; 76 1.225 0 0; 77 1.225 0 -0.266667; 78 1.47 0 0
    26. 79 1.47 0 -0.266667; 80 1.715 0 0; 81 1.715 0 -0.266667; 82 1.96 0 0
    27. 83 1.96 0 -0.266667; 84 2.205 0 0; 85 2.205 0 -0.266667; 86 2.45 0 0
    28. 87 2.45 0 -0.266667; 88 2.695 0 0; 89 2.695 0 -0.266667; 90 2.94 0 0
    29. 91 2.94 0 -0.266667; 92 3.185 0 0; 93 3.185 0 -0.266667; 94 3.43 0 0
    30. 95 3.43 0 -0.266667; 96 3.675 0 0; 97 3.675 0 -0.266667; 98 3.92 0 0
    31. 99 3.92 0 -0.266667; 100 4.165 0 0; 101 4.165 0 -0.266667; 102 4.41 0 0
    32. 103 4.41 0 -0.266667; 104 4.655 0 0; 105 4.655 0 -0.266667
    33. 106 4.9 0 -0.266667; 107 0.245 0 -0.533333; 108 0.49 0 -0.533333
    34. 109 0.735 0 -0.533333; 110 0.98 0 -0.533333; 111 1.225 0 -0.533333
    35. 112 1.47 0 -0.533333; 113 1.715 0 -0.533333; 114 1.96 0 -0.533333
    36. 115 2.205 0 -0.533333; 116 2.45 0 -0.533333; 117 2.695 0 -0.533333
    37. 118 2.94 0 -0.533333; 119 3.185 0 -0.533333; 120 3.43 0 -0.533333
    38. 121 3.675 0 -0.533333; 122 3.92 0 -0.533333; 123 4.165 0 -0.533333
```

39. 124 4.41 0 -0.533333; 125 4.655 0 -0.533333; 126 4.9 0 -0.533333
40. 127 0.245 0 -0.8; 128 0.49 0 -0.8; 129 0.735 0 -0.8; 130 0.98 0 -0.8
41. 131 1.225 0 -0.8; 132 1.47 0 -0.8; 133 1.715 0 -0.8; 134 1.96 0 -0.8
42. 135 2.205 0 -0.8; 136 2.45 0 -0.8; 137 2.695 0 -0.8; 138 2.94 0 -0.8
43. 139 3.185 0 -0.8; 140 3.43 0 -0.8; 141 3.675 0 -0.8; 142 3.92 0 -0.8
44. 143 4.165 0 -0.8; 144 4.41 0 -0.8; 145 4.655 0 -0.8; 146 -0.533333 0 3.75
45. 147 -0.533333 0 3.5; 148 -0.8 0 3.5; 149 -0.266667 0 3.75; 150 -0.266667 0 3.5
46. 151 0 0 3.5; 152 -0.533333 0 3.25; 153 -0.8 0 3.25; 154 -0.266667 0 3.25
47. 155 0 0 3.25; 156 -0.533333 0 3; 157 -0.8 0 3; 158 -0.266667 0 3; 159 0 0 3
48. 160 -0.533333 0 2.75; 161 -0.8 0 2.75; 162 -0.266667 0 2.75; 163 0 0 2.75
49. 164 -0.533333 0 2.5; 165 -0.8 0 2.5; 166 -0.266667 0 2.5; 167 0 0 2.5
50. 168 -0.533333 0 2.25; 169 -0.8 0 2.25; 170 -0.266667 0 2.25; 171 0 0 2.25
51. 172 -0.533333 0 2; 173 -0.8 0 2; 174 -0.266667 0 2; 175 0 0 2
52. 176 -0.533333 0 1.75; 177 -0.8 0 1.75; 178 -0.266667 0 1.75; 179 0 0 1.75
53. 180 -0.533333 0 1.5; 181 -0.8 0 1.5; 182 -0.266667 0 1.5; 183 0 0 1.5
54. 184 -0.533333 0 1.25; 185 -0.8 0 1.25; 186 -0.266667 0 1.25; 187 0 0 1.25
55. 188 -0.533333 0 1; 189 -0.8 0 1; 190 -0.266667 0 1; 191 0 0 1
56. 192 -0.533333 0 0.75; 193 -0.8 0 0.75; 194 -0.266667 0 0.75; 195 0 0 0.75
57. 196 -0.533333 0 0.5; 197 -0.8 0 0.5; 198 -0.266667 0 0.5; 199 0 0 0.5
58. 200 -0.533333 0 0.25; 201 -0.8 0 0.25; 202 -0.266667 0 0.25; 203 0 0 0.25
59. 204 0.245 0 3.75; 205 0.245 0 3.5; 206 0.49 0 3.75; 207 0.49 0 3.5
60. 208 0.735 0 3.75; 209 0.735 0 3.5; 210 0.98 0 3.75; 211 0.98 0 3.5
61. 212 1.225 0 3.75; 213 1.225 0 3.5; 214 1.47 0 3.75; 215 1.47 0 3.5
62. 216 1.715 0 3.75; 217 1.715 0 3.5; 218 1.96 0 3.75; 219 1.96 0 3.5
63. 220 2.205 0 3.75; 221 2.205 0 3.5; 222 2.45 0 3.75; 223 2.45 0 3.5
64. 224 2.695 0 3.75; 225 2.695 0 3.5; 226 2.94 0 3.75; 227 2.94 0 3.5
65. 228 3.185 0 3.75; 229 3.185 0 3.5; 230 3.43 0 3.75; 231 3.43 0 3.5
66. 232 3.675 0 3.75; 233 3.675 0 3.5; 234 3.92 0 3.75; 235 3.92 0 3.5
67. 236 4.165 0 3.75; 237 4.165 0 3.5; 238 4.41 0 3.75; 239 4.41 0 3.5
68. 240 4.655 0 3.75; 241 4.655 0 3.5; 242 4.9 0 3.5; 243 0.245 0 3.25
69. 244 0.49 0 3.25; 245 0.735 0 3.25; 246 0.98 0 3.25; 247 1.225 0 3.25
70. 248 1.47 0 3.25; 249 1.715 0 3.25; 250 1.96 0 3.25; 251 2.205 0 3.25
71. 252 2.45 0 3.25; 253 2.695 0 3.25; 254 2.94 0 3.25; 255 3.185 0 3.25
72. 256 3.43 0 3.25; 257 3.675 0 3.25; 258 3.92 0 3.25; 259 4.165 0 3.25
73. 260 4.41 0 3.25; 261 4.655 0 3.25; 262 4.9 0 3.25; 263 0.245 0 3; 264 0.49 0 3
74. 265 0.735 0 3; 266 0.98 0 3; 267 1.225 0 3; 268 1.47 0 3; 269 1.715 0 3
75. 270 1.96 0 3; 271 2.205 0 3; 272 2.45 0 3; 273 2.695 0 3; 274 2.94 0 3
76. 275 3.185 0 3; 276 3.43 0 3; 277 3.675 0 3; 278 3.92 0 3; 279 4.165 0 3
77. 280 4.41 0 3; 281 4.655 0 3; 282 4.9 0 3; 283 0.245 0 2.75; 284 0.49 0 2.75
78. 285 0.735 0 2.75; 286 0.98 0 2.75; 287 1.225 0 2.75; 288 1.47 0 2.75
79. 289 1.715 0 2.75; 290 1.96 0 2.75; 291 2.205 0 2.75; 292 2.45 0 2.75
80. 293 2.695 0 2.75; 294 2.94 0 2.75; 295 3.185 0 2.75; 296 3.43 0 2.75
81. 297 3.675 0 2.75; 298 3.92 0 2.75; 299 4.165 0 2.75; 300 4.41 0 2.75
82. 301 4.655 0 2.75; 302 4.9 0 2.75; 303 0.245 0 2.5; 304 0.49 0 2.5
83. 305 0.735 0 2.5; 306 0.98 0 2.5; 307 1.225 0 2.5; 308 1.47 0 2.5
84. 309 1.715 0 2.5; 310 1.96 0 2.5; 311 2.205 0 2.5; 312 2.45 0 2.5
85. 313 2.695 0 2.5; 314 2.94 0 2.5; 315 3.185 0 2.5; 316 3.43 0 2.5
86. 317 3.675 0 2.5; 318 3.92 0 2.5; 319 4.165 0 2.5; 320 4.41 0 2.5
87. 321 4.655 0 2.5; 322 4.9 0 2.5; 323 0.245 0 2.25; 324 0.49 0 2.25
88. 325 0.735 0 2.25; 326 0.98 0 2.25; 327 1.225 0 2.25; 328 1.47 0 2.25
89. 329 1.715 0 2.25; 330 1.96 0 2.25; 331 2.205 0 2.25; 332 2.45 0 2.25
90. 333 2.695 0 2.25; 334 2.94 0 2.25; 335 3.185 0 2.25; 336 3.43 0 2.25
91. 337 3.675 0 2.25; 338 3.92 0 2.25; 339 4.165 0 2.25; 340 4.41 0 2.25
92. 341 4.655 0 2.25; 342 4.9 0 2.25; 343 0.245 0 2; 344 0.49 0 2; 345 0.735 0 2
93. 346 0.98 0 2; 347 1.225 0 2; 348 1.47 0 2; 349 1.715 0 2; 350 1.96 0 2
94. 351 2.205 0 2; 352 2.45 0 2; 353 2.695 0 2; 354 2.94 0 2; 355 3.185 0 2

```
 95. 356 3.43 0 2; 357 3.675 0 2; 358 3.92 0 2; 359 4.165 0 2; 360 4.41 0 2
 96. 361 4.655 0 2; 362 4.9 0 2; 363 0.245 0 1.75; 364 0.49 0 1.75
 97. 365 0.735 0 1.75; 366 0.98 0 1.75; 367 1.225 0 1.75; 368 1.47 0 1.75
 98. 369 1.715 0 1.75; 370 1.96 0 1.75; 371 2.205 0 1.75; 372 2.45 0 1.75
 99. 373 2.695 0 1.75; 374 2.94 0 1.75; 375 3.185 0 1.75; 376 3.43 0 1.75
100. 377 3.675 0 1.75; 378 3.92 0 1.75; 379 4.165 0 1.75; 380 4.41 0 1.75
101. 381 4.655 0 1.75; 382 4.9 0 1.75; 383 0.245 0 1.5; 384 0.49 0 1.5
102. 385 0.735 0 1.5; 386 0.98 0 1.5; 387 1.225 0 1.5; 388 1.47 0 1.5
103. 389 1.715 0 1.5; 390 1.96 0 1.5; 391 2.205 0 1.5; 392 2.45 0 1.5
104. 393 2.695 0 1.5; 394 2.94 0 1.5; 395 3.185 0 1.5; 396 3.43 0 1.5
105. 397 3.675 0 1.5; 398 3.92 0 1.5; 399 4.165 0 1.5; 400 4.41 0 1.5
106. 401 4.655 0 1.5; 402 4.9 0 1.5; 403 0.245 0 1.25; 404 0.49 0 1.25
107. 405 0.735 0 1.25; 406 0.98 0 1.25; 407 1.225 0 1.25; 408 1.47 0 1.25
108. 409 1.715 0 1.25; 410 1.96 0 1.25; 411 2.205 0 1.25; 412 2.45 0 1.25
109. 413 2.695 0 1.25; 414 2.94 0 1.25; 415 3.185 0 1.25; 416 3.43 0 1.25
110. 417 3.675 0 1.25; 418 3.92 0 1.25; 419 4.165 0 1.25; 420 4.41 0 1.25
111. 421 4.655 0 1.25; 422 4.9 0 1.25; 423 0.245 0 1; 424 0.49 0 1; 425 0.735 0 1
112. 426 0.98 0 1; 427 1.225 0 1; 428 1.47 0 1; 429 1.715 0 1; 430 1.96 0 1
113. 431 2.205 0 1; 432 2.45 0 1; 433 2.695 0 1; 434 2.94 0 1; 435 3.185 0 1
114. 436 3.43 0 1; 437 3.675 0 1; 438 3.92 0 1; 439 4.165 0 1; 440 4.41 0 1
115. 441 4.655 0 1; 442 4.9 0 1; 443 0.245 0 0.75; 444 0.49 0 0.75
116. 445 0.735 0 0.75; 446 0.98 0 0.75; 447 1.225 0 0.75; 448 1.47 0 0.75
117. 449 1.715 0 0.75; 450 1.96 0 0.75; 451 2.205 0 0.75; 452 2.45 0 0.75
118. 453 2.695 0 0.75; 454 2.94 0 0.75; 455 3.185 0 0.75; 456 3.43 0 0.75
119. 457 3.675 0 0.75; 458 3.92 0 0.75; 459 4.165 0 0.75; 460 4.41 0 0.75
120. 461 4.655 0 0.75; 462 4.9 0 0.75; 463 0.245 0 0.5; 464 0.49 0 0.5
121. 465 0.735 0 0.5; 466 0.98 0 0.5; 467 1.225 0 0.5; 468 1.47 0 0.5
122. 469 1.715 0 0.5; 470 1.96 0 0.5; 471 2.205 0 0.5; 472 2.45 0 0.5
123. 473 2.695 0 0.5; 474 2.94 0 0.5; 475 3.185 0 0.5; 476 3.43 0 0.5
124. 477 3.675 0 0.5; 478 3.92 0 0.5; 479 4.165 0 0.5; 480 4.41 0 0.5
125. 481 4.655 0 0.5; 482 4.9 0 0.5; 483 0.245 0 0.25; 484 0.49 0 0.25
126. 485 0.735 0 0.25; 486 0.98 0 0.25; 487 1.225 0 0.25; 488 1.47 0 0.25
127. 489 1.715 0 0.25; 490 1.96 0 0.25; 491 2.205 0 0.25; 492 2.45 0 0.25
128. 493 2.695 0 0.25; 494 2.94 0 0.25; 495 3.185 0 0.25; 496 3.43 0 0.25
129. 497 3.675 0 0.25; 498 3.92 0 0.25; 499 4.165 0 0.25; 500 4.41 0 0.25
130. 501 4.655 0 0.25; 502 4.9 0 0.25; 503 5.16667 0 3.75; 504 5.16667 0 3.5
131. 505 5.43333 0 3.75; 506 5.43333 0 3.5; 507 5.7 0 3.5; 508 5.16667 0 3.25
132. 509 5.43333 0 3.25; 510 5.7 0 3.25; 511 5.16667 0 3; 512 5.43333 0 3
133. 513 5.7 0 3; 514 5.16667 0 2.75; 515 5.43333 0 2.75; 516 5.7 0 2.75
134. 517 5.16667 0 2.5; 518 5.43333 0 2.5; 519 5.7 0 2.5; 520 5.16667 0 2.25
135. 521 5.43333 0 2.25; 522 5.7 0 2.25; 523 5.16667 0 2; 524 5.43333 0 2
136. 525 5.7 0 2; 526 5.16667 0 1.75; 527 5.43333 0 1.75; 528 5.7 0 1.75
137. 529 5.16667 0 1.5; 530 5.43333 0 1.5; 531 5.7 0 1.5; 532 5.16667 0 1.25
138. 533 5.43333 0 1.25; 534 5.7 0 1.25; 535 5.16667 0 1; 536 5.43333 0 1
139. 537 5.7 0 1; 538 5.16667 0 0.75; 539 5.43333 0 0.75; 540 5.7 0 0.75
140. 541 5.16667 0 0.5; 542 5.43333 0 0.5; 543 5.7 0 0.5; 544 5.16667 0 0.25
141. 545 5.43333 0 0.25; 546 5.7 0 0.25; 547 5.16667 0 0; 548 5.43333 0 0
142. 549 -0.533333 0 4.35; 550 -0.533333 0 4.05; 551 -0.8 0 4.05
143. 552 -0.266667 0 4.35; 553 -0.266667 0 4.05; 554 0 0 4.05; 555 0.245 0 4.35
144. 556 0.245 0 4.05; 557 0.49 0 4.35; 558 0.49 0 4.05; 559 0.735 0 4.35
145. 560 0.735 0 4.05; 561 0.98 0 4.35; 562 0.98 0 4.05; 563 1.225 0 4.35
146. 564 1.225 0 4.05; 565 1.47 0 4.35; 566 1.47 0 4.05; 567 1.715 0 4.35
147. 568 1.715 0 4.05; 569 1.96 0 4.35; 570 1.96 0 4.05; 571 2.205 0 4.35
148. 572 2.205 0 4.05; 573 2.45 0 4.35; 574 2.45 0 4.05; 575 2.695 0 4.35
149. 576 2.695 0 4.05; 577 2.94 0 4.35; 578 2.94 0 4.05; 579 3.185 0 4.35
150. 580 3.185 0 4.05; 581 3.43 0 4.35; 582 3.43 0 4.05; 583 3.675 0 4.35
```

151. 584 3.675 0 4.05; 585 3.92 0 4.35; 586 3.92 0 4.05; 587 4.165 0 4.35
152. 588 4.165 0 4.05; 589 4.41 0 4.35; 590 4.41 0 4.05; 591 4.655 0 4.35
153. 592 4.655 0 4.05; 593 4.9 0 4.05; 594 5.16667 0 4.35; 595 5.16667 0 4.05
154. 596 5.43333 0 4.35; 597 5.43333 0 4.05; 598 5.7 0 4.05; 599 5.95217 0 4.35
155. 600 5.95217 0 4.05; 601 6.20435 0 4.35; 602 6.20435 0 4.05; 603 6.45652 0 4.35
156. 604 6.45652 0 4.05; 605 6.7087 0 4.35; 606 6.7087 0 4.05; 607 6.96087 0 4.35
157. 608 6.96087 0 4.05; 609 7.21304 0 4.35; 610 7.21304 0 4.05; 611 7.46522 0 4.35
158. 612 7.46522 0 4.05; 613 7.71739 0 4.35; 614 7.71739 0 4.05; 615 7.96957 0 4.35
159. 616 7.96957 0 4.05; 617 8.22174 0 4.35; 618 8.22174 0 4.05; 619 8.47391 0 4.35
160. 620 8.47391 0 4.05; 621 8.72609 0 4.35; 622 8.72609 0 4.05; 623 8.97826 0 4.35
161. 624 8.97826 0 4.05; 625 9.23044 0 4.35; 626 9.23044 0 4.05; 627 9.48261 0 4.35
162. 628 9.48261 0 4.05; 629 9.73478 0 4.35; 630 9.73478 0 4.05; 631 9.98696 0 4.35
163. 632 9.98696 0 4.05; 633 10.2391 0 4.35; 634 10.2391 0 4.05; 635 10.4913 0 4.35
164. 636 10.4913 0 4.05; 637 10.7435 0 4.35; 638 10.7435 0 4.05; 639 10.9957 0 4.35
165. 640 10.9957 0 4.05; 641 11.2478 0 4.35; 642 11.2478 0 4.05; 643 11.5 0 4.05
166. 644 5.95217 0 3.75; 645 6.20435 0 3.75; 646 6.45652 0 3.75; 647 6.7087 0 3.75
167. 648 6.96087 0 3.75; 649 7.21304 0 3.75; 650 7.46522 0 3.75; 651 7.71739 0 3.75
168. 652 7.96957 0 3.75; 653 8.22174 0 3.75; 654 8.47391 0 3.75; 655 8.72609 0 3.75
169. 656 8.97826 0 3.75; 657 9.23044 0 3.75; 658 9.48261 0 3.75; 659 9.73478 0 3.75
170. 660 9.98696 0 3.75; 661 10.2391 0 3.75; 662 10.4913 0 3.75; 663 10.7435 0 3.75
171. 664 10.9957 0 3.75; 665 11.2478 0 3.75; 666 -0.533333 0 5.8
172. 667 -0.533333 0 5.55833; 668 -0.8 0 5.55833; 669 -0.266667 0 5.8
173. 670 -0.266667 0 5.55833; 671 0 0 5.55833; 672 -0.533333 0 5.31667
174. 673 -0.8 0 5.31667; 674 -0.266667 0 5.31667; 675 0 0 5.31667
175. 676 -0.533333 0 5.075; 677 -0.8 0 5.075; 678 -0.266667 0 5.075; 679 0 0 5.075
176. 680 -0.533333 0 4.83333; 681 -0.8 0 4.83333; 682 -0.266667 0 4.83333
177. 683 0 0 4.83333; 684 -0.533333 0 4.59167; 685 -0.8 0 4.59167
178. 686 -0.266667 0 4.59167; 687 0 0 4.59167; 688 0.245 0 5.8; 689 0.245 0 5.55833
179. 690 0.49 0 5.8; 691 0.49 0 5.55833; 692 0.735 0 5.8; 693 0.735 0 5.55833
180. 694 0.98 0 5.8; 695 0.98 0 5.55833; 696 1.225 0 5.8; 697 1.225 0 5.55833
181. 698 1.47 0 5.8; 699 1.47 0 5.55833; 700 1.715 0 5.8; 701 1.715 0 5.55833
182. 702 1.96 0 5.8; 703 1.96 0 5.55833; 704 2.205 0 5.8; 705 2.205 0 5.55833
183. 706 2.45 0 5.8; 707 2.45 0 5.55833; 708 2.695 0 5.8; 709 2.695 0 5.55833
184. 710 2.94 0 5.8; 711 2.94 0 5.55833; 712 3.185 0 5.8; 713 3.185 0 5.55833
185. 714 3.43 0 5.8; 715 3.43 0 5.55833; 716 3.675 0 5.8; 717 3.675 0 5.55833
186. 718 3.92 0 5.8; 719 3.92 0 5.55833; 720 4.165 0 5.8; 721 4.165 0 5.55833
187. 722 4.41 0 5.8; 723 4.41 0 5.55833; 724 4.655 0 5.8; 725 4.655 0 5.55833
188. 726 4.9 0 5.55833; 727 0.245 0 5.31667; 728 0.49 0 5.31667
189. 729 0.735 0 5.31667; 730 0.98 0 5.31667; 731 1.225 0 5.31667
190. 732 1.47 0 5.31667; 733 1.715 0 5.31667; 734 1.96 0 5.31667
191. 735 2.205 0 5.31667; 736 2.45 0 5.31667; 737 2.695 0 5.31667
192. 738 2.94 0 5.31667; 739 3.185 0 5.31667; 740 3.43 0 5.31667
193. 741 3.675 0 5.31667; 742 3.92 0 5.31667; 743 4.165 0 5.31667
194. 744 4.41 0 5.31667; 745 4.655 0 5.31667; 746 4.9 0 5.31667; 747 0.245 0 5.075
195. 748 0.49 0 5.075; 749 0.735 0 5.075; 750 0.98 0 5.075; 751 1.225 0 5.075
196. 752 1.47 0 5.075; 753 1.715 0 5.075; 754 1.96 0 5.075; 755 2.205 0 5.075
197. 756 2.45 0 5.075; 757 2.695 0 5.075; 758 2.94 0 5.075; 759 3.185 0 5.075
198. 760 3.43 0 5.075; 761 3.675 0 5.075; 762 3.92 0 5.075; 763 4.165 0 5.075
199. 764 4.41 0 5.075; 765 4.655 0 5.075; 766 4.9 0 5.075; 767 0.245 0 4.83333
200. 768 0.49 0 4.83333; 769 0.735 0 4.83333; 770 0.98 0 4.83333
201. 771 1.225 0 4.83333; 772 1.47 0 4.83333; 773 1.715 0 4.83333
202. 774 1.96 0 4.83333; 775 2.205 0 4.83333; 776 2.45 0 4.83333
203. 777 2.695 0 4.83333; 778 2.94 0 4.83333; 779 3.185 0 4.83333
204. 780 3.43 0 4.83333; 781 3.675 0 4.83333; 782 3.92 0 4.83333
205. 783 4.165 0 4.83333; 784 4.41 0 4.83333; 785 4.655 0 4.83333
206. 786 4.9 0 4.83333; 787 0.245 0 4.59167; 788 0.49 0 4.59167

207. 789 0.735 0 4.59167; 790 0.98 0 4.59167; 791 1.225 0 4.59167
208. 792 1.47 0 4.59167; 793 1.715 0 4.59167; 794 1.96 0 4.59167
209. 795 2.205 0 4.59167; 796 2.45 0 4.59167; 797 2.695 0 4.59167
210. 798 2.94 0 4.59167; 799 3.185 0 4.59167; 800 3.43 0 4.59167
211. 801 3.675 0 4.59167; 802 3.92 0 4.59167; 803 4.165 0 4.59167
212. 804 4.41 0 4.59167; 805 4.655 0 4.59167; 806 4.9 0 4.59167; 807 5.16667 0 5.8
213. 808 5.16667 0 5.55833; 809 5.43333 0 5.8; 810 5.43333 0 5.55833
214. 811 5.7 0 5.55833; 812 5.16667 0 5.31667; 813 5.43333 0 5.31667
215. 814 5.7 0 5.31667; 815 5.16667 0 5.075; 816 5.43333 0 5.075; 817 5.7 0 5.075
216. 818 5.16667 0 4.83333; 819 5.43333 0 4.83333; 820 5.7 0 4.83333
217. 821 5.16667 0 4.59167; 822 5.43333 0 4.59167; 823 5.7 0 4.59167
218. 824 5.95217 0 5.8; 825 5.95217 0 5.55833; 826 6.20435 0 5.8
219. 827 6.20435 0 5.55833; 828 6.45652 0 5.8; 829 6.45652 0 5.55833
220. 830 6.7087 0 5.8; 831 6.7087 0 5.55833; 832 6.96087 0 5.8
221. 833 6.96087 0 5.55833; 834 7.21304 0 5.8; 835 7.21304 0 5.55833
222. 836 7.46522 0 5.8; 837 7.46522 0 5.55833; 838 7.71739 0 5.8
223. 839 7.71739 0 5.55833; 840 7.96957 0 5.8; 841 7.96957 0 5.55833
224. 842 8.22174 0 5.8; 843 8.22174 0 5.55833; 844 8.47391 0 5.8
225. 845 8.47391 0 5.55833; 846 8.72609 0 5.8; 847 8.72609 0 5.55833
226. 848 8.97826 0 5.8; 849 8.97826 0 5.55833; 850 9.23044 0 5.8
227. 851 9.23044 0 5.55833; 852 9.48261 0 5.8; 853 9.48261 0 5.55833
228. 854 9.73478 0 5.8; 855 9.73478 0 5.55833; 856 9.98696 0 5.8
229. 857 9.98696 0 5.55833; 858 10.2391 0 5.8; 859 10.2391 0 5.55833
230. 860 10.4913 0 5.8; 861 10.4913 0 5.55833; 862 10.7435 0 5.8
231. 863 10.7435 0 5.55833; 864 10.9957 0 5.8; 865 10.9957 0 5.55833
232. 866 11.2478 0 5.8; 867 11.2478 0 5.55833; 868 11.5 0 5.55833
233. 869 5.95217 0 5.31667; 870 6.20435 0 5.31667; 871 6.45652 0 5.31667
234. 872 6.7087 0 5.31667; 873 6.96087 0 5.31667; 874 7.21304 0 5.31667
235. 875 7.46522 0 5.31667; 876 7.71739 0 5.31667; 877 7.96957 0 5.31667
236. 878 8.22174 0 5.31667; 879 8.47391 0 5.31667; 880 8.72609 0 5.31667
237. 881 8.97826 0 5.31667; 882 9.23044 0 5.31667; 883 9.48261 0 5.31667
238. 884 9.73478 0 5.31667; 885 9.98696 0 5.31667; 886 10.2391 0 5.31667
239. 887 10.4913 0 5.31667; 888 10.7435 0 5.31667; 889 10.9957 0 5.31667
240. 890 11.2478 0 5.31667; 891 11.5 0 5.31667; 892 5.95217 0 5.075
241. 893 6.20435 0 5.075; 894 6.45652 0 5.075; 895 6.7087 0 5.075
242. 896 6.96087 0 5.075; 897 7.21304 0 5.075; 898 7.46522 0 5.075
243. 899 7.71739 0 5.075; 900 7.96957 0 5.075; 901 8.22174 0 5.075
244. 902 8.47391 0 5.075; 903 8.72609 0 5.075; 904 8.97826 0 5.075
245. 905 9.23044 0 5.075; 906 9.48261 0 5.075; 907 9.73478 0 5.075
246. 908 9.98696 0 5.075; 909 10.2391 0 5.075; 910 10.4913 0 5.075
247. 911 10.7435 0 5.075; 912 10.9957 0 5.075; 913 11.2478 0 5.075
248. 914 11.5 0 5.075; 915 5.95217 0 4.83333; 916 6.20435 0 4.83333
249. 917 6.45652 0 4.83333; 918 6.7087 0 4.83333; 919 6.96087 0 4.83333
250. 920 7.21304 0 4.83333; 921 7.46522 0 4.83333; 922 7.71739 0 4.83333
251. 923 7.96957 0 4.83333; 924 8.22174 0 4.83333; 925 8.47391 0 4.83333
252. 926 8.72609 0 4.83333; 927 8.97826 0 4.83333; 928 9.23044 0 4.83333
253. 929 9.48261 0 4.83333; 930 9.73478 0 4.83333; 931 9.98696 0 4.83333
254. 932 10.2391 0 4.83333; 933 10.4913 0 4.83333; 934 10.7435 0 4.83333
255. 935 10.9957 0 4.83333; 936 11.2478 0 4.83333; 937 11.5 0 4.83333
256. 938 5.95217 0 4.59167; 939 6.20435 0 4.59167; 940 6.45652 0 4.59167
257. 941 6.7087 0 4.59167; 942 6.96087 0 4.59167; 943 7.21304 0 4.59167
258. 944 7.46522 0 4.59167; 945 7.71739 0 4.59167; 946 7.96957 0 4.59167
259. 947 8.22174 0 4.59167; 948 8.47391 0 4.59167; 949 8.72609 0 4.59167
260. 950 8.97826 0 4.59167; 951 9.23044 0 4.59167; 952 9.48261 0 4.59167
261. 953 9.73478 0 4.59167; 954 9.98696 0 4.59167; 955 10.2391 0 4.59167
262. 956 10.4913 0 4.59167; 957 10.7435 0 4.59167; 958 10.9957 0 4.59167

263. 959 11.2478 0 4.59167; 960 11.5 0 4.59167; 961 11.7833 0 5.8
264. 962 11.7833 0 5.55833; 963 12.0667 0 5.8; 964 12.0667 0 5.55833
265. 965 12.35 0 5.55833; 966 11.7833 0 5.31667; 967 12.0667 0 5.31667
266. 968 12.35 0 5.31667; 969 11.7833 0 5.075; 970 12.0667 0 5.075
267. 971 12.35 0 5.075; 972 11.7833 0 4.83333; 973 12.0667 0 4.83333
268. 974 12.35 0 4.83333; 975 11.7833 0 4.59167; 976 12.0667 0 4.59167
269. 977 12.35 0 4.59167; 978 11.7833 0 4.35; 979 12.0667 0 4.35
270. 980 -0.533333 0 7.25; 981 -0.533333 0 7.00833; 982 -0.8 0 7.00833
271. 983 -0.266667 0 7.25; 984 -0.266667 0 7.00833; 985 0 0 7.00833
272. 986 -0.533333 0 6.76667; 987 -0.8 0 6.76667; 988 -0.266667 0 6.76667
273. 989 0 0 6.76667; 990 -0.533333 0 6.525; 991 -0.8 0 6.525
274. 992 -0.266667 0 6.525; 993 0 0 6.525; 994 -0.533333 0 6.28333
275. 995 -0.8 0 6.28333; 996 -0.266667 0 6.28333; 997 0 0 6.28333
276. 998 -0.533333 0 6.04167; 999 -0.8 0 6.04167; 1000 -0.266667 0 6.04167
277. 1001 0 0 6.04167; 1002 0.245 0 7.25; 1003 0.245 0 7.00833; 1004 0.49 0 7.25
278. 1005 0.49 0 7.00833; 1006 0.735 0 7.25; 1007 0.735 0 7.00833; 1008 0.98 0 7.25
279. 1009 0.98 0 7.00833; 1010 1.225 0 7.25; 1011 1.225 0 7.00833; 1012 1.47 0 7.25
280. 1013 1.47 0 7.00833; 1014 1.715 0 7.25; 1015 1.715 0 7.00833; 1016 1.96 0 7.25
281. 1017 1.96 0 7.00833; 1018 2.205 0 7.25; 1019 2.205 0 7.00833; 1020 2.45 0 7.25
282. 1021 2.45 0 7.00833; 1022 2.695 0 7.25; 1023 2.695 0 7.00833; 1024 2.94 0 7.25
283. 1025 2.94 0 7.00833; 1026 3.185 0 7.25; 1027 3.185 0 7.00833; 1028 3.43 0 7.25
284. 1029 3.43 0 7.00833; 1030 3.675 0 7.25; 1031 3.675 0 7.00833; 1032 3.92 0 7.25
285. 1033 3.92 0 7.00833; 1034 4.165 0 7.25; 1035 4.165 0 7.00833; 1036 4.41 0 7.25
286. 1037 4.41 0 7.00833; 1038 4.655 0 7.25; 1039 4.655 0 7.00833
287. 1040 4.9 0 7.00833; 1041 0.245 0 6.76667; 1042 0.49 0 6.76667
288. 1043 0.735 0 6.76667; 1044 0.98 0 6.76667; 1045 1.225 0 6.76667
289. 1046 1.47 0 6.76667; 1047 1.715 0 6.76667; 1048 1.96 0 6.76667
290. 1049 2.205 0 6.76667; 1050 2.45 0 6.76667; 1051 2.695 0 6.76667
291. 1052 2.94 0 6.76667; 1053 3.185 0 6.76667; 1054 3.43 0 6.76667
292. 1055 3.675 0 6.76667; 1056 3.92 0 6.76667; 1057 4.165 0 6.76667
293. 1058 4.41 0 6.76667; 1059 4.655 0 6.76667; 1060 4.9 0 6.76667
294. 1061 0.245 0 6.525; 1062 0.49 0 6.525; 1063 0.735 0 6.525; 1064 0.98 0 6.525
295. 1065 1.225 0 6.525; 1066 1.47 0 6.525; 1067 1.715 0 6.525; 1068 1.96 0 6.525
296. 1069 2.205 0 6.525; 1070 2.45 0 6.525; 1071 2.695 0 6.525; 1072 2.94 0 6.525
297. 1073 3.185 0 6.525; 1074 3.43 0 6.525; 1075 3.675 0 6.525; 1076 3.92 0 6.525
298. 1077 4.165 0 6.525; 1078 4.41 0 6.525; 1079 4.655 0 6.525; 1080 4.9 0 6.525
299. 1081 0.245 0 6.28333; 1082 0.49 0 6.28333; 1083 0.735 0 6.28333
300. 1084 0.98 0 6.28333; 1085 1.225 0 6.28333; 1086 1.47 0 6.28333
301. 1087 1.715 0 6.28333; 1088 1.96 0 6.28333; 1089 2.205 0 6.28333
302. 1090 2.45 0 6.28333; 1091 2.695 0 6.28333; 1092 2.94 0 6.28333
303. 1093 3.185 0 6.28333; 1094 3.43 0 6.28333; 1095 3.675 0 6.28333
304. 1096 3.92 0 6.28333; 1097 4.165 0 6.28333; 1098 4.41 0 6.28333
305. 1099 4.655 0 6.28333; 1100 4.9 0 6.28333; 1101 0.245 0 6.04167
306. 1102 0.49 0 6.04167; 1103 0.735 0 6.04167; 1104 0.98 0 6.04167
307. 1105 1.225 0 6.04167; 1106 1.47 0 6.04167; 1107 1.715 0 6.04167
308. 1108 1.96 0 6.04167; 1109 2.205 0 6.04167; 1110 2.45 0 6.04167
309. 1111 2.695 0 6.04167; 1112 2.94 0 6.04167; 1113 3.185 0 6.04167
310. 1114 3.43 0 6.04167; 1115 3.675 0 6.04167; 1116 3.92 0 6.04167
311. 1117 4.165 0 6.04167; 1118 4.41 0 6.04167; 1119 4.655 0 6.04167
312. 1120 4.9 0 6.04167; 1121 5.16667 0 7.25; 1122 5.16667 0 7.00833
313. 1123 5.43333 0 7.25; 1124 5.43333 0 7.00833; 1125 5.7 0 7.00833
314. 1126 5.16667 0 6.76667; 1127 5.43333 0 6.76667; 1128 5.7 0 6.76667
315. 1129 5.16667 0 6.525; 1130 5.43333 0 6.525; 1131 5.7 0 6.525
316. 1132 5.16667 0 6.28333; 1133 5.43333 0 6.28333; 1134 5.7 0 6.28333
317. 1135 5.16667 0 6.04167; 1136 5.43333 0 6.04167; 1137 5.7 0 6.04167
318. 1138 5.95217 0 7.25; 1139 5.95217 0 7.00833; 1140 6.20435 0 7.25

319. 1141 6.20435 0 7.00833; 1142 6.45652 0 7.25; 1143 6.45652 0 7.00833
320. 1144 6.7087 0 7.25; 1145 6.7087 0 7.00833; 1146 6.96087 0 7.25
321. 1147 6.96087 0 7.00833; 1148 7.21304 0 7.25; 1149 7.21304 0 7.00833
322. 1150 7.46522 0 7.25; 1151 7.46522 0 7.00833; 1152 7.71739 0 7.25
323. 1153 7.71739 0 7.00833; 1154 7.96957 0 7.25; 1155 7.96957 0 7.00833
324. 1156 8.22174 0 7.25; 1157 8.22174 0 7.00833; 1158 8.47391 0 7.25
325. 1159 8.47391 0 7.00833; 1160 8.72609 0 7.25; 1161 8.72609 0 7.00833
326. 1162 8.97826 0 7.25; 1163 8.97826 0 7.00833; 1164 9.23044 0 7.25
327. 1165 9.23044 0 7.00833; 1166 9.48261 0 7.25; 1167 9.48261 0 7.00833
328. 1168 9.73478 0 7.25; 1169 9.73478 0 7.00833; 1170 9.98696 0 7.25
329. 1171 9.98696 0 7.00833; 1172 10.2391 0 7.25; 1173 10.2391 0 7.00833
330. 1174 10.4913 0 7.25; 1175 10.4913 0 7.00833; 1176 10.7435 0 7.25
331. 1177 10.7435 0 7.00833; 1178 10.9957 0 7.25; 1179 10.9957 0 7.00833
332. 1180 11.2478 0 7.25; 1181 11.2478 0 7.00833; 1182 11.5 0 7.00833
333. 1183 5.95217 0 6.76667; 1184 6.20435 0 6.76667; 1185 6.45652 0 6.76667
334. 1186 6.7087 0 6.76667; 1187 6.96087 0 6.76667; 1188 7.21304 0 6.76667
335. 1189 7.46522 0 6.76667; 1190 7.71739 0 6.76667; 1191 7.96957 0 6.76667
336. 1192 8.22174 0 6.76667; 1193 8.47391 0 6.76667; 1194 8.72609 0 6.76667
337. 1195 8.97826 0 6.76667; 1196 9.23044 0 6.76667; 1197 9.48261 0 6.76667
338. 1198 9.73478 0 6.76667; 1199 9.98696 0 6.76667; 1200 10.2391 0 6.76667
339. 1201 10.4913 0 6.76667; 1202 10.7435 0 6.76667; 1203 10.9957 0 6.76667
340. 1204 11.2478 0 6.76667; 1205 11.5 0 6.76667; 1206 5.95217 0 6.525
341. 1207 6.20435 0 6.525; 1208 6.45652 0 6.525; 1209 6.7087 0 6.525
342. 1210 6.96087 0 6.525; 1211 7.21304 0 6.525; 1212 7.46522 0 6.525
343. 1213 7.71739 0 6.525; 1214 7.96957 0 6.525; 1215 8.22174 0 6.525
344. 1216 8.47391 0 6.525; 1217 8.72609 0 6.525; 1218 8.97826 0 6.525
345. 1219 9.23044 0 6.525; 1220 9.48261 0 6.525; 1221 9.73478 0 6.525
346. 1222 9.98696 0 6.525; 1223 10.2391 0 6.525; 1224 10.4913 0 6.525
347. 1225 10.7435 0 6.525; 1226 10.9957 0 6.525; 1227 11.2478 0 6.525
348. 1228 11.5 0 6.525; 1229 5.95217 0 6.28333; 1230 6.20435 0 6.28333
349. 1231 6.45652 0 6.28333; 1232 6.7087 0 6.28333; 1233 6.96087 0 6.28333
350. 1234 7.21304 0 6.28333; 1235 7.46522 0 6.28333; 1236 7.71739 0 6.28333
351. 1237 7.96957 0 6.28333; 1238 8.22174 0 6.28333; 1239 8.47391 0 6.28333
352. 1240 8.72609 0 6.28333; 1241 8.97826 0 6.28333; 1242 9.23044 0 6.28333
353. 1243 9.48261 0 6.28333; 1244 9.73478 0 6.28333; 1245 9.98696 0 6.28333
354. 1246 10.2391 0 6.28333; 1247 10.4913 0 6.28333; 1248 10.7435 0 6.28333
355. 1249 10.9957 0 6.28333; 1250 11.2478 0 6.28333; 1251 11.5 0 6.28333
356. 1252 5.95217 0 6.04167; 1253 6.20435 0 6.04167; 1254 6.45652 0 6.04167
357. 1255 6.7087 0 6.04167; 1256 6.96087 0 6.04167; 1257 7.21304 0 6.04167
358. 1258 7.46522 0 6.04167; 1259 7.71739 0 6.04167; 1260 7.96957 0 6.04167
359. 1261 8.22174 0 6.04167; 1262 8.47391 0 6.04167; 1263 8.72609 0 6.04167
360. 1264 8.97826 0 6.04167; 1265 9.23044 0 6.04167; 1266 9.48261 0 6.04167
361. 1267 9.73478 0 6.04167; 1268 9.98696 0 6.04167; 1269 10.2391 0 6.04167
362. 1270 10.4913 0 6.04167; 1271 10.7435 0 6.04167; 1272 10.9957 0 6.04167
363. 1273 11.2478 0 6.04167; 1274 11.5 0 6.04167; 1275 11.7833 0 7.25
364. 1276 11.7833 0 7.00833; 1277 12.0667 0 7.25; 1278 12.0667 0 7.00833
365. 1279 12.35 0 7.00833; 1280 11.7833 0 6.76667; 1281 12.0667 0 6.76667
366. 1282 12.35 0 6.76667; 1283 11.7833 0 6.525; 1284 12.0667 0 6.525
367. 1285 12.35 0 6.525; 1286 11.7833 0 6.28333; 1287 12.0667 0 6.28333
368. 1288 12.35 0 6.28333; 1289 11.7833 0 6.04167; 1290 12.0667 0 6.04167
369. 1291 12.35 0 6.04167; 1292 -0.533333 0 7.85; 1293 -0.533333 0 7.55
370. 1294 -0.8 0 7.55; 1295 -0.266667 0 7.85; 1296 -0.266667 0 7.55; 1297 0 0 7.55
371. 1298 0.245 0 7.85; 1299 0.245 0 7.55; 1300 0.49 0 7.85; 1301 0.49 0 7.55
372. 1302 0.735 0 7.85; 1303 0.735 0 7.55; 1304 0.98 0 7.85; 1305 0.98 0 7.55
373. 1306 1.225 0 7.85; 1307 1.225 0 7.55; 1308 1.47 0 7.85; 1309 1.47 0 7.55
374. 1310 1.715 0 7.85; 1311 1.715 0 7.55; 1312 1.96 0 7.85; 1313 1.96 0 7.55

375. 1314 2.205 0 7.85; 1315 2.205 0 7.55; 1316 2.45 0 7.85; 1317 2.45 0 7.55
376. 1318 2.695 0 7.85; 1319 2.695 0 7.55; 1320 2.94 0 7.85; 1321 2.94 0 7.55
377. 1322 3.185 0 7.85; 1323 3.185 0 7.55; 1324 3.43 0 7.85; 1325 3.43 0 7.55
378. 1326 3.675 0 7.85; 1327 3.675 0 7.55; 1328 3.92 0 7.85; 1329 3.92 0 7.55
379. 1330 4.165 0 7.85; 1331 4.165 0 7.55; 1332 4.41 0 7.85; 1333 4.41 0 7.55
380. 1334 4.655 0 7.85; 1335 4.655 0 7.55; 1336 4.9 0 7.55; 1337 5.16667 0 7.85
381. 1338 5.16667 0 7.55; 1339 5.43333 0 7.85; 1340 5.43333 0 7.55; 1341 5.7 0 7.55
382. 1342 5.95217 0 7.85; 1343 5.95217 0 7.55; 1344 6.20435 0 7.85
383. 1345 6.20435 0 7.55; 1346 6.45652 0 7.85; 1347 6.45652 0 7.55
384. 1348 6.7087 0 7.85; 1349 6.7087 0 7.55; 1350 6.96087 0 7.85
385. 1351 6.96087 0 7.55; 1352 7.21304 0 7.85; 1353 7.21304 0 7.55
386. 1354 7.46522 0 7.85; 1355 7.46522 0 7.55; 1356 7.71739 0 7.85
387. 1357 7.71739 0 7.55; 1358 7.96957 0 7.85; 1359 7.96957 0 7.55
388. 1360 8.22174 0 7.85; 1361 8.22174 0 7.55; 1362 8.47391 0 7.85
389. 1363 8.47391 0 7.55; 1364 8.72609 0 7.85; 1365 8.72609 0 7.55
390. 1366 8.97826 0 7.85; 1367 8.97826 0 7.55; 1368 9.23044 0 7.85
391. 1369 9.23044 0 7.55; 1370 9.48261 0 7.85; 1371 9.48261 0 7.55
392. 1372 9.73478 0 7.85; 1373 9.73478 0 7.55; 1374 9.98696 0 7.85
393. 1375 9.98696 0 7.55; 1376 10.2391 0 7.85; 1377 10.2391 0 7.55
394. 1378 10.4913 0 7.85; 1379 10.4913 0 7.55; 1380 10.7435 0 7.85
395. 1381 10.7435 0 7.55; 1382 10.9957 0 7.85; 1383 10.9957 0 7.55
396. 1384 11.2478 0 7.85; 1385 11.2478 0 7.55; 1386 11.5 0 7.55
397. 1387 -0.533333 0 11.6; 1388 -0.533333 0 11.35; 1389 -0.8 0 11.35
398. 1390 -0.266667 0 11.6; 1391 -0.266667 0 11.35; 1392 0 0 11.35
399. 1393 -0.533333 0 11.1; 1394 -0.8 0 11.1; 1395 -0.266667 0 11.1; 1396 0 0 11.1
400. 1397 -0.533333 0 10.85; 1398 -0.8 0 10.85; 1399 -0.266667 0 10.85
401. 1400 0 0 10.85; 1401 -0.533333 0 10.6; 1402 -0.8 0 10.6; 1403 -0.266667 0 10.6
402. 1404 0 0 10.6; 1405 -0.533333 0 10.35; 1406 -0.8 0 10.35
403. 1407 -0.266667 0 10.35; 1408 0 0 10.35; 1409 -0.533333 0 10.1
404. 1410 -0.8 0 10.1; 1411 -0.266667 0 10.1; 1412 0 0 10.1; 1413 -0.533333 0 9.85
405. 1414 -0.8 0 9.85; 1415 -0.266667 0 9.85; 1416 0 0 9.85; 1417 -0.533333 0 9.6
406. 1418 -0.8 0 9.6; 1419 -0.266667 0 9.6; 1420 0 0 9.6; 1421 -0.533333 0 9.35
407. 1422 -0.8 0 9.35; 1423 -0.266667 0 9.35; 1424 0 0 9.35; 1425 -0.533333 0 9.1
408. 1426 -0.8 0 9.1; 1427 -0.266667 0 9.1; 1428 0 0 9.1; 1429 -0.533333 0 8.85
409. 1430 -0.8 0 8.85; 1431 -0.266667 0 8.85; 1432 0 0 8.85; 1433 -0.533333 0 8.6
410. 1434 -0.8 0 8.6; 1435 -0.266667 0 8.6; 1436 0 0 8.6; 1437 -0.533333 0 8.35
411. 1438 -0.8 0 8.35; 1439 -0.266667 0 8.35; 1440 0 0 8.35; 1441 -0.533333 0 8.1
412. 1442 -0.8 0 8.1; 1443 -0.266667 0 8.1; 1444 0 0 8.1; 1445 0.245 0 11.6
413. 1446 0.245 0 11.35; 1447 0.49 0 11.6; 1448 0.49 0 11.35; 1449 0.735 0 11.6
414. 1450 0.735 0 11.35; 1451 0.98 0 11.6; 1452 0.98 0 11.35; 1453 1.225 0 11.6
415. 1454 1.225 0 11.35; 1455 1.47 0 11.6; 1456 1.47 0 11.35; 1457 1.715 0 11.6
416. 1458 1.715 0 11.35; 1459 1.96 0 11.6; 1460 1.96 0 11.35; 1461 2.205 0 11.6
417. 1462 2.205 0 11.35; 1463 2.45 0 11.6; 1464 2.45 0 11.35; 1465 2.695 0 11.6
418. 1466 2.695 0 11.35; 1467 2.94 0 11.6; 1468 2.94 0 11.35; 1469 3.185 0 11.6
419. 1470 3.185 0 11.35; 1471 3.43 0 11.6; 1472 3.43 0 11.35; 1473 3.675 0 11.6
420. 1474 3.675 0 11.35; 1475 3.92 0 11.6; 1476 3.92 0 11.35; 1477 4.165 0 11.6
421. 1478 4.165 0 11.35; 1479 4.41 0 11.6; 1480 4.41 0 11.35; 1481 4.655 0 11.6
422. 1482 4.655 0 11.35; 1483 4.9 0 11.35; 1484 0.245 0 11.1; 1485 0.49 0 11.1
423. 1486 0.735 0 11.1; 1487 0.98 0 11.1; 1488 1.225 0 11.1; 1489 1.47 0 11.1
424. 1490 1.715 0 11.1; 1491 1.96 0 11.1; 1492 2.205 0 11.1; 1493 2.45 0 11.1
425. 1494 2.695 0 11.1; 1495 2.94 0 11.1; 1496 3.185 0 11.1; 1497 3.43 0 11.1
426. 1498 3.675 0 11.1; 1499 3.92 0 11.1; 1500 4.165 0 11.1; 1501 4.41 0 11.1
427. 1502 4.655 0 11.1; 1503 4.9 0 11.1; 1504 0.245 0 10.85; 1505 0.49 0 10.85
428. 1506 0.735 0 10.85; 1507 0.98 0 10.85; 1508 1.225 0 10.85; 1509 1.47 0 10.85
429. 1510 1.715 0 10.85; 1511 1.96 0 10.85; 1512 2.205 0 10.85; 1513 2.45 0 10.85
430. 1514 2.695 0 10.85; 1515 2.94 0 10.85; 1516 3.185 0 10.85; 1517 3.43 0 10.85

431. 1518 3.675 0 10.85; 1519 3.92 0 10.85; 1520 4.165 0 10.85; 1521 4.41 0 10.85
432. 1522 4.655 0 10.85; 1523 4.9 0 10.85; 1524 0.245 0 10.6; 1525 0.49 0 10.6
433. 1526 0.735 0 10.6; 1527 0.98 0 10.6; 1528 1.225 0 10.6; 1529 1.47 0 10.6
434. 1530 1.715 0 10.6; 1531 1.96 0 10.6; 1532 2.205 0 10.6; 1533 2.45 0 10.6
435. 1534 2.695 0 10.6; 1535 2.94 0 10.6; 1536 3.185 0 10.6; 1537 3.43 0 10.6
436. 1538 3.675 0 10.6; 1539 3.92 0 10.6; 1540 4.165 0 10.6; 1541 4.41 0 10.6
437. 1542 4.655 0 10.6; 1543 4.9 0 10.6; 1544 0.245 0 10.35; 1545 0.49 0 10.35
438. 1546 0.735 0 10.35; 1547 0.98 0 10.35; 1548 1.225 0 10.35; 1549 1.47 0 10.35
439. 1550 1.715 0 10.35; 1551 1.96 0 10.35; 1552 2.205 0 10.35; 1553 2.45 0 10.35
440. 1554 2.695 0 10.35; 1555 2.94 0 10.35; 1556 3.185 0 10.35; 1557 3.43 0 10.35
441. 1558 3.675 0 10.35; 1559 3.92 0 10.35; 1560 4.165 0 10.35; 1561 4.41 0 10.35
442. 1562 4.655 0 10.35; 1563 4.9 0 10.35; 1564 0.245 0 10.1; 1565 0.49 0 10.1
443. 1566 0.735 0 10.1; 1567 0.98 0 10.1; 1568 1.225 0 10.1; 1569 1.47 0 10.1
444. 1570 1.715 0 10.1; 1571 1.96 0 10.1; 1572 2.205 0 10.1; 1573 2.45 0 10.1
445. 1574 2.695 0 10.1; 1575 2.94 0 10.1; 1576 3.185 0 10.1; 1577 3.43 0 10.1
446. 1578 3.675 0 10.1; 1579 3.92 0 10.1; 1580 4.165 0 10.1; 1581 4.41 0 10.1
447. 1582 4.655 0 10.1; 1583 4.9 0 10.1; 1584 0.245 0 9.85; 1585 0.49 0 9.85
448. 1586 0.735 0 9.85; 1587 0.98 0 9.85; 1588 1.225 0 9.85; 1589 1.47 0 9.85
449. 1590 1.715 0 9.85; 1591 1.96 0 9.85; 1592 2.205 0 9.85; 1593 2.45 0 9.85
450. 1594 2.695 0 9.85; 1595 2.94 0 9.85; 1596 3.185 0 9.85; 1597 3.43 0 9.85
451. 1598 3.675 0 9.85; 1599 3.92 0 9.85; 1600 4.165 0 9.85; 1601 4.41 0 9.85
452. 1602 4.655 0 9.85; 1603 4.9 0 9.85; 1604 0.245 0 9.6; 1605 0.49 0 9.6
453. 1606 0.735 0 9.6; 1607 0.98 0 9.6; 1608 1.225 0 9.6; 1609 1.47 0 9.6
454. 1610 1.715 0 9.6; 1611 1.96 0 9.6; 1612 2.205 0 9.6; 1613 2.45 0 9.6
455. 1614 2.695 0 9.6; 1615 2.94 0 9.6; 1616 3.185 0 9.6; 1617 3.43 0 9.6
456. 1618 3.675 0 9.6; 1619 3.92 0 9.6; 1620 4.165 0 9.6; 1621 4.41 0 9.6
457. 1622 4.655 0 9.6; 1623 4.9 0 9.6; 1624 0.245 0 9.35; 1625 0.49 0 9.35
458. 1626 0.735 0 9.35; 1627 0.98 0 9.35; 1628 1.225 0 9.35; 1629 1.47 0 9.35
459. 1630 1.715 0 9.35; 1631 1.96 0 9.35; 1632 2.205 0 9.35; 1633 2.45 0 9.35
460. 1634 2.695 0 9.35; 1635 2.94 0 9.35; 1636 3.185 0 9.35; 1637 3.43 0 9.35
461. 1638 3.675 0 9.35; 1639 3.92 0 9.35; 1640 4.165 0 9.35; 1641 4.41 0 9.35
462. 1642 4.655 0 9.35; 1643 4.9 0 9.35; 1644 0.245 0 9.1; 1645 0.49 0 9.1
463. 1646 0.735 0 9.1; 1647 0.98 0 9.1; 1648 1.225 0 9.1; 1649 1.47 0 9.1
464. 1650 1.715 0 9.1; 1651 1.96 0 9.1; 1652 2.205 0 9.1; 1653 2.45 0 9.1
465. 1654 2.695 0 9.1; 1655 2.94 0 9.1; 1656 3.185 0 9.1; 1657 3.43 0 9.1
466. 1658 3.675 0 9.1; 1659 3.92 0 9.1; 1660 4.165 0 9.1; 1661 4.41 0 9.1
467. 1662 4.655 0 9.1; 1663 4.9 0 9.1; 1664 0.245 0 8.85; 1665 0.49 0 8.85
468. 1666 0.735 0 8.85; 1667 0.98 0 8.85; 1668 1.225 0 8.85; 1669 1.47 0 8.85
469. 1670 1.715 0 8.85; 1671 1.96 0 8.85; 1672 2.205 0 8.85; 1673 2.45 0 8.85
470. 1674 2.695 0 8.85; 1675 2.94 0 8.85; 1676 3.185 0 8.85; 1677 3.43 0 8.85
471. 1678 3.675 0 8.85; 1679 3.92 0 8.85; 1680 4.165 0 8.85; 1681 4.41 0 8.85
472. 1682 4.655 0 8.85; 1683 4.9 0 8.85; 1684 0.245 0 8.6; 1685 0.49 0 8.6
473. 1686 0.735 0 8.6; 1687 0.98 0 8.6; 1688 1.225 0 8.6; 1689 1.47 0 8.6
474. 1690 1.715 0 8.6; 1691 1.96 0 8.6; 1692 2.205 0 8.6; 1693 2.45 0 8.6
475. 1694 2.695 0 8.6; 1695 2.94 0 8.6; 1696 3.185 0 8.6; 1697 3.43 0 8.6
476. 1698 3.675 0 8.6; 1699 3.92 0 8.6; 1700 4.165 0 8.6; 1701 4.41 0 8.6
477. 1702 4.655 0 8.6; 1703 4.9 0 8.6; 1704 0.245 0 8.35; 1705 0.49 0 8.35
478. 1706 0.735 0 8.35; 1707 0.98 0 8.35; 1708 1.225 0 8.35; 1709 1.47 0 8.35
479. 1710 1.715 0 8.35; 1711 1.96 0 8.35; 1712 2.205 0 8.35; 1713 2.45 0 8.35
480. 1714 2.695 0 8.35; 1715 2.94 0 8.35; 1716 3.185 0 8.35; 1717 3.43 0 8.35
481. 1718 3.675 0 8.35; 1719 3.92 0 8.35; 1720 4.165 0 8.35; 1721 4.41 0 8.35
482. 1722 4.655 0 8.35; 1723 4.9 0 8.35; 1724 0.245 0 8.1; 1725 0.49 0 8.1
483. 1726 0.735 0 8.1; 1727 0.98 0 8.1; 1728 1.225 0 8.1; 1729 1.47 0 8.1
484. 1730 1.715 0 8.1; 1731 1.96 0 8.1; 1732 2.205 0 8.1; 1733 2.45 0 8.1
485. 1734 2.695 0 8.1; 1735 2.94 0 8.1; 1736 3.185 0 8.1; 1737 3.43 0 8.1
486. 1738 3.675 0 8.1; 1739 3.92 0 8.1; 1740 4.165 0 8.1; 1741 4.41 0 8.1

```
487. 1742 4.655 0 8.1; 1743 4.9 0 8.1; 1744 5.16667 0 11.6; 1745 5.16667 0 11.35
488. 1746 5.43333 0 11.6; 1747 5.43333 0 11.35; 1748 5.7 0 11.35
489. 1749 5.16667 0 11.1; 1750 5.43333 0 11.1; 1751 5.7 0 11.1
490. 1752 5.16667 0 10.85; 1753 5.43333 0 10.85; 1754 5.7 0 10.85
491. 1755 5.16667 0 10.6; 1756 5.43333 0 10.6; 1757 5.7 0 10.6
492. 1758 5.16667 0 10.35; 1759 5.43333 0 10.35; 1760 5.7 0 10.35
493. 1761 5.16667 0 10.1; 1762 5.43333 0 10.1; 1763 5.7 0 10.1; 1764 5.16667 0 9.85
494. 1765 5.43333 0 9.85; 1766 5.7 0 9.85; 1767 5.16667 0 9.6; 1768 5.43333 0 9.6
495. 1769 5.7 0 9.6; 1770 5.16667 0 9.35; 1771 5.43333 0 9.35; 1772 5.7 0 9.35
496. 1773 5.16667 0 9.1; 1774 5.43333 0 9.1; 1775 5.7 0 9.1; 1776 5.16667 0 8.85
497. 1777 5.43333 0 8.85; 1778 5.7 0 8.85; 1779 5.16667 0 8.6; 1780 5.43333 0 8.6
498. 1781 5.7 0 8.6; 1782 5.16667 0 8.35; 1783 5.43333 0 8.35; 1784 5.7 0 8.35
499. 1785 5.16667 0 8.1; 1786 5.43333 0 8.1; 1787 5.7 0 8.1; 1788 -0.533333 0 12.4
500. 1789 -0.533333 0 12.1333; 1790 -0.8 0 12.1333; 1791 -0.266667 0 12.4
501. 1792 -0.266667 0 12.1333; 1793 0 0 12.1333; 1794 -0.533333 0 11.8667
502. 1795 -0.8 0 11.8667; 1796 -0.266667 0 11.8667; 1797 0 0 11.8667
503. 1798 0.245 0 12.4; 1799 0.245 0 12.1333; 1800 0.49 0 12.4; 1801 0.49 0 12.1333
504. 1802 0.735 0 12.4; 1803 0.735 0 12.1333; 1804 0.98 0 12.4; 1805 0.98 0 12.1333
505. 1806 1.225 0 12.4; 1807 1.225 0 12.1333; 1808 1.47 0 12.4; 1809 1.47 0 12.1333
506. 1810 1.715 0 12.4; 1811 1.715 0 12.1333; 1812 1.96 0 12.4; 1813 1.96 0 12.1333
507. 1814 2.205 0 12.4; 1815 2.205 0 12.1333; 1816 2.45 0 12.4; 1817 2.45 0 12.1333
508. 1818 2.695 0 12.4; 1819 2.695 0 12.1333; 1820 2.94 0 12.4; 1821 2.94 0 12.1333
509. 1822 3.185 0 12.4; 1823 3.185 0 12.1333; 1824 3.43 0 12.4; 1825 3.43 0 12.1333
510. 1826 3.675 0 12.4; 1827 3.675 0 12.1333; 1828 3.92 0 12.4; 1829 3.92 0 12.1333
511. 1830 4.165 0 12.4; 1831 4.165 0 12.1333; 1832 4.41 0 12.4; 1833 4.41 0 12.1333
512. 1834 4.655 0 12.4; 1835 4.655 0 12.1333; 1836 4.9 0 12.1333
513. 1837 0.245 0 11.8667; 1838 0.49 0 11.8667; 1839 0.735 0 11.8667
514. 1840 0.98 0 11.8667; 1841 1.225 0 11.8667; 1842 1.47 0 11.8667
515. 1843 1.715 0 11.8667; 1844 1.96 0 11.8667; 1845 2.205 0 11.8667
516. 1846 2.45 0 11.8667; 1847 2.695 0 11.8667; 1848 2.94 0 11.8667
517. 1849 3.185 0 11.8667; 1850 3.43 0 11.8667; 1851 3.675 0 11.8667
518. 1852 3.92 0 11.8667; 1853 4.165 0 11.8667; 1854 4.41 0 11.8667
519. 1855 4.655 0 11.8667; 1856 4.9 0 11.8667; 1861 5.16667 0 -0.533333
520. 1862 5.43333 0 -0.266667; 1863 5.16667 0 -0.266667; 1870 11.7833 0 3.75
521. 1871 12.0667 0 4.05; 1872 11.7833 0 4.05; 1878 11.7833 0 7.85
522. 1879 12.0667 0 7.55; 1880 11.7833 0 7.55; 1885 5.43333 0 11.8667
523. 1886 5.16667 0 12.1333; 1887 5.16667 0 11.8667; 1888 0 0.2 0; 1889 4.9 0.2 0
524. 1890 4.9 0.2 4.35; 1891 0 0.2 5.8; 1892 4.9 0.2 7.25; 1893 0 0.2 11.6
525. 1894 4.9 0.2 11.6; 1895 11.5 0.2 4.35; 1896 11.5 0.2 7.25
526. MEMBER INCIDENCES
527. 19 1 1888; 20 2 1889; 21 4 1890; 22 5 1891; 23 8 1892; 24 9 1893; 25 10 1894
528. 26 11 1895; 27 13 1896; 2452 1888 14; 2453 1889 15; 2454 1890 16; 2455 1891 17
529. 2456 1892 18; 2457 1893 19; 2458 1894 20; 2459 1895 21; 2460 1896 22
530. ELEMENT INCIDENCES SHELL
531. 97 31 56 57 58; 99 56 59 60 57; 101 59 1 61 60; 103 58 57 62 63
532. 104 57 60 64 62; 106 60 61 65 64; 108 63 62 66 39; 110 62 64 67 66
533. 111 64 65 36 67; 113 1 68 69 61; 115 68 70 71 69; 117 70 72 73 71
534. 119 72 74 75 73; 121 74 76 77 75; 123 76 78 79 77; 125 78 80 81 79
535. 127 80 82 83 81; 129 82 84 85 83; 131 84 86 87 85; 133 86 88 89 87
536. 135 88 90 91 89; 137 90 92 93 91; 139 92 94 95 93; 141 94 96 97 95
537. 143 96 98 99 97; 145 98 100 101 99; 147 100 102 103 101; 149 102 104 105 103
538. 151 104 2 106 105; 152 61 69 107 65; 153 69 71 108 107; 154 71 73 109 108
539. 155 73 75 110 109; 156 75 77 111 110; 157 77 79 112 111; 158 79 81 113 112
540. 159 81 83 114 113; 160 83 85 115 114; 161 85 87 116 115; 162 87 89 117 116
541. 163 89 91 118 117; 164 91 93 119 118; 165 93 95 120 119; 166 95 97 121 120
542. 167 97 99 122 121; 168 99 101 123 122; 169 101 103 124 123
```

543. 170 103 105 125 124; 172 105 106 126 125; 174 65 107 127 36
544. 176 107 108 128 127; 178 108 109 129 128; 180 109 110 130 129
545. 182 110 111 131 130; 184 111 112 132 131; 186 112 113 133 132
546. 188 113 114 134 133; 190 114 115 135 134; 192 115 116 136 135
547. 194 116 117 137 136; 196 117 118 138 137; 198 118 119 139 138
548. 200 119 120 140 139; 202 120 121 141 140; 204 121 122 142 141
549. 206 122 123 143 142; 208 123 124 144 143; 210 124 125 145 144
550. 211 125 126 37 145; 214 52 146 147 148; 216 146 149 150 147
551. 218 149 50 151 150; 220 148 147 152 153; 221 147 150 154 152
552. 223 150 151 155 154; 225 153 152 156 157; 226 152 154 158 156
553. 228 154 155 159 158; 230 157 156 160 161; 231 156 158 162 160
554. 233 158 159 163 162; 235 161 160 164 165; 236 160 162 166 164
555. 238 162 163 167 166; 240 165 164 168 169; 241 164 166 170 168
556. 243 166 167 171 170; 245 169 168 172 173; 246 168 170 174 172
557. 248 170 171 175 174; 250 173 172 176 177; 251 172 174 178 176
558. 253 174 175 179 178; 255 177 176 180 181; 256 176 178 182 180
559. 258 178 179 183 182; 260 181 180 184 185; 261 180 182 186 184
560. 263 182 183 187 186; 265 185 184 188 189; 266 184 186 190 188
561. 268 186 187 191 190; 270 189 188 192 193; 271 188 190 194 192
562. 273 190 191 195 194; 275 193 192 196 197; 276 192 194 198 196
563. 278 194 195 199 198; 280 197 196 200 201; 281 196 198 202 200
564. 283 198 199 203 202; 284 201 200 56 31; 285 200 202 59 56; 286 202 203 1 59
565. 288 50 204 205 151; 290 204 206 207 205; 292 206 208 209 207
566. 294 208 210 211 209; 296 210 212 213 211; 298 212 214 215 213
567. 300 214 216 217 215; 302 216 218 219 217; 304 218 220 221 219
568. 306 220 222 223 221; 308 222 224 225 223; 310 224 226 227 225
569. 312 226 228 229 227; 314 228 230 231 229; 316 230 232 233 231
570. 318 232 234 235 233; 320 234 236 237 235; 322 236 238 239 237
571. 324 238 240 241 239; 326 240 51 242 241; 327 151 205 243 155
572. 328 205 207 244 243; 329 207 209 245 244; 330 209 211 246 245
573. 331 211 213 247 246; 332 213 215 248 247; 333 215 217 249 248
574. 334 217 219 250 249; 335 219 221 251 250; 336 221 223 252 251
575. 337 223 225 253 252; 338 225 227 254 253; 339 227 229 255 254
576. 340 229 231 256 255; 341 231 233 257 256; 342 233 235 258 257
577. 343 235 237 259 258; 344 237 239 260 259; 345 239 241 261 260
578. 347 241 242 262 261; 348 155 243 263 159; 349 243 244 264 263
579. 350 244 245 265 264; 351 245 246 266 265; 352 246 247 267 266
580. 353 247 248 268 267; 354 248 249 269 268; 355 249 250 270 269
581. 356 250 251 271 270; 357 251 252 272 271; 358 252 253 273 272
582. 359 253 254 274 273; 360 254 255 275 274; 361 255 256 276 275
583. 362 256 257 277 276; 363 257 258 278 277; 364 258 259 279 278
584. 365 259 260 280 279; 366 260 261 281 280; 368 261 262 282 281
585. 369 159 263 283 163; 370 263 264 284 283; 371 264 265 285 284
586. 372 265 266 286 285; 373 266 267 287 286; 374 267 268 288 287
587. 375 268 269 289 288; 376 269 270 290 289; 377 270 271 291 290
588. 378 271 272 292 291; 379 272 273 293 292; 380 273 274 294 293
589. 381 274 275 295 294; 382 275 276 296 295; 383 276 277 297 296
590. 384 277 278 298 297; 385 278 279 299 298; 386 279 280 300 299
591. 387 280 281 301 300; 389 281 282 302 301; 390 163 283 303 167
592. 391 283 284 304 303; 392 284 285 305 304; 393 285 286 306 305
593. 394 286 287 307 306; 395 287 288 308 307; 396 288 289 309 308
594. 397 289 290 310 309; 398 290 291 311 310; 399 291 292 312 311
595. 400 292 293 313 312; 401 293 294 314 313; 402 294 295 315 314
596. 403 295 296 316 315; 404 296 297 317 316; 405 297 298 318 317
597. 406 298 299 319 318; 407 299 300 320 319; 408 300 301 321 320
598. 410 301 302 322 321; 411 167 303 323 171; 412 303 304 324 323

599. 413 304 305 325 324; 414 305 306 326 325; 415 306 307 327 326
600. 416 307 308 328 327; 417 308 309 329 328; 418 309 310 330 329
601. 419 310 311 331 330; 420 311 312 332 331; 421 312 313 333 332
602. 422 313 314 334 333; 423 314 315 335 334; 424 315 316 336 335
603. 425 316 317 337 336; 426 317 318 338 337; 427 318 319 339 338
604. 428 319 320 340 339; 429 320 321 341 340; 431 321 322 342 341
605. 432 171 323 343 175; 433 323 324 344 343; 434 324 325 345 344
606. 435 325 326 346 345; 436 326 327 347 346; 437 327 328 348 347
607. 438 328 329 349 348; 439 329 330 350 349; 440 330 331 351 350
608. 441 331 332 352 351; 442 332 333 353 352; 443 333 334 354 353
609. 444 334 335 355 354; 445 335 336 356 355; 446 336 337 357 356
610. 447 337 338 358 357; 448 338 339 359 358; 449 339 340 360 359
611. 450 340 341 361 360; 452 341 342 362 361; 453 175 343 363 179
612. 454 343 344 364 363; 455 344 345 365 364; 456 345 346 366 365
613. 457 346 347 367 366; 458 347 348 368 367; 459 348 349 369 368
614. 460 349 350 370 369; 461 350 351 371 370; 462 351 352 372 371
615. 463 352 353 373 372; 464 353 354 374 373; 465 354 355 375 374
616. 466 355 356 376 375; 467 356 357 377 376; 468 357 358 378 377
617. 469 358 359 379 378; 470 359 360 380 379; 471 360 361 381 380
618. 473 361 362 382 381; 474 179 363 383 183; 475 363 364 384 383
619. 476 364 365 385 384; 477 365 366 386 385; 478 366 367 387 386
620. 479 367 368 388 387; 480 368 369 389 388; 481 369 370 390 389
621. 482 370 371 391 390; 483 371 372 392 391; 484 372 373 393 392
622. 485 373 374 394 393; 486 374 375 395 394; 487 375 376 396 395
623. 488 376 377 397 396; 489 377 378 398 397; 490 378 379 399 398
624. 491 379 380 400 399; 492 380 381 401 400; 494 381 382 402 401
625. 495 183 383 403 187; 496 383 384 404 403; 497 384 385 405 404
626. 498 385 386 406 405; 499 386 387 407 406; 500 387 388 408 407
627. 501 388 389 409 408; 502 389 390 410 409; 503 390 391 411 410
628. 504 391 392 412 411; 505 392 393 413 412; 506 393 394 414 413
629. 507 394 395 415 414; 508 395 396 416 415; 509 396 397 417 416
630. 510 397 398 418 417; 511 398 399 419 418; 512 399 400 420 419
631. 513 400 401 421 420; 515 401 402 422 421; 516 187 403 423 191
632. 517 403 404 424 423; 518 404 405 425 424; 519 405 406 426 425
633. 520 406 407 427 426; 521 407 408 428 427; 522 408 409 429 428
634. 523 409 410 430 429; 524 410 411 431 430; 525 411 412 432 431
635. 526 412 413 433 432; 527 413 414 434 433; 528 414 415 435 434
636. 529 415 416 436 435; 530 416 417 437 436; 531 417 418 438 437
637. 532 418 419 439 438; 533 419 420 440 439; 534 420 421 441 440
638. 536 421 422 442 441; 537 191 423 443 195; 538 423 424 444 443
639. 539 424 425 445 444; 540 425 426 446 445; 541 426 427 447 446
640. 542 427 428 448 447; 543 428 429 449 448; 544 429 430 450 449
641. 545 430 431 451 450; 546 431 432 452 451; 547 432 433 453 452
642. 548 433 434 454 453; 549 434 435 455 454; 550 435 436 456 455
643. 551 436 437 457 456; 552 437 438 458 457; 553 438 439 459 458
644. 554 439 440 460 459; 555 440 441 461 460; 557 441 442 462 461
645. 558 195 443 463 199; 559 443 444 464 463; 560 444 445 465 464
646. 561 445 446 466 465; 562 446 447 467 466; 563 447 448 468 467
647. 564 448 449 469 468; 565 449 450 470 469; 566 450 451 471 470
648. 567 451 452 472 471; 568 452 453 473 472; 569 453 454 474 473
649. 570 454 455 475 474; 571 455 456 476 475; 572 456 457 477 476
650. 573 457 458 478 477; 574 458 459 479 478; 575 459 460 480 479
651. 576 460 461 481 480; 578 461 462 482 481; 579 199 463 483 203
652. 580 463 464 484 483; 581 464 465 485 484; 582 465 466 486 485
653. 583 466 467 487 486; 584 467 468 488 487; 585 468 469 489 488
654. 586 469 470 490 489; 587 470 471 491 490; 588 471 472 492 491

655. 589 472 473 493 492; 590 473 474 494 493; 591 474 475 495 494
656. 592 475 476 496 495; 593 476 477 497 496; 594 477 478 498 497
657. 595 478 479 499 498; 596 479 480 500 499; 597 480 481 501 500
658. 599 481 482 502 501; 600 203 483 68 1; 601 483 484 70 68; 602 484 485 72 70
659. 603 485 486 74 72; 604 486 487 76 74; 605 487 488 78 76; 606 488 489 80 78
660. 607 489 490 82 80; 608 490 491 84 82; 609 491 492 86 84; 610 492 493 88 86
661. 611 493 494 90 88; 612 494 495 92 90; 613 495 496 94 92; 614 496 497 96 94
662. 615 497 498 98 96; 616 498 499 100 98; 617 499 500 102 100
663. 618 500 501 104 102; 619 501 502 2 104; 621 51 503 504 242
664. 623 503 505 506 504; 625 505 46 507 506; 626 242 504 508 262
665. 627 504 506 509 508; 629 506 507 510 509; 630 262 508 511 282
666. 631 508 509 512 511; 633 509 510 513 512; 634 282 511 514 302
667. 635 511 512 515 514; 637 512 513 516 515; 638 302 514 517 322
668. 639 514 515 518 517; 641 515 516 519 518; 642 322 517 520 342
669. 643 517 518 521 520; 645 518 519 522 521; 646 342 520 523 362
670. 647 520 521 524 523; 649 521 522 525 524; 650 362 523 526 382
671. 651 523 524 527 526; 653 524 525 528 527; 654 382 526 529 402
672. 655 526 527 530 529; 657 527 528 531 530; 658 402 529 532 422
673. 659 529 530 533 532; 661 530 531 534 533; 662 422 532 535 442
674. 663 532 533 536 535; 665 533 534 537 536; 666 442 535 538 462
675. 667 535 536 539 538; 669 536 537 540 539; 670 462 538 541 482
676. 671 538 539 542 541; 673 539 540 543 542; 674 482 541 544 502
677. 675 541 542 545 544; 677 542 543 546 545; 679 502 544 547 2
678. 681 544 545 548 547; 682 545 546 26 548; 709 32 549 550 551
679. 711 549 552 553 550; 713 552 3 554 553; 714 551 550 146 52
680. 715 550 553 149 146; 716 553 554 50 149; 718 3 555 556 554
681. 720 555 557 558 556; 722 557 559 560 558; 724 559 561 562 560
682. 726 561 563 564 562; 728 563 565 566 564; 730 565 567 568 566
683. 732 567 569 570 568; 734 569 571 572 570; 736 571 573 574 572
684. 738 573 575 576 574; 740 575 577 578 576; 742 577 579 580 578
685. 744 579 581 582 580; 746 581 583 584 582; 748 583 585 586 584
686. 750 585 587 588 586; 752 587 589 590 588; 754 589 591 592 590
687. 756 591 4 593 592; 757 554 556 204 50; 758 556 558 206 204
688. 759 558 560 208 206; 760 560 562 210 208; 761 562 564 212 210
689. 762 564 566 214 212; 763 566 568 216 214; 764 568 570 218 216
690. 765 570 572 220 218; 766 572 574 222 220; 767 574 576 224 222
691. 768 576 578 226 224; 769 578 580 228 226; 770 580 582 230 228
692. 771 582 584 232 230; 772 584 586 234 232; 773 586 588 236 234
693. 774 588 590 238 236; 775 590 592 240 238; 776 592 593 51 240
694. 778 4 594 595 593; 780 594 596 597 595; 782 596 27 598 597; 783 593 595 503 51
695. 784 595 597 505 503; 785 597 598 46 505; 787 27 599 600 598
696. 789 599 601 602 600; 791 601 603 604 602; 793 603 605 606 604
697. 795 605 607 608 606; 797 607 609 610 608; 799 609 611 612 610
698. 801 611 613 614 612; 803 613 615 616 614; 805 615 617 618 616
699. 807 617 619 620 618; 809 619 621 622 620; 811 621 623 624 622
700. 813 623 625 626 624; 815 625 627 628 626; 817 627 629 630 628
701. 819 629 631 632 630; 821 631 633 634 632; 823 633 635 636 634
702. 825 635 637 638 636; 827 637 639 640 638; 829 639 641 642 640
703. 831 641 11 643 642; 833 598 600 644 46; 835 600 602 645 644
704. 837 602 604 646 645; 839 604 606 647 646; 841 606 608 648 647
705. 843 608 610 649 648; 845 610 612 650 649; 847 612 614 651 650
706. 849 614 616 652 651; 851 616 618 653 652; 853 618 620 654 653
707. 855 620 622 655 654; 857 622 624 656 655; 859 624 626 657 656
708. 861 626 628 658 657; 863 628 630 659 658; 865 630 632 660 659
709. 867 632 634 661 660; 869 634 636 662 661; 871 636 638 663 662
710. 873 638 640 664 663; 875 640 642 665 664; 876 642 643 44 665

711. 879 33 666 667 668; 881 666 669 670 667; 883 669 5 671 670
712. 885 668 667 672 673; 886 667 670 674 672; 888 670 671 675 674
713. 890 673 672 676 677; 891 672 674 678 676; 893 674 675 679 678
714. 895 677 676 680 681; 896 676 678 682 680; 898 678 679 683 682
715. 900 681 680 684 685; 901 680 682 686 684; 903 682 683 687 686
716. 904 685 684 549 32; 905 684 686 552 549; 906 686 687 3 552; 908 5 688 689 671
717. 910 688 690 691 689; 912 690 692 693 691; 914 692 694 695 693
718. 916 694 696 697 695; 918 696 698 699 697; 920 698 700 701 699
719. 922 700 702 703 701; 924 702 704 705 703; 926 704 706 707 705
720. 928 706 708 709 707; 930 708 710 711 709; 932 710 712 713 711
721. 934 712 714 715 713; 936 714 716 717 715; 938 716 718 719 717
722. 940 718 720 721 719; 942 720 722 723 721; 944 722 724 725 723
723. 946 724 6 726 725; 947 671 689 727 675; 948 689 691 728 727
724. 949 691 693 729 728; 950 693 695 730 729; 951 695 697 731 730
725. 952 697 699 732 731; 953 699 701 733 732; 954 701 703 734 733
726. 955 703 705 735 734; 956 705 707 736 735; 957 707 709 737 736
727. 958 709 711 738 737; 959 711 713 739 738; 960 713 715 740 739
728. 961 715 717 741 740; 962 717 719 742 741; 963 719 721 743 742
729. 964 721 723 744 743; 965 723 725 745 744; 967 725 726 746 745
730. 968 675 727 747 679; 969 727 728 748 747; 970 728 729 749 748
731. 971 729 730 750 749; 972 730 731 751 750; 973 731 732 752 751
732. 974 732 733 753 752; 975 733 734 754 753; 976 734 735 755 754
733. 977 735 736 756 755; 978 736 737 757 756; 979 737 738 758 757
734. 980 738 739 759 758; 981 739 740 760 759; 982 740 741 761 760
735. 983 741 742 762 761; 984 742 743 763 762; 985 743 744 764 763
736. 986 744 745 765 764; 988 745 746 766 765; 989 679 747 767 683
737. 990 747 748 768 767; 991 748 749 769 768; 992 749 750 770 769
738. 993 750 751 771 770; 994 751 752 772 771; 995 752 753 773 772
739. 996 753 754 774 773; 997 754 755 775 774; 998 755 756 776 775
740. 999 756 757 777 776; 1000 757 758 778 777; 1001 758 759 779 778
741. 1002 759 760 780 779; 1003 760 761 781 780; 1004 761 762 782 781
742. 1005 762 763 783 782; 1006 763 764 784 783; 1007 764 765 785 784
743. 1009 765 766 786 785; 1010 683 767 787 687; 1011 767 768 788 787
744. 1012 768 769 789 788; 1013 769 770 790 789; 1014 770 771 791 790
745. 1015 771 772 792 791; 1016 772 773 793 792; 1017 773 774 794 793
746. 1018 774 775 795 794; 1019 775 776 796 795; 1020 776 777 797 796
747. 1021 777 778 798 797; 1022 778 779 799 798; 1023 779 780 800 799
748. 1024 780 781 801 800; 1025 781 782 802 801; 1026 782 783 803 802
749. 1027 783 784 804 803; 1028 784 785 805 804; 1030 785 786 806 805
750. 1031 687 787 555 3; 1032 787 788 557 555; 1033 788 789 559 557
751. 1034 789 790 561 559; 1035 790 791 563 561; 1036 791 792 565 563
752. 1037 792 793 567 565; 1038 793 794 569 567; 1039 794 795 571 569
753. 1040 795 796 573 571; 1041 796 797 575 573; 1042 797 798 577 575
754. 1043 798 799 579 577; 1044 799 800 581 579; 1045 800 801 583 581
755. 1046 801 802 585 583; 1047 802 803 587 585; 1048 803 804 589 587
756. 1049 804 805 591 589; 1050 805 806 4 591; 1052 6 807 808 726
757. 1054 807 809 810 808; 1056 809 28 811 810; 1057 726 808 812 746
758. 1058 808 810 813 812; 1060 810 811 814 813; 1061 746 812 815 766
759. 1062 812 813 816 815; 1064 813 814 817 816; 1065 766 815 818 786
760. 1066 815 816 819 818; 1068 816 817 820 819; 1069 786 818 821 806
761. 1070 818 819 822 821; 1072 819 820 823 822; 1073 806 821 594 4
762. 1074 821 822 596 594; 1075 822 823 27 596; 1077 28 824 825 811
763. 1079 824 826 827 825; 1081 826 828 829 827; 1083 828 830 831 829
764. 1085 830 832 833 831; 1087 832 834 835 833; 1089 834 836 837 835
765. 1091 836 838 839 837; 1093 838 840 841 839; 1095 840 842 843 841
766. 1097 842 844 845 843; 1099 844 846 847 845; 1101 846 848 849 847

767. 1103 848 850 851 849; 1105 850 852 853 851; 1107 852 854 855 853
768. 1109 854 856 857 855; 1111 856 858 859 857; 1113 858 860 861 859
769. 1115 860 862 863 861; 1117 862 864 865 863; 1119 864 866 867 865
770. 1121 866 12 868 867; 1122 811 825 869 814; 1123 825 827 870 869
771. 1124 827 829 871 870; 1125 829 831 872 871; 1126 831 833 873 872
772. 1127 833 835 874 873; 1128 835 837 875 874; 1129 837 839 876 875
773. 1130 839 841 877 876; 1131 841 843 878 877; 1132 843 845 879 878
774. 1133 845 847 880 879; 1134 847 849 881 880; 1135 849 851 882 881
775. 1136 851 853 883 882; 1137 853 855 884 883; 1138 855 857 885 884
776. 1139 857 859 886 885; 1140 859 861 887 886; 1141 861 863 888 887
777. 1142 863 865 889 888; 1143 865 867 890 889; 1145 867 868 891 890
778. 1146 814 869 892 817; 1147 869 870 893 892; 1148 870 871 894 893
779. 1149 871 872 895 894; 1150 872 873 896 895; 1151 873 874 897 896
780. 1152 874 875 898 897; 1153 875 876 899 898; 1154 876 877 900 899
781. 1155 877 878 901 900; 1156 878 879 902 901; 1157 879 880 903 902
782. 1158 880 881 904 903; 1159 881 882 905 904; 1160 882 883 906 905
783. 1161 883 884 907 906; 1162 884 885 908 907; 1163 885 886 909 908
784. 1164 886 887 910 909; 1165 887 888 911 910; 1166 888 889 912 911
785. 1167 889 890 913 912; 1169 890 891 914 913; 1170 817 892 915 820
786. 1171 892 893 916 915; 1172 893 894 917 916; 1173 894 895 918 917
787. 1174 895 896 919 918; 1175 896 897 920 919; 1176 897 898 921 920
788. 1177 898 899 922 921; 1178 899 900 923 922; 1179 900 901 924 923
789. 1180 901 902 925 924; 1181 902 903 926 925; 1182 903 904 927 926
790. 1183 904 905 928 927; 1184 905 906 929 928; 1185 906 907 930 929
791. 1186 907 908 931 930; 1187 908 909 932 931; 1188 909 910 933 932
792. 1189 910 911 934 933; 1190 911 912 935 934; 1191 912 913 936 935
793. 1193 913 914 937 936; 1194 820 915 938 823; 1195 915 916 939 938
794. 1196 916 917 940 939; 1197 917 918 941 940; 1198 918 919 942 941
795. 1199 919 920 943 942; 1200 920 921 944 943; 1201 921 922 945 944
796. 1202 922 923 946 945; 1203 923 924 947 946; 1204 924 925 948 947
797. 1205 925 926 949 948; 1206 926 927 950 949; 1207 927 928 951 950
798. 1208 928 929 952 951; 1209 929 930 953 952; 1210 930 931 954 953
799. 1211 931 932 955 954; 1212 932 933 956 955; 1213 933 934 957 956
800. 1214 934 935 958 957; 1215 935 936 959 958; 1217 936 937 960 959
801. 1218 823 938 599 27; 1219 938 939 601 599; 1220 939 940 603 601
802. 1221 940 941 605 603; 1222 941 942 607 605; 1223 942 943 609 607
803. 1224 943 944 611 609; 1225 944 945 613 611; 1226 945 946 615 613
804. 1227 946 947 617 615; 1228 947 948 619 617; 1229 948 949 621 619
805. 1230 949 950 623 621; 1231 950 951 625 623; 1232 951 952 627 625
806. 1233 952 953 629 627; 1234 953 954 631 629; 1235 954 955 633 631
807. 1236 955 956 635 633; 1237 956 957 637 635; 1238 957 958 639 637
808. 1239 958 959 641 639; 1240 959 960 11 641; 1242 12 961 962 868
809. 1244 961 963 964 962; 1246 963 24 965 964; 1247 868 962 966 891
810. 1248 962 964 967 966; 1250 964 965 968 967; 1251 891 966 969 914
811. 1252 966 967 970 969; 1254 967 968 971 970; 1255 914 969 972 937
812. 1256 969 970 973 972; 1258 970 971 974 973; 1259 937 972 975 960
813. 1260 972 973 976 975; 1262 973 974 977 976; 1264 960 975 978 11
814. 1266 975 976 979 978; 1267 976 977 23 979; 1270 34 980 981 982
815. 1272 980 983 984 981; 1274 983 7 985 984; 1276 982 981 986 987
816. 1277 981 984 988 986; 1279 984 985 989 988; 1281 987 986 990 991
817. 1282 986 988 992 990; 1284 988 989 993 992; 1286 991 990 994 995
818. 1287 990 992 996 994; 1289 992 993 997 996; 1291 995 994 998 999
819. 1292 994 996 1000 998; 1294 996 997 1001 1000; 1295 999 998 666 33
820. 1296 998 1000 669 666; 1297 1000 1001 5 669; 1299 7 1002 1003 985
821. 1301 1002 1004 1005 1003; 1303 1004 1006 1007 1005; 1305 1006 1008 1009 1007
822. 1307 1008 1010 1011 1009; 1309 1010 1012 1013 1011; 1311 1012 1014 1015 1013

823. 1313 1014 1016 1017 1015; 1315 1016 1018 1019 1017; 1317 1018 1020 1021 1019
824. 1319 1020 1022 1023 1021; 1321 1022 1024 1025 1023; 1323 1024 1026 1027 1025
825. 1325 1026 1028 1029 1027; 1327 1028 1030 1031 1029; 1329 1030 1032 1033 1031
826. 1331 1032 1034 1035 1033; 1333 1034 1036 1037 1035; 1335 1036 1038 1039 1037
827. 1337 1038 8 1040 1039; 1338 985 1003 1041 989; 1339 1003 1005 1042 1041
828. 1340 1005 1007 1043 1042; 1341 1007 1009 1044 1043; 1342 1009 1011 1045 1044
829. 1343 1011 1013 1046 1045; 1344 1013 1015 1047 1046; 1345 1015 1017 1048 1047
830. 1346 1017 1019 1049 1048; 1347 1019 1021 1050 1049; 1348 1021 1023 1051 1050
831. 1349 1023 1025 1052 1051; 1350 1025 1027 1053 1052; 1351 1027 1029 1054 1053
832. 1352 1029 1031 1055 1054; 1353 1031 1033 1056 1055; 1354 1033 1035 1057 1056
833. 1355 1035 1037 1058 1057; 1356 1037 1039 1059 1058; 1358 1039 1040 1060 1059
834. 1359 989 1041 1061 993; 1360 1041 1042 1062 1061; 1361 1042 1043 1063 1062
835. 1362 1043 1044 1064 1063; 1363 1044 1045 1065 1064; 1364 1045 1046 1066 1065
836. 1365 1046 1047 1067 1066; 1366 1047 1048 1068 1067; 1367 1048 1049 1069 1068
837. 1368 1049 1050 1070 1069; 1369 1050 1051 1071 1070; 1370 1051 1052 1072 1071
838. 1371 1052 1053 1073 1072; 1372 1053 1054 1074 1073; 1373 1054 1055 1075 1074
839. 1374 1055 1056 1076 1075; 1375 1056 1057 1077 1076; 1376 1057 1058 1078 1077
840. 1377 1058 1059 1079 1078; 1379 1059 1060 1080 1079; 1380 993 1061 1081 997
841. 1381 1061 1062 1082 1081; 1382 1062 1063 1083 1082; 1383 1063 1064 1084 1083
842. 1384 1064 1065 1085 1084; 1385 1065 1066 1086 1085; 1386 1066 1067 1087 1086
843. 1387 1067 1068 1088 1087; 1388 1068 1069 1089 1088; 1389 1069 1070 1090 1089
844. 1390 1070 1071 1091 1090; 1391 1071 1072 1092 1091; 1392 1072 1073 1093 1092
845. 1393 1073 1074 1094 1093; 1394 1074 1075 1095 1094; 1395 1075 1076 1096 1095
846. 1396 1076 1077 1097 1096; 1397 1077 1078 1098 1097; 1398 1078 1079 1099 1098
847. 1400 1079 1080 1100 1099; 1401 997 1081 1101 1001; 1402 1081 1082 1102 1101
848. 1403 1082 1083 1103 1102; 1404 1083 1084 1104 1103; 1405 1084 1085 1105 1104
849. 1406 1085 1086 1106 1105; 1407 1086 1087 1107 1106; 1408 1087 1088 1108 1107
850. 1409 1088 1089 1109 1108; 1410 1089 1090 1110 1109; 1411 1090 1091 1111 1110
851. 1412 1091 1092 1112 1111; 1413 1092 1093 1113 1112; 1414 1093 1094 1114 1113
852. 1415 1094 1095 1115 1114; 1416 1095 1096 1116 1115; 1417 1096 1097 1117 1116
853. 1418 1097 1098 1118 1117; 1419 1098 1099 1119 1118; 1421 1099 1100 1120 1119
854. 1422 1001 1101 688 5; 1423 1101 1102 690 688; 1424 1102 1103 692 690
855. 1425 1103 1104 694 692; 1426 1104 1105 696 694; 1427 1105 1106 698 696
856. 1428 1106 1107 700 698; 1429 1107 1108 702 700; 1430 1108 1109 704 702
857. 1431 1109 1110 706 704; 1432 1110 1111 708 706; 1433 1111 1112 710 708
858. 1434 1112 1113 712 710; 1435 1113 1114 714 712; 1436 1114 1115 716 714
859. 1437 1115 1116 718 716; 1438 1116 1117 720 718; 1439 1117 1118 722 720
860. 1440 1118 1119 724 722; 1441 1119 1120 6 724; 1443 8 1121 1122 1040
861. 1445 1121 1123 1124 1122; 1447 1123 29 1125 1124; 1448 1040 1122 1126 1060
862. 1449 1122 1124 1127 1126; 1451 1124 1125 1128 1127; 1452 1060 1126 1129 1080
863. 1453 1126 1127 1130 1129; 1455 1127 1128 1131 1130; 1456 1080 1129 1132 1100
864. 1457 1129 1130 1133 1132; 1459 1130 1131 1134 1133; 1460 1100 1132 1135 1120
865. 1461 1132 1133 1136 1135; 1463 1133 1134 1137 1136; 1464 1120 1135 807 6
866. 1465 1135 1136 809 807; 1466 1136 1137 28 809; 1468 29 1138 1139 1125
867. 1470 1138 1140 1141 1139; 1472 1140 1142 1143 1141; 1474 1142 1144 1145 1143
868. 1476 1144 1146 1147 1145; 1478 1146 1148 1149 1147; 1480 1148 1150 1151 1149
869. 1482 1150 1152 1153 1151; 1484 1152 1154 1155 1153; 1486 1154 1156 1157 1155
870. 1488 1156 1158 1159 1157; 1490 1158 1160 1161 1159; 1492 1160 1162 1163 1161
871. 1494 1162 1164 1165 1163; 1496 1164 1166 1167 1165; 1498 1166 1168 1169 1167
872. 1500 1168 1170 1171 1169; 1502 1170 1172 1173 1171; 1504 1172 1174 1175 1173
873. 1506 1174 1176 1177 1175; 1508 1176 1178 1179 1177; 1510 1178 1180 1181 1179
874. 1512 1180 13 1182 1181; 1513 1125 1139 1183 1128; 1514 1139 1141 1184 1183
875. 1515 1141 1143 1185 1184; 1516 1143 1145 1186 1185; 1517 1145 1147 1187 1186
876. 1518 1147 1149 1188 1187; 1519 1149 1151 1189 1188; 1520 1151 1153 1190 1189
877. 1521 1153 1155 1191 1190; 1522 1155 1157 1192 1191; 1523 1157 1159 1193 1192
878. 1524 1159 1161 1194 1193; 1525 1161 1163 1195 1194; 1526 1163 1165 1196 1195

879. 1527 1165 1167 1197 1196; 1528 1167 1169 1198 1197; 1529 1169 1171 1199 1198
880. 1530 1171 1173 1200 1199; 1531 1173 1175 1201 1200; 1532 1175 1177 1202 1201
881. 1533 1177 1179 1203 1202; 1534 1179 1181 1204 1203; 1536 1181 1182 1205 1204
882. 1537 1128 1183 1206 1131; 1538 1183 1184 1207 1206; 1539 1184 1185 1208 1207
883. 1540 1185 1186 1209 1208; 1541 1186 1187 1210 1209; 1542 1187 1188 1211 1210
884. 1543 1188 1189 1212 1211; 1544 1189 1190 1213 1212; 1545 1190 1191 1214 1213
885. 1546 1191 1192 1215 1214; 1547 1192 1193 1216 1215; 1548 1193 1194 1217 1216
886. 1549 1194 1195 1218 1217; 1550 1195 1196 1219 1218; 1551 1196 1197 1220 1219
887. 1552 1197 1198 1221 1220; 1553 1198 1199 1222 1221; 1554 1199 1200 1223 1222
888. 1555 1200 1201 1224 1223; 1556 1201 1202 1225 1224; 1557 1202 1203 1226 1225
889. 1558 1203 1204 1227 1226; 1560 1204 1205 1228 1227; 1561 1131 1206 1229 1134
890. 1562 1206 1207 1230 1229; 1563 1207 1208 1231 1230; 1564 1208 1209 1232 1231
891. 1565 1209 1210 1233 1232; 1566 1210 1211 1234 1233; 1567 1211 1212 1235 1234
892. 1568 1212 1213 1236 1235; 1569 1213 1214 1237 1236; 1570 1214 1215 1238 1237
893. 1571 1215 1216 1239 1238; 1572 1216 1217 1240 1239; 1573 1217 1218 1241 1240
894. 1574 1218 1219 1242 1241; 1575 1219 1220 1243 1242; 1576 1220 1221 1244 1243
895. 1577 1221 1222 1245 1244; 1578 1222 1223 1246 1245; 1579 1223 1224 1247 1246
896. 1580 1224 1225 1248 1247; 1581 1225 1226 1249 1248; 1582 1226 1227 1250 1249
897. 1584 1227 1228 1251 1250; 1585 1134 1229 1252 1137; 1586 1229 1230 1253 1252
898. 1587 1230 1231 1254 1253; 1588 1231 1232 1255 1254; 1589 1232 1233 1256 1255
899. 1590 1233 1234 1257 1256; 1591 1234 1235 1258 1257; 1592 1235 1236 1259 1258
900. 1593 1236 1237 1260 1259; 1594 1237 1238 1261 1260; 1595 1238 1239 1262 1261
901. 1596 1239 1240 1263 1262; 1597 1240 1241 1264 1263; 1598 1241 1242 1265 1264
902. 1599 1242 1243 1266 1265; 1600 1243 1244 1267 1266; 1601 1244 1245 1268 1267
903. 1602 1245 1246 1269 1268; 1603 1246 1247 1270 1269; 1604 1247 1248 1271 1270
904. 1605 1248 1249 1272 1271; 1606 1249 1250 1273 1272; 1608 1250 1251 1274 1273
905. 1609 1137 1252 824 28; 1610 1252 1253 826 824; 1611 1253 1254 828 826
906. 1612 1254 1255 830 828; 1613 1255 1256 832 830; 1614 1256 1257 834 832
907. 1615 1257 1258 836 834; 1616 1258 1259 838 836; 1617 1259 1260 840 838
908. 1618 1260 1261 842 840; 1619 1261 1262 844 842; 1620 1262 1263 846 844
909. 1621 1263 1264 848 846; 1622 1264 1265 850 848; 1623 1265 1266 852 850
910. 1624 1266 1267 854 852; 1625 1267 1268 856 854; 1626 1268 1269 858 856
911. 1627 1269 1270 860 858; 1628 1270 1271 862 860; 1629 1271 1272 864 862
912. 1630 1272 1273 866 864; 1631 1273 1274 12 866; 1633 13 1275 1276 1182
913. 1635 1275 1277 1278 1276; 1637 1277 25 1279 1278; 1638 1182 1276 1280 1205
914. 1639 1276 1278 1281 1280; 1641 1278 1279 1282 1281; 1642 1205 1280 1283 1228
915. 1643 1280 1281 1284 1283; 1645 1281 1282 1285 1284; 1646 1228 1283 1286 1251
916. 1647 1283 1284 1287 1286; 1649 1284 1285 1288 1287; 1650 1251 1286 1289 1274
917. 1651 1286 1287 1290 1289; 1653 1287 1288 1291 1290; 1654 1274 1289 961 12
918. 1655 1289 1290 963 961; 1656 1290 1291 24 963; 1659 55 1292 1293 1294
919. 1661 1292 1295 1296 1293; 1663 1295 53 1297 1296; 1664 1294 1293 980 34
920. 1665 1293 1296 983 980; 1666 1296 1297 7 983; 1668 53 1298 1299 1297
921. 1670 1298 1300 1301 1299; 1672 1300 1302 1303 1301; 1674 1302 1304 1305 1303
922. 1676 1304 1306 1307 1305; 1678 1306 1308 1309 1307; 1680 1308 1310 1311 1309
923. 1682 1310 1312 1313 1311; 1684 1312 1314 1315 1313; 1686 1314 1316 1317 1315
924. 1688 1316 1318 1319 1317; 1690 1318 1320 1321 1319; 1692 1320 1322 1323 1321
925. 1694 1322 1324 1325 1323; 1696 1324 1326 1327 1325; 1698 1326 1328 1329 1327
926. 1700 1328 1330 1331 1329; 1702 1330 1332 1333 1331; 1704 1332 1334 1335 1333
927. 1706 1334 54 1336 1335; 1707 1297 1299 1002 7; 1708 1299 1301 1004 1002
928. 1709 1301 1303 1006 1004; 1710 1303 1305 1008 1006; 1711 1305 1307 1010 1008
929. 1712 1307 1309 1012 1010; 1713 1309 1311 1014 1012; 1714 1311 1313 1016 1014
930. 1715 1313 1315 1018 1016; 1716 1315 1317 1020 1018; 1717 1317 1319 1022 1020
931. 1718 1319 1321 1024 1022; 1719 1321 1323 1026 1024; 1720 1323 1325 1028 1026
932. 1721 1325 1327 1030 1028; 1722 1327 1329 1032 1030; 1723 1329 1331 1034 1032
933. 1724 1331 1333 1036 1034; 1725 1333 1335 1038 1036; 1726 1335 1336 8 1038
934. 1728 54 1337 1338 1336; 1730 1337 1339 1340 1338; 1732 1339 49 1341 1340

935. 1733 1336 1338 1121 8; 1734 1338 1340 1123 1121; 1735 1340 1341 29 1123
936. 1737 49 1342 1343 1341; 1739 1342 1344 1345 1343; 1741 1344 1346 1347 1345
937. 1743 1346 1348 1349 1347; 1745 1348 1350 1351 1349; 1747 1350 1352 1353 1351
938. 1749 1352 1354 1355 1353; 1751 1354 1356 1357 1355; 1753 1356 1358 1359 1357
939. 1755 1358 1360 1361 1359; 1757 1360 1362 1363 1361; 1759 1362 1364 1365 1363
940. 1761 1364 1366 1367 1365; 1763 1366 1368 1369 1367; 1765 1368 1370 1371 1369
941. 1767 1370 1372 1373 1371; 1769 1372 1374 1375 1373; 1771 1374 1376 1377 1375
942. 1773 1376 1378 1379 1377; 1775 1378 1380 1381 1379; 1777 1380 1382 1383 1381
943. 1779 1382 1384 1385 1383; 1781 1384 47 1386 1385; 1782 1341 1343 1138 29
944. 1783 1343 1345 1140 1138; 1784 1345 1347 1142 1140; 1785 1347 1349 1144 1142
945. 1786 1349 1351 1146 1144; 1787 1351 1353 1148 1146; 1788 1353 1355 1150 1148
946. 1789 1355 1357 1152 1150; 1790 1357 1359 1154 1152; 1791 1359 1361 1156 1154
947. 1792 1361 1363 1158 1156; 1793 1363 1365 1160 1158; 1794 1365 1367 1162 1160
948. 1795 1367 1369 1164 1162; 1796 1369 1371 1166 1164; 1797 1371 1373 1168 1166
949. 1798 1373 1375 1170 1168; 1799 1375 1377 1172 1170; 1800 1377 1379 1174 1172
950. 1801 1379 1381 1176 1174; 1802 1381 1383 1178 1176; 1803 1383 1385 1180 1178
951. 1804 1385 1386 13 1180; 1807 35 1387 1388 1389; 1809 1387 1390 1391 1388
952. 1811 1390 9 1392 1391; 1813 1389 1388 1393 1394; 1814 1388 1391 1395 1393
953. 1816 1391 1392 1396 1395; 1818 1394 1393 1397 1398; 1819 1393 1395 1399 1397
954. 1821 1395 1396 1400 1399; 1823 1398 1397 1401 1402; 1824 1397 1399 1403 1401
955. 1826 1399 1400 1404 1403; 1828 1402 1401 1405 1406; 1829 1401 1403 1407 1405
956. 1831 1403 1404 1408 1407; 1833 1406 1405 1409 1410; 1834 1405 1407 1411 1409
957. 1836 1407 1408 1412 1411; 1838 1410 1409 1413 1414; 1839 1409 1411 1415 1413
958. 1841 1411 1412 1416 1415; 1843 1414 1413 1417 1418; 1844 1413 1415 1419 1417
959. 1846 1415 1416 1420 1419; 1848 1418 1417 1421 1422; 1849 1417 1419 1423 1421
960. 1851 1419 1420 1424 1423; 1853 1422 1421 1425 1426; 1854 1421 1423 1427 1425
961. 1856 1423 1424 1428 1427; 1858 1426 1425 1429 1430; 1859 1425 1427 1431 1429
962. 1861 1427 1428 1432 1431; 1863 1430 1429 1433 1434; 1864 1429 1431 1435 1433
963. 1866 1431 1432 1436 1435; 1868 1434 1433 1437 1438; 1869 1433 1435 1439 1437
964. 1871 1435 1436 1440 1439; 1873 1438 1437 1441 1442; 1874 1437 1439 1443 1441
965. 1876 1439 1440 1444 1443; 1877 1442 1441 1292 55; 1878 1441 1443 1295 1292
966. 1879 1443 1444 53 1295; 1881 9 1445 1446 1392; 1883 1445 1447 1448 1446
967. 1885 1447 1449 1450 1448; 1887 1449 1451 1452 1450; 1889 1451 1453 1454 1452
968. 1891 1453 1455 1456 1454; 1893 1455 1457 1458 1456; 1895 1457 1459 1460 1458
969. 1897 1459 1461 1462 1460; 1899 1461 1463 1464 1462; 1901 1463 1465 1466 1464
970. 1903 1465 1467 1468 1466; 1905 1467 1469 1470 1468; 1907 1469 1471 1472 1470
971. 1909 1471 1473 1474 1472; 1911 1473 1475 1476 1474; 1913 1475 1477 1478 1476
972. 1915 1477 1479 1480 1478; 1917 1479 1481 1482 1480; 1919 1481 10 1483 1482
973. 1920 1392 1446 1484 1396; 1921 1446 1448 1485 1484; 1922 1448 1450 1486 1485
974. 1923 1450 1452 1487 1486; 1924 1452 1454 1488 1487; 1925 1454 1456 1489 1488
975. 1926 1456 1458 1490 1489; 1927 1458 1460 1491 1490; 1928 1460 1462 1492 1491
976. 1929 1462 1464 1493 1492; 1930 1464 1466 1494 1493; 1931 1466 1468 1495 1494
977. 1932 1468 1470 1496 1495; 1933 1470 1472 1497 1496; 1934 1472 1474 1498 1497
978. 1935 1474 1476 1499 1498; 1936 1476 1478 1500 1499; 1937 1478 1480 1501 1500
979. 1938 1480 1482 1502 1501; 1940 1482 1483 1503 1502; 1941 1396 1484 1504 1400
980. 1942 1484 1485 1505 1504; 1943 1485 1486 1506 1505; 1944 1486 1487 1507 1506
981. 1945 1487 1488 1508 1507; 1946 1488 1489 1509 1508; 1947 1489 1490 1510 1509
982. 1948 1490 1491 1511 1510; 1949 1491 1492 1512 1511; 1950 1492 1493 1513 1512
983. 1951 1493 1494 1514 1513; 1952 1494 1495 1515 1514; 1953 1495 1496 1516 1515
984. 1954 1496 1497 1517 1516; 1955 1497 1498 1518 1517; 1956 1498 1499 1519 1518
985. 1957 1499 1500 1520 1519; 1958 1500 1501 1521 1520; 1959 1501 1502 1522 1521
986. 1961 1502 1503 1523 1522; 1962 1400 1504 1524 1404; 1963 1504 1505 1525 1524
987. 1964 1505 1506 1526 1525; 1965 1506 1507 1527 1526; 1966 1507 1508 1528 1527
988. 1967 1508 1509 1529 1528; 1968 1509 1510 1530 1529; 1969 1510 1511 1531 1530
989. 1970 1511 1512 1532 1531; 1971 1512 1513 1533 1532; 1972 1513 1514 1534 1533
990. 1973 1514 1515 1535 1534; 1974 1515 1516 1536 1535; 1975 1516 1517 1537 1536

991. 1976 1517 1518 1538 1537; 1977 1518 1519 1539 1538; 1978 1519 1520 1540 1539
992. 1979 1520 1521 1541 1540; 1980 1521 1522 1542 1541; 1982 1522 1523 1543 1542
993. 1983 1404 1524 1544 1408; 1984 1524 1525 1545 1544; 1985 1525 1526 1546 1545
994. 1986 1526 1527 1547 1546; 1987 1527 1528 1548 1547; 1988 1528 1529 1549 1548
995. 1989 1529 1530 1550 1549; 1990 1530 1531 1551 1550; 1991 1531 1532 1552 1551
996. 1992 1532 1533 1553 1552; 1993 1533 1534 1554 1553; 1994 1534 1535 1555 1554
997. 1995 1535 1536 1556 1555; 1996 1536 1537 1557 1556; 1997 1537 1538 1558 1557
998. 1998 1538 1539 1559 1558; 1999 1539 1540 1560 1559; 2000 1540 1541 1561 1560
999. 2001 1541 1542 1562 1561; 2003 1542 1543 1563 1562; 2004 1408 1544 1564 1412
1000. 2005 1544 1545 1565 1564; 2006 1545 1546 1566 1565; 2007 1546 1547 1567 1566
1001. 2008 1547 1548 1568 1567; 2009 1548 1549 1569 1568; 2010 1549 1550 1570 1569
1002. 2011 1550 1551 1571 1570; 2012 1551 1552 1572 1571; 2013 1552 1553 1573 1572
1003. 2014 1553 1554 1574 1573; 2015 1554 1555 1575 1574; 2016 1555 1556 1576 1575
1004. 2017 1556 1557 1577 1576; 2018 1557 1558 1578 1577; 2019 1558 1559 1579 1578
1005. 2020 1559 1560 1580 1579; 2021 1560 1561 1581 1580; 2022 1561 1562 1582 1581
1006. 2024 1562 1563 1583 1582; 2025 1412 1564 1584 1416; 2026 1564 1565 1585 1584
1007. 2027 1565 1566 1586 1585; 2028 1566 1567 1587 1586; 2029 1567 1568 1588 1587
1008. 2030 1568 1569 1589 1588; 2031 1569 1570 1590 1589; 2032 1570 1571 1591 1590
1009. 2033 1571 1572 1592 1591; 2034 1572 1573 1593 1592; 2035 1573 1574 1594 1593
1010. 2036 1574 1575 1595 1594; 2037 1575 1576 1596 1595; 2038 1576 1577 1597 1596
1011. 2039 1577 1578 1598 1597; 2040 1578 1579 1599 1598; 2041 1579 1580 1600 1599
1012. 2042 1580 1581 1601 1600; 2043 1581 1582 1602 1601; 2045 1582 1583 1603 1602
1013. 2046 1416 1584 1604 1420; 2047 1584 1585 1605 1604; 2048 1585 1586 1606 1605
1014. 2049 1586 1587 1607 1606; 2050 1587 1588 1608 1607; 2051 1588 1589 1609 1608
1015. 2052 1589 1590 1610 1609; 2053 1590 1591 1611 1610; 2054 1591 1592 1612 1611
1016. 2055 1592 1593 1613 1612; 2056 1593 1594 1614 1613; 2057 1594 1595 1615 1614
1017. 2058 1595 1596 1616 1615; 2059 1596 1597 1617 1616; 2060 1597 1598 1618 1617
1018. 2061 1598 1599 1619 1618; 2062 1599 1600 1620 1619; 2063 1600 1601 1621 1620
1019. 2064 1601 1602 1622 1621; 2066 1602 1603 1623 1622; 2067 1420 1604 1624 1424
1020. 2068 1604 1605 1625 1624; 2069 1605 1606 1626 1625; 2070 1606 1607 1627 1626
1021. 2071 1607 1608 1628 1627; 2072 1608 1609 1629 1628; 2073 1609 1610 1630 1629
1022. 2074 1610 1611 1631 1630; 2075 1611 1612 1632 1631; 2076 1612 1613 1633 1632
1023. 2077 1613 1614 1634 1633; 2078 1614 1615 1635 1634; 2079 1615 1616 1636 1635
1024. 2080 1616 1617 1637 1636; 2081 1617 1618 1638 1637; 2082 1618 1619 1639 1638
1025. 2083 1619 1620 1640 1639; 2084 1620 1621 1641 1640; 2085 1621 1622 1642 1641
1026. 2087 1622 1623 1643 1642; 2088 1424 1624 1644 1428; 2089 1624 1625 1645 1644
1027. 2090 1625 1626 1646 1645; 2091 1626 1627 1647 1646; 2092 1627 1628 1648 1647
1028. 2093 1628 1629 1649 1648; 2094 1629 1630 1650 1649; 2095 1630 1631 1651 1650
1029. 2096 1631 1632 1652 1651; 2097 1632 1633 1653 1652; 2098 1633 1634 1654 1653
1030. 2099 1634 1635 1655 1654; 2100 1635 1636 1656 1655; 2101 1636 1637 1657 1656
1031. 2102 1637 1638 1658 1657; 2103 1638 1639 1659 1658; 2104 1639 1640 1660 1659
1032. 2105 1640 1641 1661 1660; 2106 1641 1642 1662 1661; 2108 1642 1643 1663 1662
1033. 2109 1428 1644 1664 1432; 2110 1644 1645 1665 1664; 2111 1645 1646 1666 1665
1034. 2112 1646 1647 1667 1666; 2113 1647 1648 1668 1667; 2114 1648 1649 1669 1668
1035. 2115 1649 1650 1670 1669; 2116 1650 1651 1671 1670; 2117 1651 1652 1672 1671
1036. 2118 1652 1653 1673 1672; 2119 1653 1654 1674 1673; 2120 1654 1655 1675 1674
1037. 2121 1655 1656 1676 1675; 2122 1656 1657 1677 1676; 2123 1657 1658 1678 1677
1038. 2124 1658 1659 1679 1678; 2125 1659 1660 1680 1679; 2126 1660 1661 1681 1680
1039. 2127 1661 1662 1682 1681; 2129 1662 1663 1683 1682; 2130 1432 1664 1684 1436
1040. 2131 1664 1665 1685 1684; 2132 1665 1666 1686 1685; 2133 1666 1667 1687 1686
1041. 2134 1667 1668 1688 1687; 2135 1668 1669 1689 1688; 2136 1669 1670 1690 1689
1042. 2137 1670 1671 1691 1690; 2138 1671 1672 1692 1691; 2139 1672 1673 1693 1692
1043. 2140 1673 1674 1694 1693; 2141 1674 1675 1695 1694; 2142 1675 1676 1696 1695
1044. 2143 1676 1677 1697 1696; 2144 1677 1678 1698 1697; 2145 1678 1679 1699 1698
1045. 2146 1679 1680 1700 1699; 2147 1680 1681 1701 1700; 2148 1681 1682 1702 1701
1046. 2150 1682 1683 1703 1702; 2151 1436 1684 1704 1440; 2152 1684 1685 1705 1704

1047. 2153 1685 1686 1706 1705; 2154 1686 1687 1707 1706; 2155 1687 1688 1708 1707
1048. 2156 1688 1689 1709 1708; 2157 1689 1690 1710 1709; 2158 1690 1691 1711 1710
1049. 2159 1691 1692 1712 1711; 2160 1692 1693 1713 1712; 2161 1693 1694 1714 1713
1050. 2162 1694 1695 1715 1714; 2163 1695 1696 1716 1715; 2164 1696 1697 1717 1716
1051. 2165 1697 1698 1718 1717; 2166 1698 1699 1719 1718; 2167 1699 1700 1720 1719
1052. 2168 1700 1701 1721 1720; 2169 1701 1702 1722 1721; 2171 1702 1703 1723 1722
1053. 2172 1440 1704 1724 1444; 2173 1704 1705 1725 1724; 2174 1705 1706 1726 1725
1054. 2175 1706 1707 1727 1726; 2176 1707 1708 1728 1727; 2177 1708 1709 1729 1728
1055. 2178 1709 1710 1730 1729; 2179 1710 1711 1731 1730; 2180 1711 1712 1732 1731
1056. 2181 1712 1713 1733 1732; 2182 1713 1714 1734 1733; 2183 1714 1715 1735 1734
1057. 2184 1715 1716 1736 1735; 2185 1716 1717 1737 1736; 2186 1717 1718 1738 1737
1058. 2187 1718 1719 1739 1738; 2188 1719 1720 1740 1739; 2189 1720 1721 1741 1740
1059. 2190 1721 1722 1742 1741; 2192 1722 1723 1743 1742; 2193 1444 1724 1298 53
1060. 2194 1724 1725 1300 1298; 2195 1725 1726 1302 1300; 2196 1726 1727 1304 1302
1061. 2197 1727 1728 1306 1304; 2198 1728 1729 1308 1306; 2199 1729 1730 1310 1308
1062. 2200 1730 1731 1312 1310; 2201 1731 1732 1314 1312; 2202 1732 1733 1316 1314
1063. 2203 1733 1734 1318 1316; 2204 1734 1735 1320 1318; 2205 1735 1736 1322 1320
1064. 2206 1736 1737 1324 1322; 2207 1737 1738 1326 1324; 2208 1738 1739 1328 1326
1065. 2209 1739 1740 1330 1328; 2210 1740 1741 1332 1330; 2211 1741 1742 1334 1332
1066. 2212 1742 1743 54 1334; 2214 10 1744 1745 1483; 2216 1744 1746 1747 1745
1067. 2218 1746 30 1748 1747; 2219 1483 1745 1749 1503; 2220 1745 1747 1750 1749
1068. 2222 1747 1748 1751 1750; 2223 1503 1749 1752 1523; 2224 1749 1750 1753 1752
1069. 2226 1750 1751 1754 1753; 2227 1523 1752 1755 1543; 2228 1752 1753 1756 1755
1070. 2230 1753 1754 1757 1756; 2231 1543 1755 1758 1563; 2232 1755 1756 1759 1758
1071. 2234 1756 1757 1760 1759; 2235 1563 1758 1761 1583; 2236 1758 1759 1762 1761
1072. 2238 1759 1760 1763 1762; 2239 1583 1761 1764 1603; 2240 1761 1762 1765 1764
1073. 2242 1762 1763 1766 1765; 2243 1603 1764 1767 1623; 2244 1764 1765 1768 1767
1074. 2246 1765 1766 1769 1768; 2247 1623 1767 1770 1643; 2248 1767 1768 1771 1770
1075. 2250 1768 1769 1772 1771; 2251 1643 1770 1773 1663; 2252 1770 1771 1774 1773
1076. 2254 1771 1772 1775 1774; 2255 1663 1773 1776 1683; 2256 1773 1774 1777 1776
1077. 2258 1774 1775 1778 1777; 2259 1683 1776 1779 1703; 2260 1776 1777 1780 1779
1078. 2262 1777 1778 1781 1780; 2263 1703 1779 1782 1723; 2264 1779 1780 1783 1782
1079. 2266 1780 1781 1784 1783; 2267 1723 1782 1785 1743; 2268 1782 1783 1786 1785
1080. 2270 1783 1784 1787 1786; 2271 1743 1785 1337 54; 2272 1785 1786 1339 1337
1081. 2273 1786 1787 49 1339; 2276 43 1788 1789 1790; 2278 1788 1791 1792 1789
1082. 2280 1791 40 1793 1792; 2282 1790 1789 1794 1795; 2283 1789 1792 1796 1794
1083. 2285 1792 1793 1797 1796; 2286 1795 1794 1387 35; 2287 1794 1796 1390 1387
1084. 2288 1796 1797 9 1390; 2290 40 1798 1799 1793; 2292 1798 1800 1801 1799
1085. 2294 1800 1802 1803 1801; 2296 1802 1804 1805 1803; 2298 1804 1806 1807 1805
1086. 2300 1806 1808 1809 1807; 2302 1808 1810 1811 1809; 2304 1810 1812 1813 1811
1087. 2306 1812 1814 1815 1813; 2308 1814 1816 1817 1815; 2310 1816 1818 1819 1817
1088. 2312 1818 1820 1821 1819; 2314 1820 1822 1823 1821; 2316 1822 1824 1825 1823
1089. 2318 1824 1826 1827 1825; 2320 1826 1828 1829 1827; 2322 1828 1830 1831 1829
1090. 2324 1830 1832 1833 1831; 2326 1832 1834 1835 1833; 2328 1834 41 1836 1835
1091. 2329 1793 1799 1837 1797; 2330 1799 1801 1838 1837; 2331 1801 1803 1839 1838
1092. 2332 1803 1805 1840 1839; 2333 1805 1807 1841 1840; 2334 1807 1809 1842 1841
1093. 2335 1809 1811 1843 1842; 2336 1811 1813 1844 1843; 2337 1813 1815 1845 1844
1094. 2338 1815 1817 1846 1845; 2339 1817 1819 1847 1846; 2340 1819 1821 1848 1847
1095. 2341 1821 1823 1849 1848; 2342 1823 1825 1850 1849; 2343 1825 1827 1851 1850
1096. 2344 1827 1829 1852 1851; 2345 1829 1831 1853 1852; 2346 1831 1833 1854 1853
1097. 2347 1833 1835 1855 1854; 2349 1835 1836 1856 1855; 2350 1797 1837 1445 9
1098. 2351 1837 1838 1447 1445; 2352 1838 1839 1449 1447; 2353 1839 1840 1451 1449
1099. 2354 1840 1841 1453 1451; 2355 1841 1842 1455 1453; 2356 1842 1843 1457 1455
1100. 2357 1843 1844 1459 1457; 2358 1844 1845 1461 1459; 2359 1845 1846 1463 1461
1101. 2360 1846 1847 1465 1463; 2361 1847 1848 1467 1465; 2362 1848 1849 1469 1467
1102. 2363 1849 1850 1471 1469; 2364 1850 1851 1473 1471; 2365 1851 1852 1475 1473

```
1103. 2366 1852 1853 1477 1475; 2367 1853 1854 1479 1477; 2368 1854 1855 1481 1479
1104. 2369 1855 1856 10 1481; 2388 126 1861 37; 2389 1863 1862 1861
1105. 2390 548 26 1862; 2391 106 1863 1861 126; 2392 2 547 1863 106
1106. 2393 547 548 1862 1863; 2408 643 1872 1870 44; 2409 11 978 1872 643
1107. 2410 978 979 1871 1872; 2411 1872 1871 1870; 2412 979 23 1871
1108. 2427 25 1277 1879; 2428 1879 1880 1878; 2429 1386 1880 1275 13
1109. 2430 1880 1879 1277 1275; 2431 47 1878 1880 1386; 2446 1856 1887 1744 10
1110. 2447 1887 1885 1746 1744; 2448 1836 1886 1887 1856; 2449 30 1746 1885
1111. 2450 1885 1887 1886; 2451 1886 1836 41
1112. START GROUP DEFINITION
1113. ELEMENT
1114. _RAFT 97 99 101 103 104 106 108 110 111 113 115 117 119 121 123 125 127 129 -
1115. 131 133 135 137 139 141 143 145 147 149 151 TO 170 172 174 176 178 180 182 -
1116. 184 186 188 190 192 194 196 198 200 202 204 206 208 210 211 214 216 218 220 -
1117. 221 223 225 226 228 230 231 233 235 236 238 240 241 243 245 246 248 250 251 -
1118. 253 255 256 258 260 261 263 265 266 268 270 271 273 275 276 278 280 281 283 -
1119. 284 TO 286 288 290 292 294 296 298 300 302 304 306 308 310 312 314 316 318 -
1120. 320 322 324 326 TO 345 347 TO 366 368 TO 387 389 TO 408 410 TO 429 -
1121. 431 TO 450 452 TO 471 473 TO 492 494 TO 513 515 TO 534 536 TO 555 -
1122. 557 TO 576 578 TO 597 599 TO 619 621 623 625 TO 627 629 TO 631 633 TO 635 -
1123. 637 TO 639 641 TO 643 645 TO 647 649 TO 651 653 TO 655 657 TO 659 -
1124. 661 TO 663 665 TO 667 669 TO 671 673 TO 675 677 679 681 682 709 711 -
1125. 713 TO 716 718 720 722 724 726 728 730 732 734 736 738 740 742 744 746 748 -
1126. 750 752 754 756 TO 776 778 780 782 TO 785 787 789 791 793 795 797 799 801 -
1127. 803 805 807 809 811 813 815 817 819 821 823 825 827 829 831 833 835 837 839 -
1128. 841 843 845 847 849 851 853 855 857 859 861 863 865 867 869 871 873 875 876 -
1129. 879
1130. _RAFT 881 883 885 886 888 890 891 893 895 896 898 900 901 903 TO 906 908 910 -
1131. 912 914 916 918 920 922 924 926 928 930 932 934 936 938 940 942 944 -
1132. 946 TO 965 967 TO 986 988 TO 1007 1009 TO 1028 1030 TO 1050 1052 1054 1056 -
1133. 1057 TO 1058 1060 TO 1062 1064 TO 1066 1068 TO 1070 1072 TO 1075 1077 1079 -
1134. 1081 1083 1085 1087 1089 1091 1093 1095 1097 1099 1101 1103 1105 1107 1109 -
1135. 1111 1113 1115 1117 1119 1121 TO 1143 1145 TO 1167 1169 TO 1191 1193 TO 1215 -
1136. 1217 TO 1240 1242 1244 1246 TO 1248 1250 TO 1252 1254 TO 1256 1258 TO 1260 -
1137. 1262 1264 1266 1267 1270 1272 1274 1276 1277 1279 1281 1282 1284 1286 1287 -
1138. 1289 1291 1292 1294 TO 1297 1299 1301 1303 1305 1307 1309 1311 1313 1315 -
1139. 1317 1319 1321 1323 1325 1327 1329 1331 1333 1335 1337 TO 1356 1358 TO 1377 -
1140. 1379 TO 1398 1400 TO 1419 1421 TO 1441 1443 1445 1447 TO 1449 1451 TO 1453 -
1141. 1455 TO 1457 1459 TO 1461 1463 TO 1466 1468 1470 1472 1474 1476 1478 1480 -
1142. 1482 1484 1486 1488 1490 1492 1494 1496 1498 1500 1502 1504 1506 1508 1510 -
1143. 1512 TO 1534 1536 TO 1558 1560 TO 1582 1584 TO 1606 1608 TO 1631 1633 1635 -
1144. 1637 TO 1639 1641 TO 1643 1645 TO 1647 1649 TO 1651 1653 TO 1656 1659 1661 -
1145. 1663 TO 1666 1668 1670 1672 1674 1676 1678 1680 1682 1684 1686 1688 1690 -
1146. 1692 1694 1696 1698 1700 1702 1704 1706 TO 1726 1728 1730 1732 TO 1735 1737 -
1147. 1739 1741 1743
1148. _RAFT 1745 1747 1749 1751 1753 1755 1757 1759 1761 1763 1765 1767 1769 1771 -
1149. 1773 1775 1777 1779 1781 TO 1804 1807 1809 1811 1813 1814 1816 1818 1819 -
1150. 1821 1823 1824 1826 1828 1829 1831 1833 1834 1836 1838 1839 1841 1843 1844 -
1151. 1846 1848 1849 1851 1853 1854 1856 1858 1859 1861 1863 1864 1866 1868 1869 -
1152. 1871 1873 1874 1876 TO 1879 1881 1883 1885 1887 1889 1891 1893 1895 1897 -
1153. 1899 1901 1903 1905 1907 1909 1911 1913 1915 1917 1919 TO 1938 1940 TO 1959 -
1154. 1961 TO 1980 1982 TO 2001 2003 TO 2022 2024 TO 2043 2045 TO 2064 -
1155. 2066 TO 2085 2087 TO 2106 2108 TO 2127 2129 TO 2148 2150 TO 2169 -
1156. 2171 TO 2190 2192 TO 2212 2214 2216 2218 TO 2220 2222 TO 2224 2226 TO 2228 -
1157. 2230 TO 2232 2234 TO 2236 2238 TO 2240 2242 TO 2244 2246 TO 2248 -
1158. 2250 TO 2252 2254 TO 2256 2258 TO 2260 2262 TO 2264 2266 TO 2268 -
```

```
1159. 2270 TO 2273 2276 2278 2280 2282 2283 2285 TO 2288 2290 2292 2294 2296 2298 -
1160. 2300 2302 2304 2306 2308 2310 2312 2314 2316 2318 2320 2322 2324 2326 2328 -
1161. 2329 TO 2347 2349 TO 2369 2388 TO 2393 2408 TO 2412 2427 TO 2431 2446 TO 2451
1162. _RAFT_MOM 97 99 103 104 106 108 110 111 115 117 119 121 123 125 127 129 131 -
1163. 133 135 137 139 141 143 145 147 149 152 TO 170 172 174 176 178 180 182 184 -
1164. 186 188 190 192 194 196 198 200 202 204 206 208 210 211 214 216 218 220 221 -
1165. 223 225 226 228 230 231 233 235 236 238 240 241 243 245 246 248 250 251 253 -
1166. 255 256 258 260 261 263 265 266 268 270 271 273 275 276 278 280 281 -
1167. 283 TO 285 288 290 292 294 296 298 300 302 304 306 308 310 312 314 316 318 -
1168. 320 322 324 326 TO 345 347 TO 366 368 TO 387 389 TO 408 410 TO 429 -
1169. 431 TO 450 452 TO 471 473 TO 492 494 TO 513 515 TO 534 536 TO 555 -
1170. 557 TO 576 578 TO 597 599 601 TO 618 621 623 625 TO 627 629 TO 631 -
1171. 633 TO 635 637 TO 639 641 TO 643 645 TO 647 649 TO 651 653 TO 655 -
1172. 657 TO 659 661 TO 663 665 TO 667 669 TO 671 673 TO 675 677 681 682 709 711 -
1173. 713 TO 716 718 720 722 724 726 728 730 732 734 736 738 740 742 744 746 748 -
1174. 750 752 754 757 TO 776 780 782 TO 785 787 789 791 793 795 797 799 801 803 -
1175. 805 807 809 811 813 815 817 819 821 823 825 827 829 833 835 837 839 841 843 -
1176. 845 847 849 851 853 855 857 859 861 863 865 867 869 871 873 875 876 879 881 -
1177. 885 886
1178. _RAFT_MOM 888 890 891 893 895 896 898 900 901 903 TO 906 910 912 914 916 918 -
1179. 920 922 924 926 928 930 932 934 936 938 940 942 944 946 TO 965 967 TO 986 -
1180. 988 TO 1007 1009 TO 1028 1030 TO 1049 1052 1054 1056 TO 1058 1060 TO 1062 -
1181. 1064 TO 1066 1068 TO 1070 1072 1074 1075 1077 1079 1081 1083 1085 1087 1089 -
1182. 1091 1093 1095 1097 1099 1101 1103 1105 1107 1109 1111 1113 1115 1117 1119 -
1183. 1121 TO 1143 1145 TO 1167 1169 TO 1191 1193 TO 1215 1217 TO 1239 1242 1244 -
1184. 1246 TO 1248 1250 TO 1252 1254 TO 1256 1258 TO 1260 1262 1266 1267 1270 1272 -
1185. 1274 1276 1277 1279 1281 1282 1284 1286 1287 1289 1291 1292 1294 TO 1296 -
1186. 1299 1301 1303 1305 1307 1309 1311 1313 1315 1317 1319 1321 1323 1325 1327 -
1187. 1329 1331 1333 1335 1338 TO 1356 1358 TO 1377 1379 TO 1398 1400 TO 1419 1421 -
1188. 1423 TO 1441 1445 1447 TO 1449 1451 TO 1453 1455 TO 1457 1459 TO 1461 1463 -
1189. 1464 TO 1466 1468 1470 1472 1474 1476 1478 1480 1482 1484 1486 1488 1490 1492 -
1190. 1494 1496 1498 1500 1502 1504 1506 1508 1510 1513 TO 1534 1536 TO 1558 1560 -
1191. 1561 TO 1582 1584 TO 1606 1608 TO 1631 1635 1637 TO 1639 1641 TO 1643 1645 -
1192. 1646 TO 1647 1649 TO 1651 1653 TO 1656 1659 1661 1663 TO 1666 1668 1670 1672 -
1193. 1674 1676 1678 1680 1682 1684 1686 1688 1690 1692 1694 1696 1698 1700 1702 -
1194. 1704 1706 TO 1725 1728 1730 1732 1734 1735 1737 1739 1741 1743 1745 1747 -
1195. 1749 1751 1753 1755 1757
1196. _RAFT_MOM 1759 1761 1763 1765 1767 1769 1771 1773 1775 1777 1779 1781 TO 1803 -
1197. 1807 1809 1813 1814 1816 1818 1819 1821 1823 1824 1826 1828 1829 1831 1833 -
1198. 1834 1836 1838 1839 1841 1843 1844 1846 1848 1849 1851 1853 1854 1856 1858 -
1199. 1859 1861 1863 1864 1866 1868 1869 1871 1873 1874 1876 TO 1879 1883 1885 -
1200. 1887 1889 1891 1893 1895 1897 1899 1901 1903 1905 1907 1909 1911 1913 1915 -
1201. 1917 1920 TO 1938 1940 TO 1959 1961 TO 1980 1982 TO 2001 2003 TO 2022 2024 -
1202. 2025 TO 2043 2045 TO 2064 2066 TO 2085 2087 TO 2106 2108 TO 2127 2129 TO 2148 -
1203. 2150 TO 2169 2171 TO 2190 2192 TO 2212 2216 2218 TO 2220 2222 TO 2224 2226 -
1204. 2227 TO 2228 2230 TO 2232 2234 TO 2236 2238 TO 2240 2242 TO 2244 2246 TO 2248 -
1205. 2250 TO 2252 2254 TO 2256 2258 TO 2260 2262 TO 2264 2266 TO 2268 -
1206. 2270 TO 2273 2276 2278 2280 2282 2283 2285 TO 2287 2290 2292 2294 2296 2298 -
1207. 2300 2302 2304 2306 2308 2310 2312 2314 2316 2318 2320 2322 2324 2326 2328 -
1208. 2329 TO 2347 2349 2351 TO 2368 2388 TO 2391 2393 2408 2410 TO 2412 2427 2428 -
1209. 2430 2431 2447 TO 2451
1210. _RAFT_SHEAR 97 103 108 110 111 117 119 121 123 125 127 129 131 133 135 137 -
1211. 139 141 143 145 147 154 TO 169 174 176 178 180 182 184 186 188 190 192 194 -
1212. 196 198 200 202 204 206 208 210 211 214 216 218 220 221 223 225 226 228 230 -
1213. 231 233 235 236 238 240 241 243 245 246 248 250 251 253 255 256 258 260 261 -
1214. 263 265 266 268 270 271 273 275 276 278 280 284 288 290 292 294 296 298 300 -
```

```
1215. 302 304 306 308 310 312 314 316 318 320 322 324 326 TO 345 347 TO 366 368 -
1216. 369 TO 387 389 TO 408 410 TO 429 431 TO 450 452 TO 471 473 TO 492 494 TO 513 -
1217. 515 TO 534 536 TO 555 557 TO 576 578 581 TO 596 602 TO 617 621 623 -
1218. 625 TO 627 629 TO 631 633 TO 635 637 TO 639 641 TO 643 645 TO 647 -
1219. 649 TO 651 653 TO 655 657 TO 659 661 TO 663 665 TO 667 669 TO 671 673 677 -
1220. 682 709 711 713 TO 716 718 720 722 724 726 728 730 732 734 736 738 740 742 -
1221. 744 746 748 750 752 757 TO 774 782 785 787 789 791 793 795 797 799 801 803 -
1222. 805 807 809 811 813 815 817 819 821 823 825 827 833 835 837 839 841 843 845 -
1223. 847 849 851 853 855 857 859 861 863 865 867 869 871 873 879 885 890 891 893 -
1224. 895 896 898 900 901 903 TO 906 912 914 916 918 920 922 924 926 928
1225. _RAFT_SHEAR 930 932 934 936 938 940 942 944 946 949 TO 965 967 TO 986 988 -
1226. 989 TO 1007 1009 TO 1027 1031 TO 1048 1052 1054 1056 TO 1058 1060 TO 1062 -
1227. 1064 TO 1066 1068 1072 1075 1077 1079 1081 1083 1085 1087 1089 1091 1093 -
1228. 1095 1097 1099 1101 1103 1105 1107 1109 1111 1113 1115 1117 1119 -
1229. 1121 TO 1143 1145 TO 1167 1169 TO 1191 1193 TO 1214 1218 TO 1238 1242 1244 -
1230. 1246 TO 1248 1250 TO 1252 1254 TO 1256 1258 1262 1267 1270 1272 1274 1276 -
1231. 1277 1279 1281 1282 1284 1286 1287 1289 1291 1295 1299 1301 1303 1305 1307 -
1232. 1309 1311 1313 1315 1317 1319 1321 1323 1325 1327 1329 1331 1333 -
1233. 1338 TO 1355 1359 TO 1377 1379 TO 1398 1400 1403 TO 1419 1421 1424 TO 1441 -
1234. 1447 1451 TO 1453 1455 TO 1457 1459 TO 1461 1463 TO 1466 1468 1470 1472 1474 -
1235. 1476 1478 1480 1482 1484 1486 1488 1490 1492 1494 1496 1498 1500 1502 1504 -
1236. 1506 1508 1513 TO 1533 1537 TO 1558 1560 TO 1582 1584 TO 1606 1608 TO 1631 -
1237. 1637 1641 TO 1643 1645 TO 1647 1649 TO 1651 1653 TO 1656 1659 1661 -
1238. 1663 TO 1666 1668 1670 1672 1674 1676 1678 1680 1682 1684 1686 1688 1690 -
1239. 1692 1694 1696 1698 1700 1702 1707 TO 1724 1732 1735 1737 1739 1741 1743 -
1240. 1745 1747 1749 1751 1753 1755 1757 1759 1761 1763 1765 1767 1769 1771 1773 -
1241. 1775 1777 1782 TO 1802 1807 1813 1818 1819 1821 1823 1824 1826 1828 1829 -
1242. 1831 1833 1834 1836 1838 1839 1841 1843 1844 1846 1848 1849 1851 1853 1854 -
1243. 1856 1858 1859
1244. _RAFT_SHEAR 1861 1863 1864 1866 1868 1869 1871 1873 1874 1876 TO 1879 1885 -
1245. 1887 1889 1891 1893 1895 1897 1899 1901 1903 1905 1907 1909 1911 1913 1915 -
1246. 1922 TO 1937 1941 TO 1959 1961 TO 1980 1982 TO 2001 2003 TO 2022 -
1247. 2024 TO 2043 2045 TO 2064 2066 TO 2085 2087 TO 2106 2108 TO 2127 -
1248. 2129 TO 2148 2150 TO 2169 2171 TO 2190 2192 TO 2212 2218 2222 TO 2224 2226 -
1249. 2227 TO 2228 2230 TO 2232 2234 TO 2236 2238 TO 2240 2242 TO 2244 2246 TO 2248 -
1250. 2250 TO 2252 2254 TO 2256 2258 TO 2260 2262 TO 2264 2266 TO 2268 -
1251. 2270 TO 2273 2276 2278 2280 2282 2286 2290 2292 2294 2296 2298 2300 2302 -
1252. 2304 2306 2308 2310 2312 2314 2316 2318 2320 2322 2324 2326 2328 -
1253. 2331 TO 2346 2352 TO 2367 2388 2390 2412 2427 2451
1254. MEMBER
1255. _C1 2452 TO 2460
1256. END GROUP DEFINITION
1257. ELEMENT PROPERTY
1258. _RAFT THICKNESS 0.4
1259. ******************************
1260. DEFINE MATERIAL START
1261. ISOTROPIC CONCRETE
1262. E 2.17185E+007
1263. POISSON 0.17
1264. DENSITY 25
1265. ALPHA 1E-005
1266. DAMP 0.05
1267. TYPE CONCRETE
1268. STRENGTH FCU 27579
1269. END DEFINE MATERIAL
1270. MEMBER PROPERTY AMERICAN
```

```
1271. 19 TO 27 2452 TO 2460 PRIS YD 0.5 ZD 0.5
1272. CONSTANTS
1273. MATERIAL CONCRETE ALL
1274. ****************************
1275. ************************************************************
1276. SUPPORTS
1277. _RAFT PLATE MAT DIRECT Y SUBGRADE 33750 PRINT
```

JOINT	INFLUENCE AREA
1	0.066090
2	0.066091
3	0.069289
4	0.069289
5	0.061827
6	0.061828
7	0.069289
8	0.069289
9	0.066095
10	0.066095
11	0.072516
12	0.064707
13	0.072516
23	0.031281
24	0.034233
25	0.031281
26	0.028519
27	0.070260
28	0.062694
29	0.070260
30	0.028520
31	0.034445
32	0.036111
33	0.032223
34	0.036111
35	0.034447
36	0.034111
37	0.028185
39	0.017778
40	0.034115
41	0.028189
43	0.017780
44	0.040162
46	0.055580
47	0.040162
49	0.055580
50	0.070354
51	0.070355
52	0.036667
53	0.070354
54	0.070355
55	0.036667
56	0.068889
57	0.071111
58	0.035556

59	0.068889
60	0.071111
61	0.068222
62	0.071111
63	0.035556
64	0.071111
65	0.068222
66	0.035556
67	0.035556
68	0.063292
69	0.065333
70	0.063292
71	0.065333
72	0.063292
73	0.065333
74	0.063292
75	0.065333
76	0.063292
77	0.065333
78	0.063292
79	0.065333
80	0.063292
81	0.065333
82	0.063292
83	0.065333
84	0.063292
85	0.065333
86	0.063292
87	0.065333
88	0.063292
89	0.065333
90	0.063292
91	0.065333
92	0.063292
93	0.065333
94	0.063292
95	0.065333
96	0.063292
97	0.065333
98	0.063292
99	0.065333
100	0.063292
101	0.065333
102	0.063292
103	0.065333
104	0.063292
105	0.065333
106	0.068223
107	0.065333
108	0.065333
109	0.065333
110	0.065333
111	0.065333
112	0.065333
113	0.065333
114	0.065333

115	0.065333
116	0.065333
117	0.065333
118	0.065333
119	0.065333
120	0.065333
121	0.065333
122	0.065333
123	0.065333
124	0.065333
125	0.065333
126	0.062297
127	0.032667
128	0.032667
129	0.032667
130	0.032667
131	0.032667
132	0.032667
133	0.032667
134	0.032667
135	0.032667
136	0.032667
137	0.032667
138	0.032667
139	0.032667
140	0.032667
141	0.032667
142	0.032667
143	0.032667
144	0.032667
145	0.032667
146	0.073333
147	0.066667
148	0.033333
149	0.073333
150	0.066667
151	0.063958
152	0.066667
153	0.033333
154	0.066667
155	0.063958
156	0.066667
157	0.033333
158	0.066667
159	0.063958
160	0.066667
161	0.033333
162	0.066667
163	0.063958
164	0.066667
165	0.033333
166	0.066667
167	0.063958
168	0.066667
169	0.033333
170	0.066667

171	0.063958
172	0.066667
173	0.033333
174	0.066667
175	0.063958
176	0.066667
177	0.033333
178	0.066667
179	0.063958
180	0.066667
181	0.033333
182	0.066667
183	0.063958
184	0.066667
185	0.033333
186	0.066667
187	0.063958
188	0.066667
189	0.033333
190	0.066667
191	0.063958
192	0.066667
193	0.033333
194	0.066667
195	0.063958
196	0.066667
197	0.033333
198	0.066667
199	0.063958
200	0.066667
201	0.033333
202	0.066667
203	0.063958
204	0.067375
205	0.061250
206	0.067375
207	0.061250
208	0.067375
209	0.061250
210	0.067375
211	0.061250
212	0.067375
213	0.061250
214	0.067375
215	0.061250
216	0.067375
217	0.061250
218	0.067375
219	0.061250
220	0.067375
221	0.061250
222	0.067375
223	0.061250
224	0.067375
225	0.061250
226	0.067375

227	0.061250
228	0.067375
229	0.061250
230	0.067375
231	0.061250
232	0.067375
233	0.061250
234	0.067375
235	0.061250
236	0.067375
237	0.061250
238	0.067375
239	0.061250
240	0.067375
241	0.061250
242	0.063959
243	0.061250
244	0.061250
245	0.061250
246	0.061250
247	0.061250
248	0.061250
249	0.061250
250	0.061250
251	0.061250
252	0.061250
253	0.061250
254	0.061250
255	0.061250
256	0.061250
257	0.061250
258	0.061250
259	0.061250
260	0.061250
261	0.061250
262	0.063959
263	0.061250
264	0.061250
265	0.061250
266	0.061250
267	0.061250
268	0.061250
269	0.061250
270	0.061250
271	0.061250
272	0.061250
273	0.061250
274	0.061250
275	0.061250
276	0.061250
277	0.061250
278	0.061250
279	0.061250
280	0.061250
281	0.061250
282	0.063959

283	0.061250
284	0.061250
285	0.061250
286	0.061250
287	0.061250
288	0.061250
289	0.061250
290	0.061250
291	0.061250
292	0.061250
293	0.061250
294	0.061250
295	0.061250
296	0.061250
297	0.061250
298	0.061250
299	0.061250
300	0.061250
301	0.061250
302	0.063959
303	0.061250
304	0.061250
305	0.061250
306	0.061250
307	0.061250
308	0.061250
309	0.061250
310	0.061250
311	0.061250
312	0.061250
313	0.061250
314	0.061250
315	0.061250
316	0.061250
317	0.061250
318	0.061250
319	0.061250
320	0.061250
321	0.061250
322	0.063959
323	0.061250
324	0.061250
325	0.061250
326	0.061250
327	0.061250
328	0.061250
329	0.061250
330	0.061250
331	0.061250
332	0.061250
333	0.061250
334	0.061250
335	0.061250
336	0.061250
337	0.061250
338	0.061250

339	0.061250
340	0.061250
341	0.061250
342	0.063959
343	0.061250
344	0.061250
345	0.061250
346	0.061250
347	0.061250
348	0.061250
349	0.061250
350	0.061250
351	0.061250
352	0.061250
353	0.061250
354	0.061250
355	0.061250
356	0.061250
357	0.061250
358	0.061250
359	0.061250
360	0.061250
361	0.061250
362	0.063959
363	0.061250
364	0.061250
365	0.061250
366	0.061250
367	0.061250
368	0.061250
369	0.061250
370	0.061250
371	0.061250
372	0.061250
373	0.061250
374	0.061250
375	0.061250
376	0.061250
377	0.061250
378	0.061250
379	0.061250
380	0.061250
381	0.061250
382	0.063959
383	0.061250
384	0.061250
385	0.061250
386	0.061250
387	0.061250
388	0.061250
389	0.061250
390	0.061250
391	0.061250
392	0.061250
393	0.061250
394	0.061250

395	0.061250
396	0.061250
397	0.061250
398	0.061250
399	0.061250
400	0.061250
401	0.061250
402	0.063959
403	0.061250
404	0.061250
405	0.061250
406	0.061250
407	0.061250
408	0.061250
409	0.061250
410	0.061250
411	0.061250
412	0.061250
413	0.061250
414	0.061250
415	0.061250
416	0.061250
417	0.061250
418	0.061250
419	0.061250
420	0.061250
421	0.061250
422	0.063959
423	0.061250
424	0.061250
425	0.061250
426	0.061250
427	0.061250
428	0.061250
429	0.061250
430	0.061250
431	0.061250
432	0.061250
433	0.061250
434	0.061250
435	0.061250
436	0.061250
437	0.061250
438	0.061250
439	0.061250
440	0.061250
441	0.061250
442	0.063959
443	0.061250
444	0.061250
445	0.061250
446	0.061250
447	0.061250
448	0.061250
449	0.061250
450	0.061250

451	0.061250
452	0.061250
453	0.061250
454	0.061250
455	0.061250
456	0.061250
457	0.061250
458	0.061250
459	0.061250
460	0.061250
461	0.061250
462	0.063959
463	0.061250
464	0.061250
465	0.061250
466	0.061250
467	0.061250
468	0.061250
469	0.061250
470	0.061250
471	0.061250
472	0.061250
473	0.061250
474	0.061250
475	0.061250
476	0.061250
477	0.061250
478	0.061250
479	0.061250
480	0.061250
481	0.061250
482	0.063959
483	0.061250
484	0.061250
485	0.061250
486	0.061250
487	0.061250
488	0.061250
489	0.061250
490	0.061250
491	0.061250
492	0.061250
493	0.061250
494	0.061250
495	0.061250
496	0.061250
497	0.061250
498	0.061250
499	0.061250
500	0.061250
501	0.061250
502	0.063959
503	0.073333
504	0.066666
505	0.073333
506	0.066666

507	0.033334
508	0.066666
509	0.066666
510	0.033334
511	0.066666
512	0.066666
513	0.033334
514	0.066666
515	0.066666
516	0.033334
517	0.066666
518	0.066666
519	0.033334
520	0.066666
521	0.066666
522	0.033334
523	0.066666
524	0.066666
525	0.033334
526	0.066666
527	0.066666
528	0.033334
529	0.066666
530	0.066666
531	0.033334
532	0.066666
533	0.066666
534	0.033334
535	0.066666
536	0.066666
537	0.033334
538	0.066666
539	0.066666
540	0.033334
541	0.066666
542	0.066666
543	0.033334
544	0.066666
545	0.066666
546	0.033334
547	0.068888
548	0.062963
549	0.072223
550	0.080000
551	0.040000
552	0.072223
553	0.080000
554	0.076750
555	0.066355
556	0.073500
557	0.066355
558	0.073500
559	0.066355
560	0.073500
561	0.066355
562	0.073500

563	0.066355
564	0.073500
565	0.066355
566	0.073500
567	0.066355
568	0.073500
569	0.066355
570	0.073500
571	0.066355
572	0.073500
573	0.066355
574	0.073500
575	0.066355
576	0.073500
577	0.066355
578	0.073500
579	0.066355
580	0.073500
581	0.066355
582	0.073500
583	0.066355
584	0.073500
585	0.066355
586	0.073500
587	0.066355
588	0.073500
589	0.066355
590	0.073500
591	0.066355
592	0.073500
593	0.076750
594	0.072222
595	0.079999
596	0.072222
597	0.080000
598	0.077826
599	0.068298
600	0.075652
601	0.068298
602	0.075653
603	0.068298
604	0.075652
605	0.068298
606	0.075652
607	0.068297
608	0.075651
609	0.068298
610	0.075653
611	0.068298
612	0.075652
613	0.068298
614	0.075652
615	0.068298
616	0.075652
617	0.068296
618	0.075651

619	0.068298
620	0.075652
621	0.068298
622	0.075652
623	0.068298
624	0.075652
625	0.068298
626	0.075653
627	0.068297
628	0.075651
629	0.068298
630	0.075652
631	0.068294
632	0.075648
633	0.068296
634	0.075651
635	0.068305
636	0.075660
637	0.068305
638	0.075660
639	0.068291
640	0.075645
641	0.068291
642	0.075645
643	0.080325
644	0.037826
645	0.037826
646	0.037826
647	0.037826
648	0.037826
649	0.037826
650	0.037826
651	0.037826
652	0.037826
653	0.037825
654	0.037826
655	0.037826
656	0.037826
657	0.037826
658	0.037826
659	0.037826
660	0.037824
661	0.037825
662	0.037830
663	0.037830
664	0.037823
665	0.037823
666	0.064445
667	0.064444
668	0.032222
669	0.064445
670	0.064444
671	0.061826
672	0.064444
673	0.032222
674	0.064444

675	0.061826
676	0.064445
677	0.032223
678	0.064445
679	0.061827
680	0.064444
681	0.032222
682	0.064444
683	0.061826
684	0.064444
685	0.032222
686	0.064444
687	0.061826
688	0.059209
689	0.059208
690	0.059209
691	0.059208
692	0.059209
693	0.059208
694	0.059209
695	0.059208
696	0.059209
697	0.059208
698	0.059209
699	0.059208
700	0.059209
701	0.059208
702	0.059209
703	0.059208
704	0.059209
705	0.059208
706	0.059209
707	0.059208
708	0.059209
709	0.059208
710	0.059209
711	0.059208
712	0.059209
713	0.059208
714	0.059209
715	0.059208
716	0.059209
717	0.059208
718	0.059209
719	0.059208
720	0.059209
721	0.059208
722	0.059209
723	0.059208
724	0.059209
725	0.059208
726	0.061826
727	0.059208
728	0.059208
729	0.059208
730	0.059208

731	0.059208
732	0.059208
733	0.059208
734	0.059208
735	0.059208
736	0.059208
737	0.059208
738	0.059208
739	0.059208
740	0.059208
741	0.059208
742	0.059208
743	0.059208
744	0.059208
745	0.059208
746	0.061826
747	0.059209
748	0.059209
749	0.059209
750	0.059209
751	0.059209
752	0.059209
753	0.059209
754	0.059209
755	0.059209
756	0.059209
757	0.059209
758	0.059209
759	0.059209
760	0.059209
761	0.059209
762	0.059209
763	0.059209
764	0.059209
765	0.059209
766	0.061828
767	0.059208
768	0.059208
769	0.059208
770	0.059208
771	0.059208
772	0.059208
773	0.059208
774	0.059208
775	0.059208
776	0.059208
777	0.059208
778	0.059208
779	0.059208
780	0.059208
781	0.059208
782	0.059208
783	0.059208
784	0.059208
785	0.059208
786	0.061826

787	0.059208
788	0.059208
789	0.059208
790	0.059208
791	0.059208
792	0.059208
793	0.059208
794	0.059208
795	0.059208
796	0.059208
797	0.059208
798	0.059208
799	0.059208
800	0.059208
801	0.059208
802	0.059208
803	0.059208
804	0.059208
805	0.059208
806	0.061826
807	0.064445
808	0.064444
809	0.064445
810	0.064444
811	0.062693
812	0.064444
813	0.064444
814	0.062693
815	0.064445
816	0.064445
817	0.062694
818	0.064444
819	0.064444
820	0.062693
821	0.064444
822	0.064444
823	0.062693
824	0.060943
825	0.060942
826	0.060943
827	0.060942
828	0.060943
829	0.060942
830	0.060943
831	0.060942
832	0.060942
833	0.060941
834	0.060943
835	0.060942
836	0.060943
837	0.060942
838	0.060943
839	0.060942
840	0.060943
841	0.060942
842	0.060942

843	0.060941
844	0.060943
845	0.060942
846	0.060943
847	0.060942
848	0.060943
849	0.060942
850	0.060943
851	0.060942
852	0.060942
853	0.060941
854	0.060943
855	0.060942
856	0.060939
857	0.060938
858	0.060942
859	0.060941
860	0.060949
861	0.060948
862	0.060949
863	0.060948
864	0.060937
865	0.060936
866	0.060937
867	0.060936
868	0.064706
869	0.060942
870	0.060942
871	0.060942
872	0.060942
873	0.060941
874	0.060942
875	0.060942
876	0.060942
877	0.060942
878	0.060941
879	0.060942
880	0.060942
881	0.060942
882	0.060942
883	0.060941
884	0.060942
885	0.060938
886	0.060941
887	0.060948
888	0.060948
889	0.060936
890	0.060936
891	0.064706
892	0.060943
893	0.060943
894	0.060943
895	0.060943
896	0.060942
897	0.060943
898	0.060943

899	0.060943
900	0.060943
901	0.060942
902	0.060943
903	0.060943
904	0.060943
905	0.060943
906	0.060942
907	0.060943
908	0.060939
909	0.060942
910	0.060949
911	0.060949
912	0.060937
913	0.060937
914	0.064707
915	0.060942
916	0.060942
917	0.060942
918	0.060942
919	0.060941
920	0.060942
921	0.060942
922	0.060942
923	0.060942
924	0.060941
925	0.060942
926	0.060942
927	0.060942
928	0.060942
929	0.060941
930	0.060942
931	0.060938
932	0.060941
933	0.060948
934	0.060948
935	0.060936
936	0.060936
937	0.064706
938	0.060942
939	0.060942
940	0.060942
941	0.060942
942	0.060941
943	0.060942
944	0.060942
945	0.060942
946	0.060942
947	0.060941
948	0.060942
949	0.060942
950	0.060942
951	0.060942
952	0.060941
953	0.060942
954	0.060938

955	0.060941
956	0.060948
957	0.060948
958	0.060936
959	0.060936
960	0.064706
961	0.068477
962	0.068476
963	0.068477
964	0.068476
965	0.034232
966	0.068476
967	0.068476
968	0.034232
969	0.068477
970	0.068477
971	0.034233
972	0.068476
973	0.068476
974	0.034232
975	0.068476
976	0.068476
977	0.034232
978	0.076741
979	0.069659
980	0.072223
981	0.064444
982	0.032222
983	0.072223
984	0.064444
985	0.061826
986	0.064444
987	0.032222
988	0.064444
989	0.061826
990	0.064445
991	0.032223
992	0.064445
993	0.061827
994	0.064444
995	0.032222
996	0.064444
997	0.061826
998	0.064444
999	0.032222
1000	0.064444
1001	0.061826
1002	0.066355
1003	0.059208
1004	0.066355
1005	0.059208
1006	0.066355
1007	0.059208
1008	0.066355
1009	0.059208
1010	0.066355

1011	0.059208
1012	0.066355
1013	0.059208
1014	0.066355
1015	0.059208
1016	0.066355
1017	0.059208
1018	0.066355
1019	0.059208
1020	0.066355
1021	0.059208
1022	0.066355
1023	0.059208
1024	0.066355
1025	0.059208
1026	0.066355
1027	0.059208
1028	0.066355
1029	0.059208
1030	0.066355
1031	0.059208
1032	0.066355
1033	0.059208
1034	0.066355
1035	0.059208
1036	0.066355
1037	0.059208
1038	0.066355
1039	0.059208
1040	0.061826
1041	0.059208
1042	0.059208
1043	0.059208
1044	0.059208
1045	0.059208
1046	0.059208
1047	0.059208
1048	0.059208
1049	0.059208
1050	0.059208
1051	0.059208
1052	0.059208
1053	0.059208
1054	0.059208
1055	0.059208
1056	0.059208
1057	0.059208
1058	0.059208
1059	0.059208
1060	0.061826
1061	0.059209
1062	0.059209
1063	0.059209
1064	0.059209
1065	0.059209
1066	0.059209

1067	0.059209
1068	0.059209
1069	0.059209
1070	0.059209
1071	0.059209
1072	0.059209
1073	0.059209
1074	0.059209
1075	0.059209
1076	0.059209
1077	0.059209
1078	0.059209
1079	0.059209
1080	0.061828
1081	0.059208
1082	0.059208
1083	0.059208
1084	0.059208
1085	0.059208
1086	0.059208
1087	0.059208
1088	0.059208
1089	0.059208
1090	0.059208
1091	0.059208
1092	0.059208
1093	0.059208
1094	0.059208
1095	0.059208
1096	0.059208
1097	0.059208
1098	0.059208
1099	0.059208
1100	0.061826
1101	0.059208
1102	0.059208
1103	0.059208
1104	0.059208
1105	0.059208
1106	0.059208
1107	0.059208
1108	0.059208
1109	0.059208
1110	0.059208
1111	0.059208
1112	0.059208
1113	0.059208
1114	0.059208
1115	0.059208
1116	0.059208
1117	0.059208
1118	0.059208
1119	0.059208
1120	0.061826
1121	0.072222
1122	0.064444

1123	0.072222
1124	0.064444
1125	0.062693
1126	0.064444
1127	0.064444
1128	0.062693
1129	0.064445
1130	0.064445
1131	0.062694
1132	0.064444
1133	0.064444
1134	0.062693
1135	0.064444
1136	0.064444
1137	0.062693
1138	0.068298
1139	0.060942
1140	0.068298
1141	0.060942
1142	0.068298
1143	0.060942
1144	0.068298
1145	0.060942
1146	0.068297
1147	0.060941
1148	0.068298
1149	0.060942
1150	0.068298
1151	0.060942
1152	0.068298
1153	0.060942
1154	0.068298
1155	0.060942
1156	0.068296
1157	0.060941
1158	0.068298
1159	0.060942
1160	0.068298
1161	0.060942
1162	0.068298
1163	0.060942
1164	0.068298
1165	0.060942
1166	0.068297
1167	0.060941
1168	0.068298
1169	0.060942
1170	0.068294
1171	0.060938
1172	0.068296
1173	0.060941
1174	0.068305
1175	0.060948
1176	0.068305
1177	0.060948
1178	0.068291

1179	0.060936
1180	0.068291
1181	0.060936
1182	0.064706
1183	0.060942
1184	0.060942
1185	0.060942
1186	0.060942
1187	0.060941
1188	0.060942
1189	0.060942
1190	0.060942
1191	0.060942
1192	0.060941
1193	0.060942
1194	0.060942
1195	0.060942
1196	0.060942
1197	0.060941
1198	0.060942
1199	0.060938
1200	0.060941
1201	0.060948
1202	0.060948
1203	0.060936
1204	0.060936
1205	0.064706
1206	0.060943
1207	0.060943
1208	0.060943
1209	0.060943
1210	0.060942
1211	0.060943
1212	0.060943
1213	0.060943
1214	0.060943
1215	0.060942
1216	0.060943
1217	0.060943
1218	0.060943
1219	0.060943
1220	0.060942
1221	0.060943
1222	0.060939
1223	0.060942
1224	0.060949
1225	0.060949
1226	0.060937
1227	0.060937
1228	0.064707
1229	0.060942
1230	0.060942
1231	0.060942
1232	0.060942
1233	0.060941
1234	0.060942

1235	0.060942
1236	0.060942
1237	0.060942
1238	0.060941
1239	0.060942
1240	0.060942
1241	0.060942
1242	0.060942
1243	0.060941
1244	0.060942
1245	0.060938
1246	0.060941
1247	0.060948
1248	0.060948
1249	0.060936
1250	0.060936
1251	0.064706
1252	0.060942
1253	0.060942
1254	0.060942
1255	0.060942
1256	0.060941
1257	0.060942
1258	0.060942
1259	0.060942
1260	0.060942
1261	0.060941
1262	0.060942
1263	0.060942
1264	0.060942
1265	0.060942
1266	0.060941
1267	0.060942
1268	0.060938
1269	0.060941
1270	0.060948
1271	0.060948
1272	0.060936
1273	0.060936
1274	0.064706
1275	0.076741
1276	0.068476
1277	0.069659
1278	0.068476
1279	0.034232
1280	0.068476
1281	0.068476
1282	0.034232
1283	0.068477
1284	0.068477
1285	0.034233
1286	0.068476
1287	0.068476
1288	0.034232
1289	0.068476
1290	0.068476

1291	0.034232
1292	0.073333
1293	0.080000
1294	0.040000
1295	0.073333
1296	0.080000
1297	0.076750
1298	0.067375
1299	0.073500
1300	0.067375
1301	0.073500
1302	0.067375
1303	0.073500
1304	0.067375
1305	0.073500
1306	0.067375
1307	0.073500
1308	0.067375
1309	0.073500
1310	0.067375
1311	0.073500
1312	0.067375
1313	0.073500
1314	0.067375
1315	0.073500
1316	0.067375
1317	0.073500
1318	0.067375
1319	0.073500
1320	0.067375
1321	0.073500
1322	0.067375
1323	0.073500
1324	0.067375
1325	0.073500
1326	0.067375
1327	0.073500
1328	0.067375
1329	0.073500
1330	0.067375
1331	0.073500
1332	0.067375
1333	0.073500
1334	0.067375
1335	0.073500
1336	0.076750
1337	0.073333
1338	0.079999
1339	0.073333
1340	0.080000
1341	0.077826
1342	0.037826
1343	0.075652
1344	0.037826
1345	0.075653
1346	0.037826

1347	0.075652
1348	0.037826
1349	0.075652
1350	0.037825
1351	0.075651
1352	0.037826
1353	0.075653
1354	0.037826
1355	0.075652
1356	0.037826
1357	0.075652
1358	0.037826
1359	0.075652
1360	0.037825
1361	0.075651
1362	0.037826
1363	0.075652
1364	0.037826
1365	0.075652
1366	0.037826
1367	0.075652
1368	0.037826
1369	0.075653
1370	0.037825
1371	0.075651
1372	0.037826
1373	0.075652
1374	0.037824
1375	0.075648
1376	0.037825
1377	0.075651
1378	0.037830
1379	0.075660
1380	0.037830
1381	0.075660
1382	0.037822
1383	0.075645
1384	0.037822
1385	0.075645
1386	0.080325
1387	0.068893
1388	0.066667
1389	0.033333
1390	0.068893
1391	0.066667
1392	0.063958
1393	0.066667
1394	0.033333
1395	0.066667
1396	0.063958
1397	0.066667
1398	0.033333
1399	0.066667
1400	0.063958
1401	0.066667
1402	0.033333

1403	0.066667
1404	0.063958
1405	0.066667
1406	0.033333
1407	0.066667
1408	0.063958
1409	0.066667
1410	0.033333
1411	0.066667
1412	0.063958
1413	0.066667
1414	0.033333
1415	0.066667
1416	0.063958
1417	0.066667
1418	0.033333
1419	0.066667
1420	0.063958
1421	0.066667
1422	0.033333
1423	0.066667
1424	0.063958
1425	0.066667
1426	0.033333
1427	0.066667
1428	0.063958
1429	0.066667
1430	0.033333
1431	0.066667
1432	0.063958
1433	0.066667
1434	0.033333
1435	0.066667
1436	0.063958
1437	0.066667
1438	0.033333
1439	0.066667
1440	0.063958
1441	0.066667
1442	0.033333
1443	0.066667
1444	0.063958
1445	0.063296
1446	0.061250
1447	0.063296
1448	0.061250
1449	0.063296
1450	0.061250
1451	0.063296
1452	0.061250
1453	0.063296
1454	0.061250
1455	0.063296
1456	0.061250
1457	0.063296
1458	0.061250

1459	0.063296
1460	0.061250
1461	0.063296
1462	0.061250
1463	0.063296
1464	0.061250
1465	0.063296
1466	0.061250
1467	0.063296
1468	0.061250
1469	0.063296
1470	0.061250
1471	0.063296
1472	0.061250
1473	0.063296
1474	0.061250
1475	0.063296
1476	0.061250
1477	0.063296
1478	0.061250
1479	0.063296
1480	0.061250
1481	0.063296
1482	0.061250
1483	0.063959
1484	0.061250
1485	0.061250
1486	0.061250
1487	0.061250
1488	0.061250
1489	0.061250
1490	0.061250
1491	0.061250
1492	0.061250
1493	0.061250
1494	0.061250
1495	0.061250
1496	0.061250
1497	0.061250
1498	0.061250
1499	0.061250
1500	0.061250
1501	0.061250
1502	0.061250
1503	0.063959
1504	0.061250
1505	0.061250
1506	0.061250
1507	0.061250
1508	0.061250
1509	0.061250
1510	0.061250
1511	0.061250
1512	0.061250
1513	0.061250
1514	0.061250

1515	0.061250
1516	0.061250
1517	0.061250
1518	0.061250
1519	0.061250
1520	0.061250
1521	0.061250
1522	0.061250
1523	0.063959
1524	0.061250
1525	0.061250
1526	0.061250
1527	0.061250
1528	0.061250
1529	0.061250
1530	0.061250
1531	0.061250
1532	0.061250
1533	0.061250
1534	0.061250
1535	0.061250
1536	0.061250
1537	0.061250
1538	0.061250
1539	0.061250
1540	0.061250
1541	0.061250
1542	0.061250
1543	0.063959
1544	0.061250
1545	0.061250
1546	0.061250
1547	0.061250
1548	0.061250
1549	0.061250
1550	0.061250
1551	0.061250
1552	0.061250
1553	0.061250
1554	0.061250
1555	0.061250
1556	0.061250
1557	0.061250
1558	0.061250
1559	0.061250
1560	0.061250
1561	0.061250
1562	0.061250
1563	0.063959
1564	0.061250
1565	0.061250
1566	0.061250
1567	0.061250
1568	0.061250
1569	0.061250
1570	0.061250

1571	0.061250
1572	0.061250
1573	0.061250
1574	0.061250
1575	0.061250
1576	0.061250
1577	0.061250
1578	0.061250
1579	0.061250
1580	0.061250
1581	0.061250
1582	0.061250
1583	0.063959
1584	0.061250
1585	0.061250
1586	0.061250
1587	0.061250
1588	0.061250
1589	0.061250
1590	0.061250
1591	0.061250
1592	0.061250
1593	0.061250
1594	0.061250
1595	0.061250
1596	0.061250
1597	0.061250
1598	0.061250
1599	0.061250
1600	0.061250
1601	0.061250
1602	0.061250
1603	0.063959
1604	0.061250
1605	0.061250
1606	0.061250
1607	0.061250
1608	0.061250
1609	0.061250
1610	0.061250
1611	0.061250
1612	0.061250
1613	0.061250
1614	0.061250
1615	0.061250
1616	0.061250
1617	0.061250
1618	0.061250
1619	0.061250
1620	0.061250
1621	0.061250
1622	0.061250
1623	0.063959
1624	0.061250
1625	0.061250
1626	0.061250

1627	0.061250
1628	0.061250
1629	0.061250
1630	0.061250
1631	0.061250
1632	0.061250
1633	0.061250
1634	0.061250
1635	0.061250
1636	0.061250
1637	0.061250
1638	0.061250
1639	0.061250
1640	0.061250
1641	0.061250
1642	0.061250
1643	0.063959
1644	0.061250
1645	0.061250
1646	0.061250
1647	0.061250
1648	0.061250
1649	0.061250
1650	0.061250
1651	0.061250
1652	0.061250
1653	0.061250
1654	0.061250
1655	0.061250
1656	0.061250
1657	0.061250
1658	0.061250
1659	0.061250
1660	0.061250
1661	0.061250
1662	0.061250
1663	0.063959
1664	0.061250
1665	0.061250
1666	0.061250
1667	0.061250
1668	0.061250
1669	0.061250
1670	0.061250
1671	0.061250
1672	0.061250
1673	0.061250
1674	0.061250
1675	0.061250
1676	0.061250
1677	0.061250
1678	0.061250
1679	0.061250
1680	0.061250
1681	0.061250
1682	0.061250

1683	0.063959
1684	0.061250
1685	0.061250
1686	0.061250
1687	0.061250
1688	0.061250
1689	0.061250
1690	0.061250
1691	0.061250
1692	0.061250
1693	0.061250
1694	0.061250
1695	0.061250
1696	0.061250
1697	0.061250
1698	0.061250
1699	0.061250
1700	0.061250
1701	0.061250
1702	0.061250
1703	0.063959
1704	0.061250
1705	0.061250
1706	0.061250
1707	0.061250
1708	0.061250
1709	0.061250
1710	0.061250
1711	0.061250
1712	0.061250
1713	0.061250
1714	0.061250
1715	0.061250
1716	0.061250
1717	0.061250
1718	0.061250
1719	0.061250
1720	0.061250
1721	0.061250
1722	0.061250
1723	0.063959
1724	0.061250
1725	0.061250
1726	0.061250
1727	0.061250
1728	0.061250
1729	0.061250
1730	0.061250
1731	0.061250
1732	0.061250
1733	0.061250
1734	0.061250
1735	0.061250
1736	0.061250
1737	0.061250
1738	0.061250

1739	0.061250
1740	0.061250
1741	0.061250
1742	0.061250
1743	0.063959
1744	0.068893
1745	0.066666
1746	0.062966
1747	0.066666
1748	0.033334
1749	0.066666
1750	0.066666
1751	0.033334
1752	0.066666
1753	0.066666
1754	0.033334
1755	0.066666
1756	0.066666
1757	0.033334
1758	0.066666
1759	0.066666
1760	0.033334
1761	0.066666
1762	0.066666
1763	0.033334
1764	0.066666
1765	0.066666
1766	0.033334
1767	0.066666
1768	0.066666
1769	0.033334
1770	0.066666
1771	0.066666
1772	0.033334
1773	0.066666
1774	0.066666
1775	0.033334
1776	0.066666
1777	0.066666
1778	0.033334
1779	0.066666
1780	0.066666
1781	0.033334
1782	0.066666
1783	0.066666
1784	0.033334
1785	0.066666
1786	0.066666
1787	0.033334
1788	0.035560
1789	0.071107
1790	0.035553
1791	0.035560
1792	0.071107
1793	0.068218
1794	0.071107

1795	0.035553
1796	0.071107
1797	0.068218
1798	0.032671
1799	0.065329
1800	0.032671
1801	0.065329
1802	0.032671
1803	0.065329
1804	0.032671
1805	0.065329
1806	0.032671
1807	0.065329
1808	0.032671
1809	0.065329
1810	0.032671
1811	0.065329
1812	0.032671
1813	0.065329
1814	0.032671
1815	0.065329
1816	0.032671
1817	0.065329
1818	0.032671
1819	0.065329
1820	0.032671
1821	0.065329
1822	0.032671
1823	0.065329
1824	0.032671
1825	0.065329
1826	0.032671
1827	0.065329
1828	0.032671
1829	0.065329
1830	0.032671
1831	0.065329
1832	0.032671
1833	0.065329
1834	0.032671
1835	0.065329
1836	0.062292
1837	0.065329
1838	0.065329
1839	0.065329
1840	0.065329
1841	0.065329
1842	0.065329
1843	0.065329
1844	0.065329
1845	0.065329
1846	0.065329
1847	0.065329
1848	0.065329
1849	0.065329
1850	0.065329

1851	0.065329
1852	0.065329
1853	0.065329
1854	0.065329
1855	0.065329
1856	0.068218
1861	0.041481
1862	0.041481
1863	0.065185
1870	0.035417
1871	0.049590
1872	0.077920
1878	0.035417
1879	0.049590
1880	0.077920
1885	0.041482
1886	0.041476
1887	0.065182
TOTAL	112.084976

```
1278. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
1279. JOINT LOAD
1280. 14 FX 0 FY -50 FZ 0
1281. 15 FX 0 FY -30 FZ 0
1282. 17 FX 0 FY -140 FZ 0
1283. 19 FX 0 FY -40 FZ 0
1284. 20 FX 0 FY -30 FZ 0
1285. 18 FX 0 FY -120 FZ 0
1286. 16 FX 0 FY -120 FZ 0
1287. 22 FX 0 FY -10 FZ 0
1288. 21 FX 0 FY -10 FZ 0
1289. SELFWEIGHT Y -1
1290. ***
1291. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
1292. JOINT LOAD
1293. 14 FX 0 FY -40 FZ 0
1294. 15 FX 0 FY -50 FZ 0
1295. 17 FX 0 FY -30 FZ 0
1296. 19 FX 0 FY -20 FZ 0
1297. 20 FX 0 FY -40 FZ 0
1298. 18 FX 0 FY -70 FZ 0
1299. 16 FX 0 FY -70 FZ 0
1300. 22 FX 0 FY -20 FZ 0
1301. 21 FX 0 FY -20 FZ 0
1302. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
1303. JOINT LOAD
1304. 14 FX 10 FY 10 FZ 0
1305. 15 FX 10 FY -10 FZ 0
1306. 17 FX 0 FY 0 FZ 0
1307. 19 FX 0 FY 10 FZ 0
1308. 20 FX 0 FY -10 FZ 0
1309. 18 FX 20 FY 20 FZ 0
1310. 16 FX 20 FY 20 FZ 0
1311. 22 FX 10 FY -20 FZ 0
1312. 21 FX 10 FY -20 FZ 0
1313. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
1314. JOINT LOAD
```

```
1315. 14 FX -10 FY -10 FZ 0
1316. 15 FX -10 FY 10 FZ 0
1317. 17 FX 0 FY 0 FZ 0
1318. 19 FX 0 FY -10 FZ 0
1319. 20 FX 0 FY 10 FZ 0
1320. 18 FX -20 FY -20 FZ 0
1321. 16 FX -20 FY -20 FZ 0
1322. 22 FX -10 FY 20 FZ 0
1323. 21 FX -10 FY 20 FZ 0
1324. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
1325. JOINT LOAD
1326. 14 FX 0 FY 20 FZ 20
1327. 15 FX 0 FY 0 FZ 0
1328. 17 FX 0 FY -20 FZ 20
1329. 19 FX 0 FY 0 FZ 0
1330. 20 FX 0 FY 0 FZ 0
1331. 18 FX 0 FY -80 FZ 30
1332. 16 FX 0 FY 80 FZ 30
1333. 22 FX 0 FY -20 FZ 10
1334. 21 FX 0 FY 20 FZ 10
1335. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
1336. JOINT LOAD
1337. 14 FX 0 FY -20 FZ -20
1338. 15 FX 0 FY 0 FZ 0
1339. 17 FX 0 FY 20 FZ -20
1340. 19 FX 0 FY 0 FZ 0
1341. 20 FX 0 FY 0 FZ 0
1342. 18 FX 0 FY 80 FZ -30
1343. 16 FX 0 FY -80 FZ -30
1344. 22 FX 0 FY 20 FZ -10
1345. 21 FX 0 FY -20 FZ -10
1346. LOAD 7 LOADTYPE SOIL  TITLE SOIL LOAD (HL)
1347. ELEMENT LOAD
1348. _RAFT PR -3.6
1349. ******************************************************
1350. ******************************************************
1351. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
1352. ****************************************
1353. LOAD COMB 101 1.4DL
1354. 1 1.4
1355. LOAD COMB 102 1.4DL+1.7LL
1356. 1 1.4 2 1.7
1357. LOAD COMB 103 1.2DL+1.0LL+1.6WLX1
1358. 1 1.2 2 1.0 3 1.6
1359. LOAD COMB 104 1.2DL+1.0LL+1.6WLX2
1360. 1 1.2 2 1.0 4 1.6
1361. LOAD COMB 105 1.2DL+1.0LL+1.6WLZ1
1362. 1 1.2 2 1.0 5 1.6
1363. LOAD COMB 106 1.2DL+1.0LL+1.6WLZ2
1364. 1 1.2 2 1.0 6 1.6
1365. LOAD COMB 107 0.9DL+1.6HL+1.6WLX1
1366. 1 0.9 7 1.6 3 1.6
1367. LOAD COMB 108 0.9DL+1.6HL+1.6WLX2
1368. 1 0.9 7 1.6 4 1.6
1369. LOAD COMB 109 0.9DL+1.6HL+1.6WLZ1
1370. 1 0.9 7 1.6 5 1.6
```

```
1371. LOAD COMB 110 0.9DL+1.6HL+1.6WLZ2
1372. 1 0.9 7 1.6 6 1.6
1373. ***************
1374. ********************************************************************************
1375. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
1376. ************************************************************
1377. LOAD COMB 201 1.0DL+1.0HL+1.0LL
1378. 1 1.0 7 1.0 2 1.0
1379. LOAD COMB 202 1.0DL+1.0HL+0.75LL
1380. 1 1.0 7 1.0 2 0.75
1381. LOAD COMB 203 1.0DL+1.0HL+1.0WLX1
1382. 1 1.0 7 1.0 3 1.0
1383. LOAD COMB 204 1.0DL+1.0HL+1.0WLX2
1384. 1 1.0 7 1.0 4 1.0
1385. LOAD COMB 205 1.0DL+1.0HL+1.0WLZ1
1386. 1 1.0 7 1.0 5 1.0
1387. LOAD COMB 206 1.0DL+1.0HL+1.0WLZ2
1388. 1 1.0 7 1.0 6 1.0
1389. LOAD COMB 207 1.0DL+1.0HL+0.75LL+0.75WLX1
1390. 1 1.0 7 1.0 2 0.75 3 0.75
1391. LOAD COMB 208 1.0DL+1.0HL+0.75LL+0.75WLX2
1392. 1 1.0 7 1.0 2 0.75 4 0.75
1393. LOAD COMB 209 1.0DL+1.0HL+0.75LL+0.75WLZ1
1394. 1 1.0 7 1.0 2 0.75 5 0.75
1395. LOAD COMB 210 1.0DL+1.0HL+0.75LL+0.75WLZ2
1396. 1 1.0 7 1.0 2 0.75 6 0.75
1397. LOAD COMB 211 0.6DL+1.0HL+1.0WLX1
1398. 1 0.6 7 1.0 3 1.0
1399. LOAD COMB 212 0.6DL+1.0HL+1.0WLX2
1400. 1 0.6 7 1.0 4 1.0
1401. LOAD COMB 213 0.6DL+1.0HL+1.0WLZ1
1402. 1 0.6 7 1.0 5 1.0
1403. LOAD COMB 214 0.6DL+1.0HL+1.0WLZ2
1404. 1 0.6 7 1.0 6 1.0
1405. ********
1406. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**1873**	**NUMBER OF MEMBERS**	**18**
NUMBER OF PLATES	**1760**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**1855**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1864/ 53/ 168 DOF
TOTAL PRIMARY LOAD CASES = 7, TOTAL DEGREES OF FREEDOM = 5673
TOTAL LOAD COMBINATION CASES = 24 SO FAR.
SIZE OF STIFFNESS MATRIX = 954 DOUBLE KILO-WORDS

STAAD SPACE **-- PAGE NO. 60**

REQRD/AVAIL. DISK SPACE = 31.9/ 111871.2 MB

```
1407. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 13,2020 TIME= 9:24:21 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 2 STAAD INPUT & OUTPUT FILE FOR F2 DESIGN

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5           *
*           Version  20.07.10.64                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    APR 13, 2020                 *
*           Time=     9:26: 3                     *
*                                                *
*      USER ID: LT Construction                  *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F2 DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 09-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 3.2 0 0; 3 0 0 7.15; 4 3.2 0 7.15; 5 -0.65 0 0; 6 -0.65 0 7.15
     9. 7 3.85 0 0; 8 3.85 0 7.15; 9 0 0 -0.5; 10 3.2 0 -0.5; 11 -0.65 0 -0.5
    10. 12 3.85 0 -0.5; 13 0 0 7.65; 14 3.2 0 7.65; 15 -0.65 0 7.65; 16 3.85 0 7.65
    11. 17 -0.25 0 0; 18 -0.25 0 -0.25; 19 -0.65 0 -0.25; 20 0 0 -0.25
    12. 21 -0.25 0 -0.5; 22 3.45 0 0; 23 3.45 0 -0.25; 24 3.2 0 -0.25; 25 3.85 0 -0.25
    13. 26 3.45 0 -0.5; 27 -0.25 0 7.65; 28 -0.25 0 7.4; 29 -0.65 0 7.4; 30 0 0 7.4
    14. 31 -0.25 0 7.15; 32 3.45 0 7.65; 33 3.45 0 7.4; 34 3.2 0 7.4; 35 3.85 0 7.4
    15. 36 3.45 0 7.15; 37 0.246154 0 0; 38 0.246154 0 -0.25; 39 0.492308 0 0
    16. 40 0.492308 0 -0.25; 41 0.738462 0 0; 42 0.738462 0 -0.25; 43 0.984615 0 0
    17. 44 0.984615 0 -0.25; 45 1.23077 0 0; 46 1.23077 0 -0.25; 47 1.47692 0 0
    18. 48 1.47692 0 -0.25; 49 1.72308 0 0; 50 1.72308 0 -0.25; 51 1.96923 0 0
    19. 52 1.96923 0 -0.25; 53 2.21538 0 0; 54 2.21538 0 -0.25; 55 2.46154 0 0
    20. 56 2.46154 0 -0.25; 57 2.70769 0 0; 58 2.70769 0 -0.25; 59 2.95385 0 0
    21. 60 2.95385 0 -0.25; 61 0.246154 0 -0.5; 62 0.492308 0 -0.5; 63 0.738462 0 -0.5
    22. 64 0.984615 0 -0.5; 65 1.23077 0 -0.5; 66 1.47692 0 -0.5; 67 1.72308 0 -0.5
    23. 68 1.96923 0 -0.5; 69 2.21538 0 -0.5; 70 2.46154 0 -0.5; 71 2.70769 0 -0.5
    24. 72 2.95385 0 -0.5; 73 -0.25 0 6.90345; 74 -0.65 0 6.90345; 75 0 0 6.90345
    25. 76 -0.25 0 6.6569; 77 -0.65 0 6.6569; 78 0 0 6.6569; 79 -0.25 0 6.15035
    26. 80 -0.65 0 6.15035; 81 0 0 6.15035; 148 -0.25 0 0.99965; 149 -0.65 0 0.99965
    27. 150 0 0 0.99965; 151 -0.25 0 0.493102; 152 -0.65 0 0.493102; 153 0 0 0.493102
    28. 154 -0.25 0 0.246551; 155 -0.65 0 0.246551; 156 0 0 0.246551
    29. 157 3.45 0 6.90345; 158 3.2 0 6.90345; 159 3.85 0 6.90345; 160 3.45 0 6.6569
    30. 161 3.2 0 6.6569; 162 3.85 0 6.6569; 163 3.45 0 6.15035; 164 3.2 0 6.15035
    31. 165 3.85 0 6.15035; 232 3.45 0 0.99965; 233 3.2 0 0.99965; 234 3.85 0 0.99965
    32. 235 3.45 0 0.493102; 236 3.2 0 0.493102; 237 3.85 0 0.493102
    33. 238 3.45 0 0.246551; 239 3.2 0 0.246551; 240 3.85 0 0.246551
    34. 241 0.246154 0 7.65; 242 0.246154 0 7.4; 243 0.492308 0 7.65
    35. 244 0.492308 0 7.4; 245 0.738462 0 7.65; 246 0.738462 0 7.4
    36. 247 0.984615 0 7.65; 248 0.984615 0 7.4; 249 1.23077 0 7.65; 250 1.23077 0 7.4
    37. 251 1.47692 0 7.65; 252 1.47692 0 7.4; 253 1.72308 0 7.65; 254 1.72308 0 7.4
    38. 255 1.96923 0 7.65; 256 1.96923 0 7.4; 257 2.21538 0 7.65; 258 2.21538 0 7.4
```

39. 259 2.46154 0 7.65; 260 2.46154 0 7.4; 261 2.70769 0 7.65; 262 2.70769 0 7.4
40. 263 2.95385 0 7.65; 264 2.95385 0 7.4; 265 0.246154 0 7.15
41. 266 0.492308 0 7.15; 267 0.738462 0 7.15; 268 0.984615 0 7.15
42. 269 1.23077 0 7.15; 270 1.47692 0 7.15; 271 1.72308 0 7.15; 272 1.96923 0 7.15
43. 273 2.21538 0 7.15; 274 2.46154 0 7.15; 275 2.70769 0 7.15; 276 2.95385 0 7.15
44. 277 0 2.75 0; 278 3.2 2.75 0; 279 0 2.75 7.15; 280 3.2 2.75 7.15
45. 281 0.246154 0 6.90345; 282 0.492308 0 6.90345; 283 0.738462 0 6.90345
46. 284 0.984615 0 6.90345; 285 1.23077 0 6.90345; 286 1.47692 0 6.90345
47. 287 1.72308 0 6.90345; 288 1.96923 0 6.90345; 289 2.21538 0 6.90345
48. 290 2.46154 0 6.90345; 291 2.70769 0 6.90345; 292 2.95385 0 6.90345
49. 293 0.246154 0 6.6569; 294 0.492308 0 6.6569; 295 0.738462 0 6.6569
50. 296 0.984615 0 6.6569; 297 1.23077 0 6.6569; 298 1.47692 0 6.6569
51. 299 1.72308 0 6.6569; 300 1.96923 0 6.6569; 301 2.21538 0 6.6569
52. 302 2.46154 0 6.6569; 303 2.70769 0 6.6569; 304 2.95385 0 6.6569
53. 305 0.246154 0 6.15035; 306 0.492308 0 6.15035; 307 0.738462 0 6.15035
54. 308 0.984615 0 6.15035; 309 1.23077 0 6.15035; 310 1.47692 0 6.15035
55. 311 1.72308 0 6.15035; 312 1.96923 0 6.15035; 313 2.21538 0 6.15035
56. 314 2.46154 0 6.15035; 315 2.70769 0 6.15035; 316 2.95385 0 6.15035
57. 581 0.246154 0 0.99965; 582 0.492308 0 0.99965; 583 0.738462 0 0.99965
58. 584 0.984615 0 0.99965; 585 1.23077 0 0.99965; 586 1.47692 0 0.99965
59. 587 1.72308 0 0.99965; 588 1.96923 0 0.99965; 589 2.21538 0 0.99965
60. 590 2.46154 0 0.99965; 591 2.70769 0 0.99965; 592 2.95385 0 0.99965
61. 593 0.246154 0 0.493102; 594 0.492308 0 0.493102; 595 0.738462 0 0.493102
62. 596 0.984615 0 0.493102; 597 1.23077 0 0.493102; 598 1.47692 0 0.493102
63. 599 1.72308 0 0.493102; 600 1.96923 0 0.493102; 601 2.21538 0 0.493102
64. 602 2.46154 0 0.493102; 603 2.70769 0 0.493102; 604 2.95385 0 0.493102
65. 605 0.246154 0 0.246551; 606 0.492308 0 0.246551; 607 0.738462 0 0.246551
66. 608 0.984615 0 0.246551; 609 1.23077 0 0.246551; 610 1.47692 0 0.246551
67. 611 1.72308 0 0.246551; 612 1.96923 0 0.246551; 613 2.21538 0 0.246551
68. 614 2.46154 0 0.246551; 615 2.70769 0 0.246551; 616 2.95385 0 0.246551
69. 617 0 0.2 0; 618 3.2 0.2 0; 619 0 0.2 7.15; 620 3.2 0.2 7.15; 621 -0.25 0 -1.
70. 622 -0.65 0 -1; 623 0 0 -1; 624 3.45 0 -1; 625 3.2 0 -1; 626 3.85 0 -1.
71. 627 0.246154 0 -1; 628 0.492308 0 -1; 629 0.738462 0 -1; 630 0.984615 0 -1.
72. 631 1.23077 0 -1; 632 1.47692 0 -1; 633 1.72308 0 -1; 634 1.96923 0 -1.
73. 635 2.21538 0 -1; 636 2.46154 0 -1; 637 2.70769 0 -1; 638 2.95385 0 -1.
74. 639 -0.65 0 8.15; 640 -0.25 0 8.15; 641 0 0 8.15; 642 3.2 0 8.15
75. 643 3.45 0 8.15; 644 3.85 0 8.15; 645 0.246154 0 8.15; 646 0.492308 0 8.15
76. 647 0.738462 0 8.15; 648 0.984615 0 8.15; 649 1.23077 0 8.15
77. 650 1.47692 0 8.15; 651 1.72308 0 8.15; 652 1.96923 0 8.15; 653 2.21538 0 8.15
78. 654 2.46154 0 8.15; 655 2.70769 0 8.15; 656 2.95385 0 8.15
79. 657 -0.25 0 6.40363; 658 -0.65 0 6.40363; 659 0 0 6.40363
80. 660 -0.25 0 0.746376; 661 -0.65 0 0.746376; 662 0 0 0.746376
81. 663 3.45 0 6.40363; 664 3.2 0 6.40363; 665 3.85 0 6.40363; 666 3.45 0 0.746376
82. 667 3.2 0 0.746376; 668 3.85 0 0.746376; 669 0.246154 0 6.40363
83. 670 0.492308 0 6.40363; 671 0.738462 0 6.40363; 672 0.984615 0 6.40363
84. 673 1.23077 0 6.40363; 674 1.47692 0 6.40363; 675 1.72308 0 6.40363
85. 676 1.96923 0 6.40363; 677 2.21538 0 6.40363; 678 2.46154 0 6.40363
86. 679 2.70769 0 6.40363; 680 2.95385 0 6.40363; 681 0.246154 0 0.746376
87. 682 0.492308 0 0.746376; 683 0.738462 0 0.746376; 684 0.984615 0 0.746376
88. 685 1.23077 0 0.746376; 686 1.47692 0 0.746376; 687 1.72308 0 0.746376
89. 688 1.96923 0 0.746376; 689 2.21538 0 0.746376; 690 2.46154 0 0.746376
90. 691 2.70769 0 0.746376; 692 2.95385 0 0.746376; 693 -0.25 0 -0.75
91. 694 -0.65 0 -0.75; 695 0 0 -0.75; 696 3.45 0 -0.75; 697 3.2 0 -0.75
92. 698 3.85 0 -0.75; 699 0.246154 0 -0.75; 700 0.492308 0 -0.75
93. 701 0.738462 0 -0.75; 702 0.984615 0 -0.75; 703 1.23077 0 -0.75
94. 704 1.47692 0 -0.75; 705 1.72308 0 -0.75; 706 1.96923 0 -0.75

```
 95. 707 2.21538 0 -0.75; 708 2.46154 0 -0.75; 709 2.70769 0 -0.75
 96. 710 2.95385 0 -0.75; 711 -0.25 0 7.9; 712 -0.65 0 7.9; 713 0 0 7.9
 97. 714 3.45 0 7.9; 715 3.2 0 7.9; 716 3.85 0 7.9; 717 0.246154 0 7.9
 98. 718 0.492308 0 7.9; 719 0.738462 0 7.9; 720 0.984615 0 7.9; 721 1.23077 0 7.9
 99. 722 1.47692 0 7.9; 723 1.72308 0 7.9; 724 1.96923 0 7.9; 725 2.21538 0 7.9
100. 726 2.46154 0 7.9; 727 2.70769 0 7.9; 728 2.95385 0 7.9; 729 -0.45 0 0
101. 730 -0.45 0 -0.25; 731 -0.45 0 -0.5; 732 3.65 0 0; 733 3.65 0 -0.25
102. 734 3.65 0 -0.5; 735 -0.45 0 7.65; 736 -0.45 0 7.4; 737 -0.45 0 7.15
103. 738 3.65 0 7.65; 739 3.65 0 7.4; 740 3.65 0 7.15; 741 -0.45 0 6.90345
104. 742 -0.45 0 6.6569; 743 -0.45 0 0.493102; 744 -0.45 0 0.246551
105. 745 3.65 0 6.90345; 746 3.65 0 6.6569; 747 3.65 0 0.493102
106. 748 3.65 0 0.246551; 749 -0.45 0 6.40363; 750 -0.45 0 6.15035
107. 751 -0.45 0 0.99965; 752 -0.45 0 0.746376; 753 3.65 0 6.40363
108. 754 3.65 0 6.15035; 755 3.65 0 0.99965; 756 3.65 0 0.746376; 757 -0.45 0 -0.75
109. 758 -0.45 0 -1; 759 3.65 0 -0.75; 760 3.65 0 -1; 761 -0.45 0 8.15
110. 762 -0.45 0 7.9; 763 3.65 0 8.15; 764 3.65 0 7.9
111. MEMBER INCIDENCES
112. 394 1 617; 395 2 618; 396 3 619; 397 4 620; 775 617 277; 776 618 278
113. 777 619 279; 778 620 280
114. ELEMENT INCIDENCES SHELL
115. 38 17 1 20 18; 41 18 20 9 21; 44 2 22 23 24; 48 24 23 26 10; 54 27 13 30 28
116. 57 28 30 3 31; 60 14 32 33 34; 64 34 33 36 4; 67 1 37 38 20; 69 37 39 40 38
117. 71 39 41 42 40; 73 41 43 44 42; 75 43 45 46 44; 77 45 47 48 46; 79 47 49 50 48
118. 81 49 51 52 50; 83 51 53 54 52; 85 53 55 56 54; 87 55 57 58 56; 89 57 59 60 58
119. 90 59 2 24 60; 92 20 38 61 9; 94 38 40 62 61; 96 40 42 63 62; 98 42 44 64 63
120. 100 44 46 65 64; 102 46 48 66 65; 104 48 50 67 66; 106 50 52 68 67
121. 108 52 54 69 68; 110 54 56 70 69; 112 56 58 71 70; 114 58 60 72 71
122. 115 60 24 10 72; 119 31 3 75 73; 123 73 75 78 76; 227 151 153 156 154
123. 229 154 156 1 17; 231 4 36 157 158; 235 158 157 160 161; 339 236 235 238 239
124. 342 239 238 22 2; 345 13 241 242 30; 347 241 243 244 242; 349 243 245 246 244
125. 351 245 247 248 246; 353 247 249 250 248; 355 249 251 252 250
126. 357 251 253 254 252; 359 253 255 256 254; 361 255 257 258 256
127. 363 257 259 260 258; 365 259 261 262 260; 367 261 263 264 262
128. 368 263 14 34 264; 370 30 242 265 3; 372 242 244 266 265; 374 244 246 267 266
129. 376 246 248 268 267; 378 248 250 269 268; 380 250 252 270 269
130. 382 252 254 271 270; 384 254 256 272 271; 386 256 258 273 272
131. 388 258 260 274 273; 390 260 262 275 274; 392 262 264 276 275
132. 393 264 34 4 276; 398 3 265 281 75; 399 265 266 282 281; 400 266 267 283 282
133. 401 267 268 284 283; 402 268 269 285 284; 403 269 270 286 285
134. 404 270 271 287 286; 405 271 272 288 287; 406 272 273 289 288
135. 407 273 274 290 289; 408 274 275 291 290; 409 275 276 292 291
136. 410 276 4 158 292; 411 75 281 293 78; 412 281 282 294 293; 413 282 283 295 294
137. 414 283 284 296 295; 415 284 285 297 296; 416 285 286 298 297
138. 417 286 287 299 298; 418 287 288 300 299; 419 288 289 301 300
139. 420 289 290 302 301; 421 290 291 303 302; 422 291 292 304 303
140. 423 292 158 161 304; 749 153 593 605 156; 750 593 594 606 605
141. 751 594 595 607 606; 752 595 596 608 607; 753 596 597 609 608
142. 754 597 598 610 609; 755 598 599 611 610; 756 599 600 612 611
143. 757 600 601 613 612; 758 601 602 614 613; 759 602 603 615 614
144. 760 603 604 616 615; 761 604 236 239 616; 762 156 605 37 1; 763 605 606 39 37
145. 764 606 607 41 39; 765 607 608 43 41; 766 608 609 45 43; 767 609 610 47 45
146. 768 610 611 49 47; 769 611 612 51 49; 770 612 613 53 51; 771 613 614 55 53
147. 772 614 615 57 55; 773 615 616 59 57; 774 616 239 2 59; 815 76 78 659 657
148. 816 657 659 81 79; 819 148 150 662 660; 820 660 662 153 151
149. 821 161 160 663 664; 822 664 663 163 164; 825 233 232 666 667
150. 826 667 666 235 236; 829 78 293 669 659; 830 659 669 305 81
```

```
151. 831 293 294 670 669; 832 669 670 306 305; 833 294 295 671 670
152. 834 670 671 307 306; 835 295 296 672 671; 836 671 672 308 307
153. 837 296 297 673 672; 838 672 673 309 308; 839 297 298 674 673
154. 840 673 674 310 309; 841 298 299 675 674; 842 674 675 311 310
155. 843 299 300 676 675; 844 675 676 312 311; 845 300 301 677 676
156. 846 676 677 313 312; 847 301 302 678 677; 848 677 678 314 313
157. 849 302 303 679 678; 850 678 679 315 314; 851 303 304 680 679
158. 852 679 680 316 315; 853 304 161 664 680; 854 680 664 164 316
159. 855 150 581 681 662; 856 662 681 593 153; 857 581 582 682 681
160. 858 681 682 594 593; 859 582 583 683 682; 860 682 683 595 594
161. 861 583 584 684 683; 862 683 684 596 595; 863 584 585 685 684
162. 864 684 685 597 596; 865 585 586 686 685; 866 685 686 598 597
163. 867 586 587 687 686; 868 686 687 599 598; 869 587 588 688 687
164. 870 687 688 600 599; 871 588 589 689 688; 872 688 689 601 600
165. 873 589 590 690 689; 874 689 690 602 601; 875 590 591 691 690
166. 876 690 691 603 602; 877 591 592 692 691; 878 691 692 604 603
167. 879 592 233 667 692; 880 692 667 236 604; 883 21 9 695 693
168. 884 693 695 623 621; 885 10 26 696 697; 886 697 696 624 625; 889 9 61 699 695
169. 890 695 699 627 623; 891 61 62 700 699; 892 699 700 628 627; 893 62 63 701 700
170. 894 700 701 629 628; 895 63 64 702 701; 896 701 702 630 629; 897 64 65 703 702
171. 898 702 703 631 630; 899 65 66 704 703; 900 703 704 632 631; 901 66 67 705 704
172. 902 704 705 633 632; 903 67 68 706 705; 904 705 706 634 633; 905 68 69 707 706
173. 906 706 707 635 634; 907 69 70 708 707; 908 707 708 636 635; 909 70 71 709 708
174. 910 708 709 637 636; 911 71 72 710 709; 912 709 710 638 637; 913 72 10 697 710
175. 914 710 697 625 638; 917 640 641 713 711; 918 711 713 13 27
176. 919 642 643 714 715; 920 715 714 32 14; 923 641 645 717 713
177. 924 713 717 241 13; 925 645 646 718 717; 926 717 718 243 241
178. 927 646 647 719 718; 928 718 719 245 243; 929 647 648 720 719
179. 930 719 720 247 245; 931 648 649 721 720; 932 720 721 249 247
180. 933 649 650 722 721; 934 721 722 251 249; 935 650 651 723 722
181. 936 722 723 253 251; 937 651 652 724 723; 938 723 724 255 253
182. 939 652 653 725 724; 940 724 725 257 255; 941 653 654 726 725
183. 942 725 726 259 257; 943 654 655 727 726; 944 726 727 261 259
184. 945 655 656 728 727; 946 727 728 263 261; 947 656 642 715 728
185. 948 728 715 14 263; 949 5 729 730 19; 950 729 17 18 730; 951 19 730 731 11
186. 952 730 18 21 731; 953 22 732 733 23; 954 732 7 25 733; 955 23 733 734 26
187. 956 733 25 12 734; 957 15 735 736 29; 958 735 27 28 736; 959 29 736 737 6
188. 960 736 28 31 737; 961 32 738 739 33; 962 738 16 35 739; 963 33 739 740 36
189. 964 739 35 8 740; 965 6 737 741 74; 966 737 31 73 741; 967 74 741 742 77
190. 968 741 73 76 742; 969 152 743 744 155; 970 743 151 154 744; 971 155 744 729 5
191. 972 744 154 17 729; 973 36 740 745 157; 974 740 8 159 745; 975 157 745 746 160
192. 976 745 159 162 746; 977 235 747 748 238; 978 747 237 240 748
193. 979 238 748 732 22; 980 748 240 7 732; 981 77 742 749 658; 982 742 76 657 749
194. 983 658 749 750 80; 984 749 657 79 750; 985 149 751 752 661
195. 986 751 148 660 752; 987 661 752 743 152; 988 752 660 151 743
196. 989 160 746 753 663; 990 746 162 665 753; 991 663 753 754 163
197. 992 753 665 165 754; 993 232 755 756 666; 994 755 234 668 756
198. 995 666 756 747 235; 996 756 668 237 747; 997 11 731 757 694
199. 998 731 21 693 757; 999 694 757 758 622; 1000 757 693 621 758
200. 1001 26 734 759 696; 1002 734 12 698 759; 1003 696 759 760 624
201. 1004 759 698 626 760; 1005 639 761 762 712; 1006 761 640 711 762
202. 1007 712 762 735 15; 1008 762 711 27 735; 1009 643 763 764 714
203. 1010 763 644 716 764; 1011 714 764 738 32; 1012 764 716 16 738
204. START GROUP DEFINITION
205. ELEMENT
206. _RAFT 38 41 44 48 54 57 60 64 67 69 71 73 75 77 79 81 83 85 87 89 90 92 94 -
```

```
207. 96 98 100 102 104 106 108 110 112 114 115 119 123 227 229 231 235 339 342 -
208. 345 347 349 351 353 355 357 359 361 363 365 367 368 370 372 374 376 378 380 -
209. 382 384 386 388 390 392 393 398 TO 423 749 TO 774 815 816 819 TO 822 825 -
210. 826 829 TO 880 883 TO 886 889 TO 914 917 TO 920 923 TO 1012
211. _RAFT_MOM 41 48 54 60 69 71 73 75 77 79 81 83 85 87 89 92 94 96 98 100 102 -
212. 104 106 108 110 112 114 115 123 227 235 339 345 347 349 351 353 355 357 359 -
213. 361 363 365 367 368 372 374 376 378 380 382 384 386 388 390 392 399 TO 409 -
214. 411 TO 423 749 TO 761 763 TO 773 815 816 819 TO 822 825 826 829 TO 880 883 -
215. 884 TO 886 889 TO 914 917 TO 920 923 TO 1012
216. _RAFT_SHEAR 71 73 75 77 79 81 83 85 87 96 98 100 102 104 106 108 110 112 349 -
217. 351 353 355 357 359 361 363 365 374 376 378 380 382 384 386 388 390 -
218. 400 TO 408 413 TO 421 751 TO 759 764 TO 772 815 816 819 TO 822 825 826 829 -
219. 830 TO 880 883 TO 886 889 TO 914 917 TO 920 923 TO 949 951 954 956 957 959 -
220. 962 964 965 967 969 971 974 976 978 980 TO 1012
221. END GROUP DEFINITION
222. ELEMENT PROPERTY
223. _RAFT THICKNESS 0.4
224. DEFINE MATERIAL START
225. ISOTROPIC CONCRETE
226. E 2.17185E+007
227. POISSON 0.17
228. DENSITY 25
229. ALPHA 1E-005
230. DAMP 0.05
231. TYPE CONCRETE
232. STRENGTH FCU 27579
233. END DEFINE MATERIAL
234. MEMBER PROPERTY AMERICAN
235. 394 TO 397 775 TO 778 PRIS YD 0.5 ZD 0.5
236. CONSTANTS
237. MATERIAL CONCRETE MEMB 38 41 44 48 54 57 60 64 67 69 71 73 75 77 79 81 83 -
238. 85 87 89 90 92 94 96 98 100 102 104 106 108 110 112 114 115 119 123 227 229 -
239. 231 235 339 342 345 347 349 351 353 355 357 359 361 363 365 367 368 370 372 -
240. 374 376 378 380 382 384 386 388 390 392 TO 423 749 TO 778 815 816 -
241. 819 TO 822 825 826 829 TO 880 883 TO 886 889 TO 914 917 TO 920 923 TO 1012
242. **********************
243. ************************************************************
244. SUPPORTS
245. _RAFT PLATE MAT DIRECT Y SUBGRADE 33750 PRINT
```

JOINT	INFLUENCE AREA
1	0.061591
2	0.061591
3	0.061591
4	0.061591
5	0.024828
6	0.024828
7	0.024828
8	0.024827
9	0.062019
10	0.062019
11	0.025000
12	0.025000
13	0.062019

14	0.062019
15	0.025000
16	0.025000
17	0.055862
18	0.056250
19	0.025000
20	0.062019
21	0.056250
22	0.055862
23	0.056250
24	0.062019
25	0.025000
26	0.056250
27	0.056250
28	0.056250
29	0.025000
30	0.062019
31	0.055862
32	0.056250
33	0.056250
34	0.062019
35	0.025000
36	0.055862
37	0.061114
38	0.061538
39	0.061114
40	0.061539
41	0.061114
42	0.061538
43	0.061114
44	0.061539
45	0.061114
46	0.061538
47	0.061114
48	0.061539
49	0.061114
50	0.061539
51	0.061113
52	0.061537
53	0.061114
54	0.061539
55	0.061114
56	0.061539
57	0.061114
58	0.061539
59	0.061114
60	0.061539
61	0.061538
62	0.061539
63	0.061538
64	0.061539
65	0.061538
66	0.061539
67	0.061539
68	0.061537
69	0.061539

70	0.061539
71	0.061539
72	0.061539
73	0.055474
74	0.024655
75	0.061163
76	0.056230
77	0.024991
78	0.061997
79	0.028494
80	0.012664
81	0.031416
148	0.028493
149	0.012664
150	0.031416
151	0.056230
152	0.024991
153	0.061998
154	0.055474
155	0.024655
156	0.061164
157	0.055474
158	0.061163
159	0.024655
160	0.056230
161	0.061996
162	0.024991
163	0.028494
164	0.031416
165	0.012664
232	0.028493
233	0.031415
234	0.012664
235	0.056230
236	0.061997
237	0.024991
238	0.055474
239	0.061163
240	0.024655
241	0.061538
242	0.061538
243	0.061538
244	0.061538
245	0.061538
246	0.061538
247	0.061538
248	0.061538
249	0.061538
250	0.061538
251	0.061539
252	0.061539
253	0.061539
254	0.061539
255	0.061537
256	0.061537
257	0.061539

258	0.061539
259	0.061539
260	0.061539
261	0.061539
262	0.061539
263	0.061539
264	0.061539
265	0.061114
266	0.061114
267	0.061114
268	0.061114
269	0.061114
270	0.061114
271	0.061114
272	0.061113
273	0.061114
274	0.061114
275	0.061114
276	0.061114
281	0.060689
282	0.060689
283	0.060689
284	0.060689
285	0.060689
286	0.060690
287	0.060690
288	0.060688
289	0.060690
290	0.060690
291	0.060690
292	0.060690
293	0.061516
294	0.061516
295	0.061516
296	0.061516
297	0.061516
298	0.061517
299	0.061517
300	0.061515
301	0.061517
302	0.061517
303	0.061517
304	0.061517
305	0.031173
306	0.031173
307	0.031173
308	0.031173
309	0.031173
310	0.031173
311	0.031173
312	0.031172
313	0.031173
314	0.031173
315	0.031173
316	0.031173
581	0.031172

582	0.031172
583	0.031172
584	0.031172
585	0.031172
586	0.031172
587	0.031172
588	0.031172
589	0.031172
590	0.031172
591	0.031172
592	0.031172
593	0.061517
594	0.061517
595	0.061517
596	0.061517
597	0.061517
598	0.061517
599	0.061517
600	0.061516
601	0.061517
602	0.061517
603	0.061517
604	0.061517
605	0.060690
606	0.060690
607	0.060689
608	0.060690
609	0.060689
610	0.060690
611	0.060690
612	0.060689
613	0.060690
614	0.060690
615	0.060690
616	0.060690
621	0.028125
622	0.012500
623	0.031010
624	0.028125
625	0.031009
626	0.012500
627	0.030769
628	0.030769
629	0.030769
630	0.030769
631	0.030769
632	0.030769
633	0.030769
634	0.030769
635	0.030769
636	0.030769
637	0.030769
638	0.030769
639	0.012500
640	0.028125
641	0.031010

642	0.031009
643	0.028125
644	0.012500
645	0.030769
646	0.030769
647	0.030769
648	0.030769
649	0.030769
650	0.030769
651	0.030769
652	0.030769
653	0.030769
654	0.030769
655	0.030769
656	0.030769
657	0.056987
658	0.025327
659	0.062832
660	0.056987
661	0.025327
662	0.062831
663	0.056987
664	0.062831
665	0.025327
666	0.056987
667	0.062831
668	0.025327
669	0.062345
670	0.062345
671	0.062344
672	0.062345
673	0.062344
674	0.062345
675	0.062345
676	0.062344
677	0.062345
678	0.062345
679	0.062345
680	0.062345
681	0.062344
682	0.062344
683	0.062344
684	0.062344
685	0.062344
686	0.062345
687	0.062345
688	0.062343
689	0.062345
690	0.062345
691	0.062345
692	0.062345
693	0.056250
694	0.025000
695	0.062019
696	0.056250
697	0.062019

698	0.025000
699	0.061538
700	0.061539
701	0.061538
702	0.061539
703	0.061538
704	0.061539
705	0.061539
706	0.061537
707	0.061539
708	0.061539
709	0.061539
710	0.061539
711	0.056250
712	0.025000
713	0.062019
714	0.056250
715	0.062019
716	0.025000
717	0.061539
718	0.061539
719	0.061538
720	0.061539
721	0.061538
722	0.061539
723	0.061539
724	0.061538
725	0.061539
726	0.061539
727	0.061539
728	0.061539
729	0.049655
730	0.050000
731	0.050000
732	0.049655
733	0.050000
734	0.050000
735	0.050000
736	0.050000
737	0.049655
738	0.050000
739	0.050000
740	0.049655
741	0.049310
742	0.049982
743	0.049982
744	0.049310
745	0.049310
746	0.049982
747	0.049982
748	0.049310
749	0.050655
750	0.025328
751	0.025327
752	0.050655
753	0.050655

```
       754          0.025328
       755          0.025327
       756          0.050655
       757          0.050000
       758          0.025000
       759          0.050000
       760          0.025000
       761          0.025000
       762          0.050000
       763          0.025000
       764          0.050000
    TOTAL          17.996841
246. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
           MASTER/SLAVE OR IF UNCONNECTED JOINTS.
247. JOINT LOAD
248. 279 FX 0 FY -20 FZ 0
249. 280 FX 0 FY -20 FZ 0
250. 277 FX 0 FY -20 FZ 0
251. 278 FX 0 FY -20 FZ 0
252. SELFWEIGHT Y -1 LIST ALL
253. ***
254. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
255. JOINT LOAD
256. 279 FX 0 FY -50 FZ 0
257. 280 FX 0 FY -30 FZ 0
258. 277 FX -10 FY -40 FZ 10
259. 278 FX 10 FY -30 FZ -10
260. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
261. JOINT LOAD
262. 279 FX 10 FY 70 FZ -10
263. 280 FX 10 FY -70 FZ -10
264. 277 FX 10 FY 60 FZ -10
265. 278 FX 10 FY -60 FZ -10
266. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
267. JOINT LOAD
268. 279 FX -10 FY -70 FZ 10
269. 280 FX -10 FY 70 FZ 10
270. 277 FX -10 FY -60 FZ 10
271. 278 FX -10 FY 60 FZ 10
272. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
273. JOINT LOAD
274. 279 FX 10 FY -10 FZ 10
275. 280 FX 10 FY -10 FZ 0
276. 277 FX 10 FY 10 FZ 10
277. 278 FX 10 FY 10 FZ 10
278. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
279. JOINT LOAD
280. 279 FX -10 FY 10 FZ -10
281. 280 FX -10 FY 10 FZ 0
282. 277 FX -10 FY -10 FZ -10
283. 278 FX -10 FY -10 FZ -10
284. LOAD 7 LOADTYPE SOIL  TITLE SOIL LOAD (HL)
285. ELEMENT LOAD
286. _RAFT PR -19.8
287. ******************************************************
```

```
288. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
289. LOAD 101 1.4DL
290. REPEAT LOAD
291. 1 1.4
292. LOAD 102 1.4DL+1.7LL
293. REPEAT LOAD
294. 1 1.4 2 1.7
295. LOAD 103 1.2DL+1.0LL+1.6WLX1
296. REPEAT LOAD
297. 1 1.2 2 1.0 3 1.6
298. LOAD 104 1.2DL+1.0LL+1.6WLX2
299. REPEAT LOAD
300. 1 1.2 2 1.0 4 1.6
301. LOAD 105 1.2DL+1.0LL+1.6WLZ1
302. REPEAT LOAD
303. 1 1.2 2 1.0 5 1.6
304. LOAD 106 1.2DL+1.0LL+1.6WLZ2
305. REPEAT LOAD
306. 1 1.2 2 1.0 6 1.6
307. LOAD 107 0.9DL+1.6HL+1.6WLX1
308. REPEAT LOAD
309. 1 0.9 7 1.6 3 1.6
310. LOAD 108 0.9DL+1.6HL+1.6WLX2
311. REPEAT LOAD
312. 1 0.9 7 1.6 4 1.6
313. LOAD 109 0.9DL+1.6HL+1.6WLZ1
314. REPEAT LOAD
315. 1 0.9 7 1.6 5 1.6
316. LOAD 110 0.9DL+1.6HL+1.6WLZ2
317. REPEAT LOAD
318. 1 0.9 7 1.6 6 1.6
319. ***************
320. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**368**	**NUMBER OF MEMBERS**	**8**
NUMBER OF PLATES	**304**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**360**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 352/ 17/ 54 DOF
TOTAL PRIMARY LOAD CASES = 17, TOTAL DEGREES OF FREEDOM = 1128
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 61 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 14.5/ 111851.2 MB

321. CHANGE
322. SUPPORTS
323. _RAFT PLATE MAT DIRECT Y SUBGRADE 33750 PRINT COMPRESSION

JOINT	INFLUENCE AREA
1	0.061591
2	0.061591
3	0.061591
4	0.061591
5	0.024828
6	0.024828
7	0.024828
8	0.024827
9	0.062019
10	0.062019
11	0.025000
12	0.025000
13	0.062019
14	0.062019
15	0.025000
16	0.025000
17	0.055862
18	0.056250
19	0.025000
20	0.062019
21	0.056250
22	0.055862
23	0.056250
24	0.062019
25	0.025000
26	0.056250
27	0.056250
28	0.056250
29	0.025000
30	0.062019
31	0.055862
32	0.056250
33	0.056250
34	0.062019
35	0.025000
36	0.055862
37	0.061114
38	0.061538
39	0.061114
40	0.061539
41	0.061114
42	0.061538
43	0.061114
44	0.061539
45	0.061114
46	0.061538
47	0.061114
48	0.061539

49	0.061114
50	0.061539
51	0.061113
52	0.061537
53	0.061114
54	0.061539
55	0.061114
56	0.061539
57	0.061114
58	0.061539
59	0.061114
60	0.061539
61	0.061538
62	0.061539
63	0.061538
64	0.061539
65	0.061538
66	0.061539
67	0.061539
68	0.061537
69	0.061539
70	0.061539
71	0.061539
72	0.061539
73	0.055474
74	0.024655
75	0.061163
76	0.056230
77	0.024991
78	0.061997
79	0.028494
80	0.012664
81	0.031416
148	0.028493
149	0.012664
150	0.031416
151	0.056230
152	0.024991
153	0.061998
154	0.055474
155	0.024655
156	0.061164
157	0.055474
158	0.061163
159	0.024655
160	0.056230
161	0.061996
162	0.024991
163	0.028494
164	0.031416
165	0.012664
232	0.028493
233	0.031415
234	0.012664
235	0.056230
236	0.061997

237	0.024991
238	0.055474
239	0.061163
240	0.024655
241	0.061538
242	0.061538
243	0.061538
244	0.061538
245	0.061538
246	0.061538
247	0.061538
248	0.061538
249	0.061538
250	0.061538
251	0.061539
252	0.061539
253	0.061539
254	0.061539
255	0.061537
256	0.061537
257	0.061539
258	0.061539
259	0.061539
260	0.061539
261	0.061539
262	0.061539
263	0.061539
264	0.061539
265	0.061114
266	0.061114
267	0.061114
268	0.061114
269	0.061114
270	0.061114
271	0.061114
272	0.061113
273	0.061114
274	0.061114
275	0.061114
276	0.061114
281	0.060689
282	0.060689
283	0.060689
284	0.060689
285	0.060689
286	0.060690
287	0.060690
288	0.060688
289	0.060690
290	0.060690
291	0.060690
292	0.060690
293	0.061516
294	0.061516
295	0.061516
296	0.061516

297	0.061516
298	0.061517
299	0.061517
300	0.061515
301	0.061517
302	0.061517
303	0.061517
304	0.061517
305	0.031173
306	0.031173
307	0.031173
308	0.031173
309	0.031173
310	0.031173
311	0.031173
312	0.031172
313	0.031173
314	0.031173
315	0.031173
316	0.031173
581	0.031172
582	0.031172
583	0.031172
584	0.031172
585	0.031172
586	0.031172
587	0.031172
588	0.031172
589	0.031172
590	0.031172
591	0.031172
592	0.031172
593	0.061517
594	0.061517
595	0.061517
596	0.061517
597	0.061517
598	0.061517
599	0.061517
600	0.061516
601	0.061517
602	0.061517
603	0.061517
604	0.061517
605	0.060690
606	0.060690
607	0.060689
608	0.060690
609	0.060689
610	0.060690
611	0.060690
612	0.060689
613	0.060690
614	0.060690
615	0.060690
616	0.060690

621	0.028125
622	0.012500
623	0.031010
624	0.028125
625	0.031009
626	0.012500
627	0.030769
628	0.030769
629	0.030769
630	0.030769
631	0.030769
632	0.030769
633	0.030769
634	0.030769
635	0.030769
636	0.030769
637	0.030769
638	0.030769
639	0.012500
640	0.028125
641	0.031010
642	0.031009
643	0.028125
644	0.012500
645	0.030769
646	0.030769
647	0.030769
648	0.030769
649	0.030769
650	0.030769
651	0.030769
652	0.030769
653	0.030769
654	0.030769
655	0.030769
656	0.030769
657	0.056987
658	0.025327
659	0.062832
660	0.056987
661	0.025327
662	0.062831
663	0.056987
664	0.062831
665	0.025327
666	0.056987
667	0.062831
668	0.025327
669	0.062345
670	0.062345
671	0.062344
672	0.062345
673	0.062344
674	0.062345
675	0.062345
676	0.062344

677	0.062345
678	0.062345
679	0.062345
680	0.062345
681	0.062344
682	0.062344
683	0.062344
684	0.062344
685	0.062344
686	0.062345
687	0.062345
688	0.062343
689	0.062345
690	0.062345
691	0.062345
692	0.062345
693	0.056250
694	0.025000
695	0.062019
696	0.056250
697	0.062019
698	0.025000
699	0.061538
700	0.061539
701	0.061538
702	0.061539
703	0.061538
704	0.061539
705	0.061539
706	0.061537
707	0.061539
708	0.061539
709	0.061539
710	0.061539
711	0.056250
712	0.025000
713	0.062019
714	0.056250
715	0.062019
716	0.025000
717	0.061539
718	0.061539
719	0.061538
720	0.061539
721	0.061538
722	0.061539
723	0.061539
724	0.061538
725	0.061539
726	0.061539
727	0.061539
728	0.061539
729	0.049655
730	0.050000
731	0.050000
732	0.049655

733	0.050000
734	0.050000
735	0.050000
736	0.050000
737	0.049655
738	0.050000
739	0.050000
740	0.049655
741	0.049310
742	0.049982
743	0.049982
744	0.049310
745	0.049310
746	0.049982
747	0.049982
748	0.049310
749	0.050655
750	0.025328
751	0.025327
752	0.050655
753	0.050655
754	0.025328
755	0.025327
756	0.050655
757	0.050000
758	0.025000
759	0.050000
760	0.025000
761	0.025000
762	0.050000
763	0.025000
764	0.050000
TOTAL	17.996841

```
324. ********************************************************************************
325. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
326. ************************************************************
327. LOAD 201 1.0DL+1.0HL+1.0LL
328. REPEAT LOAD
329. 1 1.0 7 1.0 2 1.0
330. LOAD 202 1.0DL+1.0HL+0.75LL
331. REPEAT LOAD
332. 1 1.0 7 1.0 2 0.75
333. LOAD 203 1.0DL+1.0HL+1.0WLX1
334. REPEAT LOAD
335. 1 1.0 7 1.0 3 1.0
336. LOAD 204 1.0DL+1.0HL+1.0WLX2
337. REPEAT LOAD
338. 1 1.0 7 1.0 4 1.0
339. LOAD 205 1.0DL+1.0HL+1.0WLZ1
340. REPEAT LOAD
341. 1 1.0 7 1.0 5 1.0
342. LOAD 206 1.0DL+1.0HL+1.0WLZ2
343. REPEAT LOAD
344. 1 1.0 7 1.0 6 1.0
345. LOAD 207 1.0DL+1.0HL+0.75LL+0.75WLX1
346. REPEAT LOAD
```

```
347. 1 1.0 7 1.0 2 0.75 3 0.75
348. LOAD 208 1.0DL+1.0HL+0.75LL+0.75WLX2
349. REPEAT LOAD
350. 1 1.0 7 1.0 2 0.75 4 0.75
351. LOAD 209 1.0DL+1.0HL+0.75LL+0.75WLZ1
352. REPEAT LOAD
353. 1 1.0 7 1.0 2 0.75 5 0.75
354. LOAD 210 1.0DL+1.0HL+0.75LL+0.75WLZ2
355. REPEAT LOAD
356. 1 1.0 7 1.0 2 0.75 6 0.75
357. LOAD 211 0.6DL+1.0HL+1.0WLX1
358. REPEAT LOAD
359. 1 0.6 7 1.0 3 1.0
360. LOAD 212 0.6DL+1.0HL+1.0WLX2
361. REPEAT LOAD
362. 1 0.6 7 1.0 4 1.0
363. LOAD 213 0.6DL+1.0HL+1.0WLZ1
364. REPEAT LOAD
365. 1 0.6 7 1.0 5 1.0
366. LOAD 214 0.6DL+1.0HL+1.0WLZ2
367. REPEAT LOAD
368. 1 0.6 7 1.0 6 1.0
369. ********
370. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  201



**NOTE-Tension/Compression converged after  1 iterations, Case=  202



**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  203



**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=  204



**NOTE-Tension/Compression converged after  1 iterations, Case=  205
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 211

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

371. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 13,2020 TIME= 9:26:47 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 3 STAAD INPUT & OUTPUT FILE FOR F2 STABILITY

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5           *
*           Version  20.07.10.64                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    APR 13, 2020                 *
*           Time=     9:29:42                     *
*                                                *
*      USER ID: LT Construction                  *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F2 STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 09-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 3.2 0 0; 3 0 0 7.15; 4 3.2 0 7.15; 277 0 2.75 0; 278 3.2 2.75 0
     9. 279 0 2.75 7.15; 280 3.2 2.75 7.15
    10. MEMBER INCIDENCES
    11. 394 1 277; 395 2 278; 396 3 279; 397 4 280
    12. DEFINE MATERIAL START
    13. ISOTROPIC CONCRETE
    14. E 2.17185E+007
    15. POISSON 0.17
    16. DENSITY 25
    17. ALPHA 1E-005
    18. DAMP 0.05
    19. TYPE CONCRETE
    20. STRENGTH FCU 27579
    21. END DEFINE MATERIAL
    22. MEMBER PROPERTY AMERICAN
    23. 394 TO 397 PRIS YD 0.5 ZD 0.5
    24. CONSTANTS
    25. MATERIAL CONCRETE ALL
    26. **********************
    27. ************************************************************
    28. SUPPORTS
    29. 1 TO 4 FIXED
    30. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
   **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
              MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    31. JOINT LOAD
    32. 279 FX 0 FY -20 FZ 0
    33. 280 FX 0 FY -20 FZ 0
    34. 277 FX 0 FY -20 FZ 0
    35. 278 FX 0 FY -20 FZ 0
    36. ***
```

```
37. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
38. JOINT LOAD
39. 279 FX 0 FY -50 FZ 0
40. 280 FX 0 FY -30 FZ 0
41. 277 FX -10 FY -40 FZ 10
42. 278 FX 10 FY -30 FZ -10
43. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD + X ( WLX1 )
44. JOINT LOAD
45. 279 FX 10 FY 70 FZ -10
46. 280 FX 10 FY -70 FZ -10
47. 277 FX 10 FY 60 FZ -10
48. 278 FX 10 FY -60 FZ -10
49. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD - X ( WLX2 )
50. JOINT LOAD
51. 279 FX -10 FY -70 FZ 10
52. 280 FX -10 FY 70 FZ 10
53. 277 FX -10 FY -60 FZ 10
54. 278 FX -10 FY 60 FZ 10
55. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD + Z ( WLZ1 )
56. JOINT LOAD
57. 279 FX 10 FY -10 FZ 10
58. 280 FX 10 FY -10 FZ 0
59. 277 FX 10 FY 10 FZ 10
60. 278 FX 10 FY 10 FZ 10
61. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD - Z ( WLZ2 )
62. JOINT LOAD
63. 279 FX -10 FY 10 FZ -10
64. 280 FX -10 FY 10 FZ 0
65. 277 FX -10 FY -10 FZ -10
66. 278 FX -10 FY -10 FZ -10
67. ********************************************************************************
68. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
69. ************************************************************
70. LOAD COMB 201 1.0DL+1.0LL
71. 1 1.0 2 1.0
72. LOAD COMB 202 1.0DL+0.75LL
73. 1 1.0 2 0.75
74. LOAD COMB 203 1.0DL+1.0WLX1
75. 1 1.0 3 1.0
76. LOAD COMB 204 1.0DL+1.0WLX2
77. 1 1.0 4 1.0
78. LOAD COMB 205 1.0DL+1.0WLZ1
79. 1 1.0 5 1.0
80. LOAD COMB 206 1.0DL+1.0WLZ2
81. 1 1.0 6 1.0
82. LOAD COMB 207 1.0DL+0.75LL+0.75WLX1
83. 1 1.0 2 0.75 3 0.75
84. LOAD COMB 208 1.0DL+0.75LL+0.75WLX2
85. 1 1.0 2 0.75 4 0.75
86. LOAD COMB 209 1.0DL+0.75LL+0.75WLZ1
87. 1 1.0 2 0.75 5 0.75
88. LOAD COMB 210 1.0DL+0.75LL+0.75WLZ2
89. 1 1.0 2 0.75 6 0.75
90. LOAD COMB 211 0.6DL+1.0WLX1
91. 1 0.6 3 1.0
92. LOAD COMB 212 0.6DL+1.0WLX2
```

```
 93. 1 0.6 4 1.0
 94. LOAD COMB 213 0.6DL+1.0WLZ1
 95. 1 0.6 5 1.0
 96. LOAD COMB 214 0.6DL+1.0WLZ2
 97. 1 0.6 6 1.0
 98. ********
 99. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**8**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**4**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 4/ 4/ 6 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 14 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 111846.8 MB

```
100. FINISH

          ********** END OF THE STAAD.Pro RUN **********

            **** DATE= APR 13,2020   TIME=  9:29:48 ****
```

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                    *
************************************************************
```